

Trade Restrictiveness Indices with Discriminatory Tariffs

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This paper extends the Trade Restrictiveness Index (TRI) to preferential tariffs. Using data for 2000–2020, a period of rapid expansion of Preferential Trade Agreements (PTAs), we compute 2,406 annual TRIs for 136 countries in an Armington model with large countries. Once terms-of-trade effects are taken into account, the global import-weighted TRI lies well below the average tariff, and the TRI for preferential tariffs declines faster than the TRI for MFN tariffs as the share of trade under PTAs grows. Across countries, average tariffs track the TRI poorly: the largest importers have TRIs far below their average tariffs, while many small economies have TRIs above them. Assuming that countries are small overstates the TRI, most of all for large importers.

JEL: F13, F14

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1. Introduction

The measurement of trade policy restrictiveness has long been a pivotal topic of international trade, offering critical insights into how tariffs and non-tariff barriers (NTBs) affect economic welfare, market efficiency, and global trade liberalization. Policymakers often rely on aggregated measures of tariffs to evaluate the welfare implications of trade policies. However, this task is particularly challenging due to the need to aggregate tariffs across more than 5,000 products and 200 trading partners (Anderson and Neary 2005), the complexity introduced by large-country effects that influence global prices, and the dynamic nature of trade policy, characterized by the proliferation of preferential trade agreements (PTAs), which create asymmetrical tariff structures. Simplistic indicators, such as simple or import-weighted average tariffs, are insufficient to address these complexities. As noted by Goldberg and Pavcnik (2016), studies analyzing trade policy impacts often rely on aggregate tariffs, and such aggregation must ideally be guided by theory to ensure accuracy.

The Trade Restrictiveness Index (TRI), introduced by Anderson and Neary, has become a widely used metric for this purpose (Anderson and Neary 1994). The TRI quantifies trade policy restrictiveness by determining the uniform tariff that replicates the welfare impact of a country's observed tariff structure (Anderson and Neary 1994, 1996). This theoretically grounded measure has significantly enhanced our understanding of how trade policies influence market distortions and welfare outcomes.

Despite its theoretical appeal, empirical applications of the TRI remain limited. Kee, Nicita, and Olarreaga (2009) employed data from 2000 to 2004 to estimate the TRI for over 100 countries, focusing on MFN tariffs and NTBs under a small-country framework. This approach assumes that individual countries cannot influence world prices, a simplification that overlooks the terms-of-trade effects arising when large economies strategically use tariffs to influence global prices (Bagwell and Staiger 1999). Soderbery (2017) extended the TRI framework to account for large-country effects and incorporate product varieties, using data from 1990 to 2007 for 182 countries and 1,087 HS4 products. However, even this work remained limited to MFN tariffs. More recently, Lake et al. have further extended this literature by incorporating NTBs into the TRI under the large-country assumption.

Those works have focused on MFN (Most-Favored Nation) tariffs, often ignoring preferential trade agreements (PTAs), which have reshaped the global trade landscape in the past two decades. The early 2000s were characterized by two key developments. First, as shown in Figure 1A, average MFN tariffs declined significantly between 2000 and 2005, dropping from approximately 9% to 7%, and have remained relatively stable around 6% since then. Second, preferential trade agreements (PTAs) and the resulting non-MFN tariffs have significantly reshaped global trade policies (Baldwin 2016). As illustrated in Figure 1B, the share of trade conducted under non-MFN tariffs rose sharply from just over 30% in 2000 to over 50% today. The proliferation of PTAs allowed member countries to trade at preferential tariffs, often zero, introducing asymmetrical tariff structures. These differentiated tariffs create trade distortions that complicate welfare analysis, yet their role remains underexplored in the existing TRI literature.

In this paper, we address critical gaps in the literature by extending the TRI framework to explicitly account for non-MFN tariffs, product varieties, and the large-country assumption, providing a more comprehensive measure of trade policy restrictiveness in the modern era of PTAs. First, we incorporate non-MFN tariffs and product varieties, which are intrinsic to PTAs and have introduced significant asymmetries in global trade. PTAs allow member countries to trade with each other at preferential tariffs, often far below MFN rates or even at

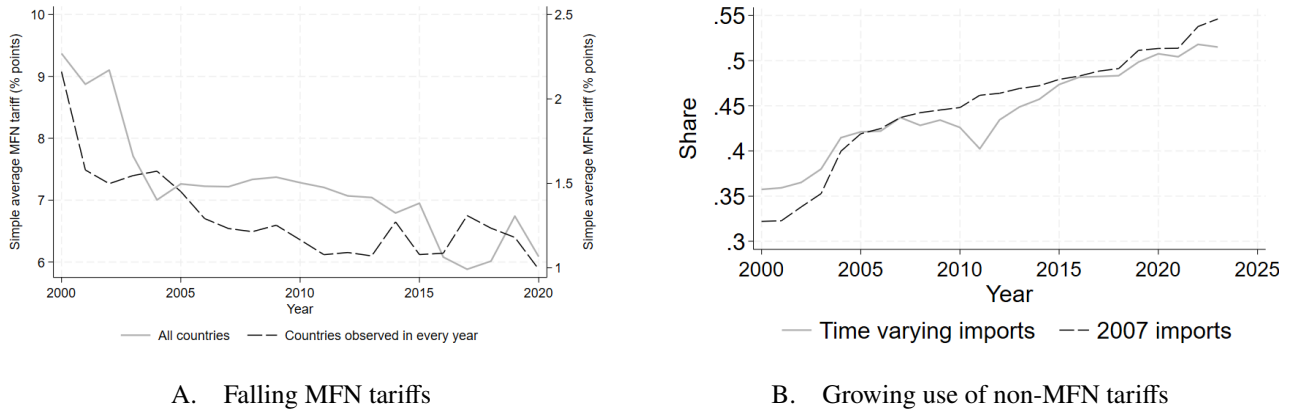


Figure 1: MFN and Non-MFN Tariffs, 2000–2020

Notes. In panel A, the solid line represents all recorded data, regardless of whether the country had complete observations between 2000 and 2020; the dashed line includes only countries with complete records over this period.

zero. These differentiated tariffs distort trade flows by creating uneven playing fields between PTA members and non-members. By explicitly modeling non-MFN tariffs, our analysis captures the welfare impacts of this asymmetry, addressing a critical dimension that has been largely overlooked in prior TRI studies.

Second, we account for the large-country effects that are amplified in the context of PTAs. Large economies such as the United States, the European Union, and China often drive the formation of PTAs and use these agreements to strategically shape global trade. By leveraging their market power, these economies can influence world prices, creating terms-of-trade effects that alter both global and domestic welfare outcomes. Our extended TRI framework captures these strategic interactions, offering a more realistic analysis of how large economies use PTAs to influence global trade dynamics.

Finally, this study utilizes a comprehensive dataset spanning from 2000 to 2020, a period characterized by the rapid proliferation of PTAs and significant changes in global trade policy. The early 2000s witnessed sharp declines in average MFN tariffs, coupled with a dramatic increase in the share of trade conducted under non-MFN tariffs. By covering this critical period, our analysis not only highlights the evolving role of PTAs but also reveals their long-term implications for trade liberalization and welfare.

The rest of this article is organized as follows: Section 2 establishes a trade model and details the logic behind calculating the TRI under the large-country assumption, accounting for product varieties. Section 3 introduces the dataset used in this study. Section 4 applies our model to the data and presents the results. Section 5 concludes with a summary of this work.

2. The model

In this section, we model import demand and export supply under the assumption of a large country, incorporating non-MFN tariffs and product varieties. The normalized random taste shock on the import side is identified using real-world data, as are the key parameters required for calculating the Trade Restrictiveness Index (TRI). Equilibrium prices—both delivered and shipped—are then determined, ensuring a balance between export supply and import demand at varying tariff levels. We provide first- and second-order derivatives of equilibrium prices and social welfare with respect to tariffs, which form the basis for deriving the conditions necessary for

calculating the TRI under large-country assumptions. Additionally, we show that the TRI formula used under small-country assumptions, as referenced in Kee's work (Kee, Nicita, and Olarreaga 2009), is a special case of our findings. Furthermore, we identify the tariff level that maximizes welfare at different price levels, although this tariff level is unstable. To address this, we present the conditions for stable equilibrium, where optimal tariffs and prices can be determined. However, since nonlinear equations govern these conditions, deriving explicit solutions or numerical values is challenging. Thus, we propose an iterative approach in a multi-period setting, facilitating convergence toward equilibrium values for tariffs and prices in large-country trade dynamics.

2.1. Estimating TRIs

The TRI quantifies the overall welfare distortions caused by a country's trade policy. It addresses the question: what uniform tariff rate, applied to all imports in place of the existing trade protection structure, would maintain the same level of national welfare? Formally, as articulated by Feenstra (1995), the TRI represents the uniform tariff equivalent that encapsulates the overall effect of a country's trade barriers on welfare, as expressed in the following equation:

$$(1) \quad \sum_g W_g^I(\text{TRI}^I) = \sum_g W_g^I(\bar{T}_g^I) = \bar{W}^I,$$

where $\bar{T}_g^I = [\bar{\tau}_{g1}^I, \bar{\tau}_{g2}^I, \dots, \bar{\tau}_{gV}^I]^\top$ is the vector of observed tariff lines applied to product g in country I , with each element $\bar{\tau}_{gv}^I$ denoting the current tariff imposed on variety v . $W_g^I(\bar{T}_g^I)$ represents the aggregated welfare associated with imports of good g in country I , considering all varieties under the current tariff rates. Similarly, $\bar{W}^I$ denotes the overall aggregated welfare in the country I corresponding to its existing tariff structure.

A second-order Taylor expansion is employed around the zero tariff levels to derive the TRI¹. Formally, the welfare associated with imports of good g under a tariff vector T_g^I is expressed as:

$$(2) \quad W_g^I(T_g^I) \approx W_g^I(\mathbf{0}) + [\nabla W_g^I(\mathbf{0})] T_g^I + \frac{1}{2} (T_g^I)^\top H(W_g^I(\mathbf{0})) T_g^I,$$

where $W_g^I(\mathbf{0})$ represents the aggregated welfare associated with imports of good g in country I under zero tariff rates (free trade). The term $\nabla W_g^I(\mathbf{0})$ is the gradient vector of welfare evaluated at the zero tariff level, given by:

$$\nabla W_g^I(\mathbf{0}) = \left[\frac{\partial W_g^I}{\partial \tau_{g1}^I} \Big|_{T_g^I=\mathbf{0}}, \frac{\partial W_g^I}{\partial \tau_{g2}^I} \Big|_{T_g^I=\mathbf{0}}, \dots, \frac{\partial W_g^I}{\partial \tau_{gV}^I} \Big|_{T_g^I=\mathbf{0}} \right].$$

The term $H(W_g^I(\mathbf{0}))$ is the Hessian matrix of welfare evaluated at the zero tariff level, which is expressed as:

¹The method employed in this paper allows for a second-order Taylor expansion around any tariff level. An intuitive choice for the expansion point should carry economic significance. Kee, Nicita, and Olarreaga (2009) choose the zero-tariff level (free trade) as their expansion point, which under the small-country assumption aligns with the optimal tariff level (where the welfare derivative with respect to the tariff is zero). In this paper, we adopt a large-country framework, where the optimal tariff level differs from the free trade scenario. Nevertheless, we perform the Taylor expansion around the zero-tariff point primarily because approximation error decreases as the expansion point approaches the observed tariff levels. In our dataset, the mean absolute gap between zero and observed tariffs is 4.50% for non-MFN tariffs and 6.57% for MFN tariffs, which is smaller than the corresponding mean absolute gaps between the maximum welfare tariff and observed tariffs (5.68% and 7.42% for non-MFN and MFN tariffs, respectively).

$$H(W_g^I(\mathbf{0})) = \begin{bmatrix} \frac{\partial^2 W^I}{\partial \tau_{g1}^I \partial \tau_{g1}^I} \Big|_{T_g^I=0} & \frac{\partial^2 W^I}{\partial \tau_{g1}^I \partial \tau_{g2}^I} \Big|_{T_g^I=0} & \cdots & \frac{\partial^2 W^I}{\partial \tau_{g1}^I \partial \tau_{gV}^I} \Big|_{T_g^I=0} \\ \frac{\partial^2 W^I}{\partial \tau_{g2}^I \partial \tau_{g1}^I} \Big|_{T_g^I=0} & \frac{\partial^2 W^I}{\partial \tau_{g2}^I \partial \tau_{g2}^I} \Big|_{T_g^I=0} & \cdots & \frac{\partial^2 W^I}{\partial \tau_{g2}^I \partial \tau_{gV}^I} \Big|_{T_g^I=0} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 W^I}{\partial \tau_{gV}^I \partial \tau_{g1}^I} \Big|_{T_g^I=0} & \frac{\partial^2 W^I}{\partial \tau_{gV}^I \partial \tau_{g2}^I} \Big|_{T_g^I=0} & \cdots & \frac{\partial^2 W^I}{\partial \tau_{gV}^I \partial \tau_{gV}^I} \Big|_{T_g^I=0} \end{bmatrix}.$$

The first term, $W_g^I(\mathbf{0})$, in Equation 2 is not significant in the calculation of TRI, because when substituting Equation 2 into Equation 1, $W_g^I(\mathbf{0})$ appears on both sides of the equation and thus cancels out. This simplification allows the focus to be placed on the remaining terms. The second term in Equation 2 captures the direct impact of tariff changes on welfare, primarily driven by redistribution and revenue effects.²

Substituting Equation 2 into Equation 1 yields a quadratic function of TRI^I , the solution of which is given by:

$$(3) \quad \text{TRI}^I = \frac{-b^I + \sqrt{(b^I)^2 + 4a^I c^I}}{2a^I},$$

where

$$a^I = \frac{1}{2} \sum_g \sum_v \sum_k \frac{\partial^2 W^I}{\partial \tau_{gv}^I \partial \tau_{gk}^I} \Big|_{T_g^I=0}$$

is negative,

$$b^I = \sum_g \sum_v \frac{\partial W_g^I}{\partial \tau_{gv}^I} \Big|_{T_g^I=0}$$

is positive, and

$$c^I = \sum_g \left[(\nabla W_g^I(\mathbf{0})) \bar{T}_g^I + \frac{1}{2} (\bar{T}_g^I)^\top H(W_g^I(\mathbf{0})) \bar{T}_g^I \right]$$

is also positive.

It is important to note that the TRI^I is uniquely determined by the parameters a^I , b^I , and c^I , as well as the current tariff levels. Assuming that current tariff levels are observable, the estimation of the TRI^I reduces to identifying the values of a^I , b^I , and c^I . These parameters, in turn, are uniquely determined by $\nabla W_g^I(\mathbf{0})$ and $H(W_g^I(\mathbf{0}))$. Therefore, a necessary condition for calculating the TRI^I is the estimation of the first- and second-order derivatives of welfare with respect to tariffs at the zero-tariff level.

²This term being non-zero (and negative) highlights a critical distinction of this study compared to [Kee, Nicita, and Olarreaga \(2009\)](#). In Equation (15) of [Kee, Nicita, and Olarreaga \(2009\)](#), the first-order derivative of welfare with respect to tariffs under free trade is zero, $\frac{\partial W}{\partial \tau} \Big|_{\tau=0} = 0$. This result arises from the small-country assumption, where the importing country is unable to influence international prices. Consequently, tariffs do not generate terms-of-trade gains, and welfare is maximized at zero tariffs. Beyond this point, welfare declines monotonically with increasing tariffs due to resource misallocation, diminished consumer surplus, and rising deadweight loss, as reflected in $\frac{\partial W}{\partial \tau} \leq 0$ (with equality only at $\tau = 0$) and $\frac{\partial^2 W}{\partial \tau^2} < 0$.

Under the large-country assumption, however, the welfare function incorporates an additional term capturing the terms-of-trade effect, which reflects import price reductions induced by tariffs. At low tariff levels, this terms-of-trade effect dominates, leading to welfare gains. However, as tariffs increase beyond the optimal level, resource misallocation intensifies, causing welfare to decline. Consequently, the welfare function exhibits an “inverted-U” shape, with welfare initially rising due to terms-of-trade improvements and eventually falling as distortionary effects prevail.

2.2. Welfare Derivatives with Tariffs

In the previous subsection, we established a necessary condition for deriving TRI^I, namely, the estimation of the gradient vector and Hessian matrix of welfare with respect to tariffs at the zero-tariff level (i.e., the first- and second-order derivatives). In this subsection, we present the explicit expressions for these elements.

We begin by adopting the generic welfare function from Feenstra (1995). Under standard assumptions regarding the market conduct of domestic firms and the separability of welfare across goods, Feenstra (1995) derives the marginal effect of a tariff change on welfare as:

$$\frac{\partial W^I}{\partial \tau_g^I} = \tau_g^I p_g^{I*} \frac{\partial x_g^I}{\partial \tau_g^I} - x_g^I \frac{\partial p_g^{I*}}{\partial \tau_g^I},$$

where all terms are scalars because Feenstra's analysis does not account for product variety. Specifically, τ_g^I represents the tariff rate imposed by importing country I on good g , p_g^{I*} denotes the international price at that tariff level, and x_g^I is the corresponding import quantity.

When accounting for product variety, interactions between tariff changes on one variety and the prices of other varieties introduce cross-price or price spillover effects, significantly complicating the analysis. To systematically capture these effects, it is more intuitive and analytically rigorous to express the relationship in matrix form. Recasting the equation in terms of trade values and trade elasticities, we rewrite it as:

$$(4) \quad \nabla W_g^I(\mathbf{T}_g^I) = \frac{\partial W^I}{\partial \mathbf{T}_g^I} = \mathbf{T}_g^I \circ \mathbf{P}_g^{I*} \circ \mathbf{X}_g^I \circ \frac{\partial \ln(\mathbf{X}_g^I)}{\partial \ln(\mathbf{P}_g^I)} \frac{\partial \ln(\mathbf{P}_g^I)}{\partial \mathbf{T}_g^I} - \mathbf{X}_g^I \frac{\partial \mathbf{P}_g^{I*}}{\partial \mathbf{T}_g^I},$$

where $\mathbf{P}_g^{I*}$, $\mathbf{P}_g^I$, and $\mathbf{X}_g^I$ denote column vectors representing, respectively, the world prices, delivered prices, and import quantities of product g under various varieties. The $\circ$ symbol represents element-wise multiplication.

From Equation (4), it is evident that the gradient vector of welfare with respect to tariffs is determined by tariff levels ($\mathbf{T}_g^I$), price levels ($\mathbf{P}_g^{I*}$, $\mathbf{P}_g^I$), import quantities ($\mathbf{X}_g^I$), the trade elasticity matrix ($\partial \ln(\mathbf{X}_g^I) / \partial \ln(\mathbf{P}_g^I)$), and the first-order derivative of prices with respect to tariffs ($\partial \mathbf{P}_g^{I*} / \partial \mathbf{T}_g^I$). Furthermore, taking the derivative of Equation (4) with respect to tariffs yields the Hessian matrix of welfare, $H(W_g^I(\mathbf{T}_g^I))$. Straightforward mathematical calculations show that this Hessian matrix depends on the tariff levels, trade elasticities, and observed price levels, import quantities, as well as the first- and second-order derivatives of prices with respect to tariffs.

A practical challenge arises because we can only observe current and historical tariff levels along with their associated prices and import quantities. In the framework of Kee, Nicita, and Olarreaga (2009), which assumes a small-country model, import tariffs do not affect world prices, implying that world prices under free trade are the same as observed prices. However, this assumption does not hold in our framework, where tariffs in a large-country setting reduce world prices. As a result, observed prices reflect historical and current tariff policies, not the counterfactual free-trade price levels. To compute the TRI, we must rely on a structural model to estimate counterfactual values, including the free-trade world prices ($\mathbf{P}_g^{I*} |_{\mathbf{T}_g^I=0}$), free-trade import quantities ($\mathbf{X}_g^I |_{\mathbf{T}_g^I=0}$), and the first- and second-order derivatives of prices with respect to tariffs ($\partial \mathbf{P}_g^{I*} / \partial \mathbf{T}_g^I |_{\mathbf{T}_g^I=0}$, $\partial^2 \mathbf{P}_g^{I*} / \partial (\mathbf{T}_g^I)^2 |_{\mathbf{T}_g^I=0}$). Additionally, we require estimates of the trade elasticity matrix for imported goods.

2.3. The Trade Model

Building on the discussion in the previous subsections, estimating the TRI requires counterfactual values, including free-trade world prices, free-trade import quantities, and price derivatives to tariffs. To ensure consistency with the core assumptions of new trade theory while maintaining the ability to estimate key model parameters, we adopt a quantitative framework based on [Soderbery \(2017\)](#). In this setup, The demand for each imported variety v of good g is given by

$$(5) \quad x_{gv}^I = \xi_X^I \phi_g^I b_{gv}^I (p_{gv}^I)^{-\sigma_g^I} (\mathcal{P}_g^I)^{\sigma_g^I - 1},$$

where

$$(6) \quad \mathcal{P}_g = \left(\sum_v b_{gv}^I (p_{gv}^I)^{1-\sigma_g^I} \right)^{\frac{1}{1-\sigma_g^I}}$$

is the standard CES price index over the delivered prices p_{gv}^I , and ξ_X^I and ϕ_g^I are the fractions of consumer income spent on imported good and on imported varieties of good, respectively. b_{gv}^I is a random taste shock. σ_g^I is the import price elasticity. The form of export supply is

$$(7) \quad p_{gv}^{I*} = \exp(\eta_{gv}^I) (x_{gv}^I)^{\omega_g^I},$$

where ω_g^I is the inverse export supply elasticity from the view of importers and η_{gv}^I is the unobservable variety-specific supply shocks. The variable p_{gv}^{I*} denotes the shipped price in this context. The relationship between the *shipped price* and the delivered price can be formalized by the equation:

$$(8) \quad p_{gv}^I = p_{gv}^{I*} (1 + \tau_{gv}^I),$$

where τ_{gv}^I is the destination–variety-specific import tariff. In this case, τ_{gv}^I directly corresponds to the Non-MFN tariff framework. This setup allows us to analyze how tariffs, specific to the destination and product variety, impact the transition from shipped to delivered prices, incorporating the Non-MFN tariff structure into the pricing dynamics.

To compute welfare, we must first derive the equilibrium price level that satisfies the export-import balance condition. By substituting Equations (5) and (8) into Equation (7) and eliminating x_{gv}^I and p_{gv}^I , we arrive at the market-clearing shipped price:

$$(9) \quad p_{gv}^{I*} = \left(\varphi_{gv} (1 + \tau_{gv}^I)^{-\sigma_g^I} (\mathcal{P}_g^I)^{\sigma_g^I - 1} \right)^{\frac{\omega_g^I}{1 + \omega_g^I \sigma_g^I}},$$

where $\varphi_{gv} = e^{\eta_{gv}^I / \omega_g^I} \xi_X^I \phi_g^I b_{gv}^I$ represents an unobservable, variety-specific shock independent of tariffs. The market-clearing shipped price p_{gv}^{I*} is jointly determined by the destination–variety-specific import tariff τ_{gv}^I , demand elasticity σ_g^I , and supply-side parameters, including unobservable supply shocks η_{gv}^I and inverse supply elasticity ω_g^I . Additionally, the price index $\mathcal{P}_g^I$ is an endogenous variable, co-determined with p_{gv}^{I*} , highlighting

the interdependence between market structure and equilibrium prices.

This formulation elegantly integrates the structural parameters of the market and the impact of tariffs into the determination of equilibrium prices. However, several challenges arise when directly applying this equation to compute the TRI. First, many parameters and shocks in the equation are unobservable. Second, p_{gv}^{I*} is presented in an implicit form, where the price index on the right-hand side and the shipped price on the left-hand side are jointly determined. As a result, this equation represents a system of V nonlinear equations across varieties. While intuition suggests the existence of a unique solution, formal verification is currently lacking. Moreover, the implicit nature of this formulation complicates the derivation of price derivatives with respect to tariffs, which are critical for TRI calculation.

In the next subsection, we address these issues systematically. First, we use observed data to infer shocks and calibrate the unobserved parameters. We then demonstrate that p_{gv}^{I*} can be uniquely determined from the structural parameters, shocks, and import tariffs via Equation (9), and we provide its explicit expression. Once the market-clearing shipped price is determined, the corresponding market-clearing import quantity and delivered price can be computed using Equations (5) and (8), respectively.

2.4. Identifying Counterfactual Market Equilibrium Prices at Discretionary Tariff Levels

In real-world trade economies, we typically observe a single set of import tariffs, market-clearing prices, and import quantities over a given period. This poses a challenge, as we cannot simultaneously observe the market-clearing prices and import quantities under alternative tariff scenarios. In this subsection, we demonstrate that within the current structural model, given the observed set of tariffs, market-clearing prices, and import quantities, it is possible to identify the counterfactual market-clearing prices and import quantities under any hypothetical tariff level. This allows for a deeper understanding of how tariff changes would affect market outcomes, providing a robust framework for policy evaluation and welfare analysis.

In this subsection, we begin by leveraging observed data on prices, import tariffs, import quantities, and import and export elasticities to recover consumer taste shocks and the necessary structural parameters. Using these parameters, we derive the market-clearing equilibrium prices and quantities for any given tariff level. Additionally, we present explicit expressions for these variables under the special case of zero tariffs, as the market-clearing prices and trade volumes in this scenario are critical inputs for calculating the TRI.

2.4.1. Normalizing random taste shock

To streamline the analysis and emphasize the key economic mechanisms driving the results, I first normalize the taste shocks. This approach reflects the intuition that relative preferences, rather than their absolute levels, drive economic outcomes of interest. Moreover, normalization ensures consistent identification of the model's parameters, facilitating comparisons across different scenarios while maintaining computational tractability and interoperability.³

³This intuition can be easily verified. By performing an equivalent transformation on Equation (5), we obtain the expression

$$p_{gv}^{I*} = \left(e^{\eta_{gv}/\omega_g^I} \xi_X^I \phi_g^I (1 + \tau_{gv}^I)^{-\sigma_g^I} \left(\sum_m \frac{b_{gm}^I}{b_{gv}^I} (p_{gm}^I)^{1-\sigma_g^I} \right) \right)^{\frac{\omega_g^I}{1+\omega_g^I \sigma_g^I}}.$$

This expression reveals that the system is determined by the ratio of taste shocks between any two varieties.

Define $\bar{p}_{gv}^I$, $\bar{p}_{gv}^{I*}$, $\bar{x}_{gv}^I$, and $\bar{\tau}_{gv}^I$ as the observed market-clearing delivered prices, market-clearing shipped prices, corresponding import quantities, and tariff rates, respectively. In Equation (5), the only variety-specific parameter is the taste shock. To isolate and normalize the taste shock, non-variety-specific parameters ξ_X^I and ϕ_g^I are eliminated by taking the ratio between two varieties. Using the first variety as the base,⁴ we divide Equation (5) (using observed data) by the equation for the first variety,

$$\bar{x}_{g1}^I = \xi_X^I \phi_g^I b_{g1}^I (\bar{p}_{g1}^I)^{-\sigma_g^I} (\bar{\mathcal{P}}_g^I)^{\sigma_g^I - 1}.$$

This yields the normalized random taste shock:

$$B_{gv}^I = \frac{\bar{x}_{gv}^I}{\bar{x}_{g1}^I} \left(\frac{\bar{p}_{gv}^I}{\bar{p}_{g1}^I} \right)^{\sigma_g^I},$$

where $B_{gv}^I = b_{gv}^I / b_{g1}^I$, which can be calculated using observed market-clearing shipped prices and corresponding import quantities.

2.4.2. Identifying Additional Essential Parameters for Market Clearing Prices

Beyond the random taste shock, the unobservable variety-specific shock φ_{gv} in Equation (9) also includes a joint variety-specific shock that must be identified, denoted as

$$\delta_{gv}^I = e^{\eta_{gv}} \xi_X^I \phi_g^I.$$

By substituting Equations (6) and (8) into Equation (7), and eliminating p_{gv}^{I*} and $\mathcal{P}_g$, the expression is simplified. After rearranging the equation and incorporating the normalized taste shock, we derive the identified expression for δ_{gv}^I as follows:

$$(10) \quad \delta_{gv}^I = e^{-1/\omega_g^I} (B_{gv}^I)^{-1} (\bar{p}_{gv}^I)^{\sigma_g^I + \frac{1}{\omega_g^I}} (1 + \bar{\tau}_{gv}^I)^{-\frac{1}{\omega_g^I}} \sum_m B_{gm}^I (\bar{p}_{gm}^I)^{1 - \sigma_g^I},$$

where δ_{gv}^I can be uniquely calculated using observed delivered prices, tariff levels, and the previously normalized random taste shock. At this stage, assuming known import and export elasticities, all parameters necessary to solve for the market-clearing price level in Equation (9) are now identified. In principle, given any tariff level, the corresponding market-clearing price can be determined.

2.4.3. The Explicit Expression of Market Clearing Prices

In this part, we focus on deriving the explicit expression for the market-clearing equilibrium price, as this serves as a cornerstone for calculating the corresponding import quantities under equilibrium conditions. Once this expression is established, it enables the computation of various essential derivatives with respect to tariffs and facilitates counterfactual analysis, such as equilibrium prices and import quantities under different tariff levels.

⁴In practice, any variety's taste shock can serve as the base, as we are solely concerned with the ratio.

By substituting Equations (6) and (8) into Equation (7), and eliminating p_{gv}^{I*} and $\mathcal{P}_g$, the following equation is obtained:

$$(11) \quad \delta_{gv}^I e^{1/\omega_g^I} B_{gv}^I (p_{gv}^I)^{-\left(\sigma_g^I + \frac{1}{\omega_g^I}\right)} (1 + \tau_{gv}^I)^{\frac{1}{\omega_g^I}} = \sum_m B_{gm}^I (p_{gm}^I)^{1-\sigma_g^I},$$

Equation (11) represents a system of V nonlinear equations with V unknowns, p_{gm}^I . Despite its nonlinearity, the uniqueness of the solution can still be established. Dividing the equations corresponding to two varieties, k and v , eliminates terms involving the price index and import elasticities, yielding:

$$(12) \quad \frac{p_{gk}^I}{p_{gv}^I} = \left[\frac{e^{\eta_{gk}} B_{gk}^I (1 + \tau_{gk}^I)^{\frac{1}{\omega_g^I}}}{e^{\eta_{gv}} B_{gv}^I (1 + \tau_{gv}^I)^{\frac{1}{\omega_g^I}}} \right]^{\frac{1}{\sigma_g^I + 1/\omega_g^I}}.$$

This equation indicates that the price ratio between two varieties of the same good g depends on the ratio of their export shocks, normalized taste shocks, and trade costs $1 + \tau$.

Substituting Equation (12) into Equation (11), we derive the explicit expression for the equilibrium delivered price:

$$(13) \quad p_{gv}^I = \left[\frac{\left(\delta_{gv} B_{gv}^I (1 + \tau_{gv}^I)^{\frac{1}{\omega_g^I}} \right)^{\frac{1+1/\omega_g^I}{\sigma_g^I + 1/\omega_g^I}}}{\sum_m B_{gm}^I \left(\delta_{gm} B_{gm}^I (1 + \tau_{gm}^I)^{\frac{1}{\omega_g^I}} \right)^{\frac{1-\sigma_g^I}{\sigma_g^I + 1/\omega_g^I}}} \right]^{\frac{\omega_g^I}{1+\omega_g^I}}.$$

Equation (13) illustrates that the market-clearing price is uniquely determined by external shocks (supply shocks and normalized taste shocks), structural parameters (import and export elasticities), and tariff levels. This can be regarded as a “Market Response Function (MRF),” capturing how equilibrium prices adjust to changes in tariff policies.

The market-clearing shipped price, p_{gv}^{I*} , can be determined using Equation (8). Although variety-specific supply shocks remain unobservable, the corresponding import quantities can still be computed. Substituting $\tau_{gv}^I = \bar{\tau}_{gv}^I$ into Equation (7), we obtain:

$$(14) \quad \bar{p}_{gv}^{I*} = \exp(\eta_{gv}) (\bar{x}_{gv}^I)^{\omega_g^I}.$$

Dividing Equation (7) by Equation (14) and rearranging, the import quantity is given by:

$$(15) \quad x_{gv}^I = \left(\frac{p_{gv}^{I*}}{\bar{p}_{gv}^{I*}} \right)^{\frac{1}{\omega_g^I}} \bar{x}_{gv}^I.$$

Equations (8), (13), and (15) collectively provide counterfactual outcomes under any given tariff level. For

instance, in the extreme case of no tariff barriers, the tariff-free market-clearing delivered price is expressed as:

$$(16) \quad \dot{p}_{gv}^I = p_{gv}^I|_{\tau_{gv}^I=0} = \left[\frac{(\delta_{gv} B_{gv}^I)^{\frac{1+1/\omega_g^I}{\sigma_g^I+1/\omega_g^I}}}{\sum_m B_{gm}^I (\delta_{gm} B_{gm}^I)^{\frac{1-\sigma_g^I}{\sigma_g^I+1/\omega_g^I}}} \right]^{\frac{\omega_g^I}{1+\omega_g^I}}.$$

The corresponding tariff-free market-clearing quantity is:

$$(17) \quad \dot{x}_{gv}^I = \left(\frac{\dot{p}_{gv}^{I*}}{\bar{p}_{gv}^{I*}} \right)^{\frac{1}{\omega_g^I}} \bar{x}_{gv}^I.$$

2.5. The Derivatives to Tariffs

Building on the preceding discussion, we know that computing the TRI requires counterfactual values under a free-trade scenario, which we derived in the previous subsection. Beyond this, we also need to calculate the first- and second-order derivatives of prices with respect to tariffs. These derivatives are essential for determining the welfare gradient vector and Hessian matrix. This subsection focuses on deriving these critical derivatives⁵.

By differentiating Equation (9) or Equation (13) with respect to tariffs, we obtain the first- and second-order derivatives of prices. Substituting these derivatives into the expressions for the welfare gradient vector and Hessian matrix yields the following results. Define

$$f_k^{gv} = \frac{\partial p_{gv}^{I*}}{\partial \tau_{gk}^I} \Big|_{\tau_g^I=0} \quad \text{and} \quad f_{km}^{gv} = \frac{\partial^2 p_{gv}^{I*}}{\partial \tau_{gk}^I \partial \tau_{gm}^I} \Big|_{\tau_g^I=0},$$

such that:

$$(18) \quad f_k^{gv} = \begin{cases} \Lambda_{gvv}^1 \dot{p}_{gv}^{I*}, & \text{if } v = k, \\ \Lambda_{gvk}^2 \dot{p}_{gv}^{I*}, & \text{if } v \neq k, \end{cases}$$

and

$$(19) \quad f_{km}^{gv} = \begin{cases} \Lambda_{gvv}^1 (\Lambda_{gvv}^1 - 1) \dot{p}_{gv}^{I*}, & \text{if } v = k = m, \\ \Lambda_{gvv}^1 \Lambda_{gvm}^2 \dot{p}_{gv}^{I*}, & \text{if } v = k \neq m, \\ \Lambda_{gvk}^2 \Lambda_{gvm}^2 \dot{p}_{gv}^{I*}, & \text{if } v \neq k \text{ and } v \neq m, \end{cases}$$

where:

$$\Lambda_{gvk}^1 = -\frac{\frac{1}{\omega_g^I} (1 - \sigma_g^I)}{\left(1 + \frac{1}{\omega_g^I}\right) \left(\sigma_g^I + \frac{1}{\omega_g^I}\right)} \Psi_{gk} - \frac{\sigma_g^I}{\sigma_g^I + \frac{1}{\omega_g^I}},$$

⁵The detailed derivation is shown in Appendix A.2

and

$$\Lambda_{g^vk}^2 = -\frac{\frac{1}{\omega_g^I}(1 - \sigma_g^I)}{\left(1 + \frac{1}{\omega_g^I}\right)\left(\sigma_g^I + \frac{1}{\omega_g^I}\right)}\Psi_{gk},$$

with:

$$\Psi_{gm} = \frac{B_{gm}^I(\dot{p}_{gm}^I)^{1-\sigma_g^I}}{\sum_m B_{gm}^I(\dot{p}_{gm}^I)^{1-\sigma_g^I}}.$$

To facilitate the representation of welfare derivatives, define

$$h_k^{gv} = \frac{\partial x_{gv}^I}{\partial \tau_{gk}^I} \Big|_{\tau_g^I=0}.$$

The expression for h_k^{gv} is:

$$(20) \quad h_k^{gv} = \begin{cases} -\sigma_g^I \left(f_k^{gv} \frac{1}{\dot{p}_{gv}^{I*}} + 1 \right), & \text{if } v = k, \\ -\sigma_g^I f_k^{gv} \frac{1}{\dot{p}_{gv}^{I*}}, & \text{if } v \neq k. \end{cases}$$

Define

$$F_v^g = \frac{\partial W^I}{\partial \tau_{gv}^I} \Big|_{\tau_g^I=0},$$

which represents the welfare gradient vector. As noted, Soderbery's framework neglects important cross-variety effects, as shown in his Equation (2). Specifically, he overlooks terms involving $\partial \log(p_{gv}^*) / \partial \tau_{gk}^I$ for $v \neq k$. Correcting for these effects, we derive:

$$(21) \quad F_v^g = -\sum_m f_v^{gm} \dot{x}_{gm}^I.$$

For the Hessian matrix, define

$$F_{vk}^g = \frac{\partial^2 W^I}{\partial \tau_{gv}^I \partial \tau_{gk}^I} \Big|_{\tau_g^I=0},$$

such that:

$$(22) \quad F_{vk}^g = \begin{cases} -\sigma_g^I [\dot{p}_{gv}^{I*} \dot{x}_{gv}^I + f_k^{gk} \dot{x}_{gk}^I] - \sum_m [f_{vk}^{gm} \dot{x}_{gm}^I + f_v^{gv} h_k^{gm} \dot{x}_{gm}^I], & \text{if } v = k, \\ -\sigma_g^I f_k^{gk} \dot{x}_{gk}^I - \sum_m [f_{vk}^{gm} \dot{x}_{gm}^I + f_v^{gv} h_k^{gm} \dot{x}_{gm}^I], & \text{if } v \neq k. \end{cases}$$

2.6. Welfare-Maximizing Equilibrium

In the preceding subsections, we analyzed market-clearing prices, import levels, and the marginal effects of import tariffs. Building on this foundation, determining the welfare-maximizing equilibrium becomes crucial. This equilibrium represents the optimal allocation of resources between consumers and producers, serving as a benchmark for evaluating the efficiency of trade policies. Moreover, it provides critical insights for international trade negotiations by identifying policy configurations that enhance global welfare through mutual gains.

To derive this equilibrium, we first define the *Government Reaction Function* (GRF) as follows:

DEFINITION 1. *The Government Reaction Function (GRF) specifies the tariff policy that maximizes welfare under the prevailing market structure, given current prices, import quantities, elasticities, preferences, and other relevant factors.*

According to this definition, the GRF is characterized by the condition that the welfare gradient vector, or equivalently Equation (4), equals $\mathbf{0}$. Solving this equation yields the explicit form of the GRF:

$$(23) \quad \text{diag}(\mathbf{T}_g^I) = -\frac{1}{\sigma_g} (\mathbf{I} + \Lambda^g)^{-1} \Lambda^g \mathbf{I},$$

where $\mathbf{I}$ denotes the identity matrix, and the coefficients Λ_{gvk} are defined as:

$$(24) \quad \Lambda_{gvk} = \begin{cases} \Lambda_{gvk}^1, & \text{if } v = k, \\ \Lambda_{gvk}^2, & \text{if } v \neq k, \end{cases}$$

with the matrix Λ^g structured as:

$$(25) \quad \Lambda^g = \begin{bmatrix} \Lambda_{g11} & \Lambda_{g21} & \cdots & \Lambda_{gV1} \\ \Lambda_{g12} & \Lambda_{g22} & \cdots & \Lambda_{gV2} \\ \vdots & \vdots & \ddots & \vdots \\ \Lambda_{g1V} & \Lambda_{g2V} & \cdots & \Lambda_{gVV} \end{bmatrix}.$$

With the GRF established, we next define the *Welfare-Maximizing Equilibrium* (WME):

DEFINITION 2. *The Welfare-Maximizing Equilibrium (WME) is defined as a set of prices and tariff levels $(\tilde{\mathbf{P}}_g^I, \tilde{\mathbf{T}}_g^I)$, under which both market and government responses are optimized, such that markets clear, and the government achieves its welfare-maximization objective.*

Formally, the WME corresponds to a vector of prices and tariffs that simultaneously satisfies Equations (13) and (23). Solving this system yields:

$$\tilde{\mathbf{T}}_g^I = \frac{\omega_g}{\sigma_g}.$$

This remarkably simple expression reveals that the optimal tariff level depends solely on the inverse of the product of import and export elasticities. Importantly, it is independent of other factors such as preferences for different varieties, export shocks, or consumption shares.

2.7. The TRI under the Small Country Assumption

In this subsection, we demonstrate the broad applicability of our TRI framework by showing that it reduces to the well-known results in the literature under the small country assumption without variety differentiation. A simple example is the case considered in [Kee, Nicita, and Olarreaga \(2009\)](#), where the TRI under a small country assumption aligns with their conclusions.

Under the small country assumption, the inverse export supply elasticity $\omega_g^I \rightarrow 0$ for each good g of a variety

v . Additionally, when variety differentiation is absent, we set $V = 1$ for each good g . In this scenario, we have:

$$(26) \quad \Lambda_{gvk}^1 = - \frac{\sigma_g^I + (1 - \sigma_g^I)\Psi_{gk} + \sigma_g^I(1 - \sigma_g^I) \sum_{v \neq k} \frac{\Psi_{gv}}{\sigma_g^I + \infty}}{(\sigma_g^I + \infty) \left[1 + (1 - \sigma_g^I) \sum_{v=1}^V \frac{\Psi_{gv}}{\sigma_g^I + \infty} \right]} = 0,$$

and similarly,

$$(27) \quad \Lambda_{gvk}^2 = \Lambda_{gvk}^1 + \frac{\sigma_g^I}{\sigma_g^I + \infty} = 0.$$

Substituting $\Lambda_{gvk}^1 = \Lambda_{gvk}^2 = 0$ into Equations (18) and (19), we find that $f_k^{gv} = f_{km}^{gv} = 0$. Furthermore, substituting these results into Equations (21) and (22), we obtain:

$$(28) \quad F_k^g = 0,$$

and

$$(29) \quad F_{km}^g = \begin{cases} -\hat{x}_{gk}^* \sigma_g^I \hat{p}_{gk}^*, & \text{if } k = m, \\ 0, & \text{if } k \neq m. \end{cases}$$

These results imply that $\nabla W_g(\mathbf{T}_g) = [0, 0, \dots, 0]^\top$, and the Hessian matrix is diagonal:

$$H_g(\mathbf{T}_g) = \begin{bmatrix} F_{11}^g & 0 & \dots & 0 \\ 0 & F_{22}^g & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & F_{VV}^g \end{bmatrix}.$$

When the number of varieties $V = 1$, the Hessian reduces to $H_g(\mathbf{T}_g) = F_{11}^g = -\hat{x}_{g1}^* \sigma_g^I \hat{p}_{g1}^*$. Since $V = 1$, the variety subscript is dropped, and substituting $\nabla W_g(\mathbf{T}_g)$ and $H_g(\mathbf{T}_g)$ into Equation (3) yields:

$$(30) \quad \text{TRI} = \left(\frac{\sum_g \hat{x}_g^* \sigma_g^I \hat{p}_g^* T_g^2}{\sum_g \hat{x}_g^* \sigma_g^I \hat{p}_g^*} \right)^{1/2}.$$

This result is identical to Equation (16) in [Kee, Nicita, and Olarreaga \(2009\)](#), confirming that our TRI framework generalizes their approach and reduces to their findings under the small country assumption.

3. Data

Our analysis utilizes a comprehensive dataset spanning from 2000 to 2020, compiled from four key sources. First, bilateral trade data is sourced from the *BACI* database (CEPII)⁶, which covers over 200 countries and more

⁶CEPII (2020). BACI: International Trade Database at the Product-Level. Available at: [CEPII Website](#).

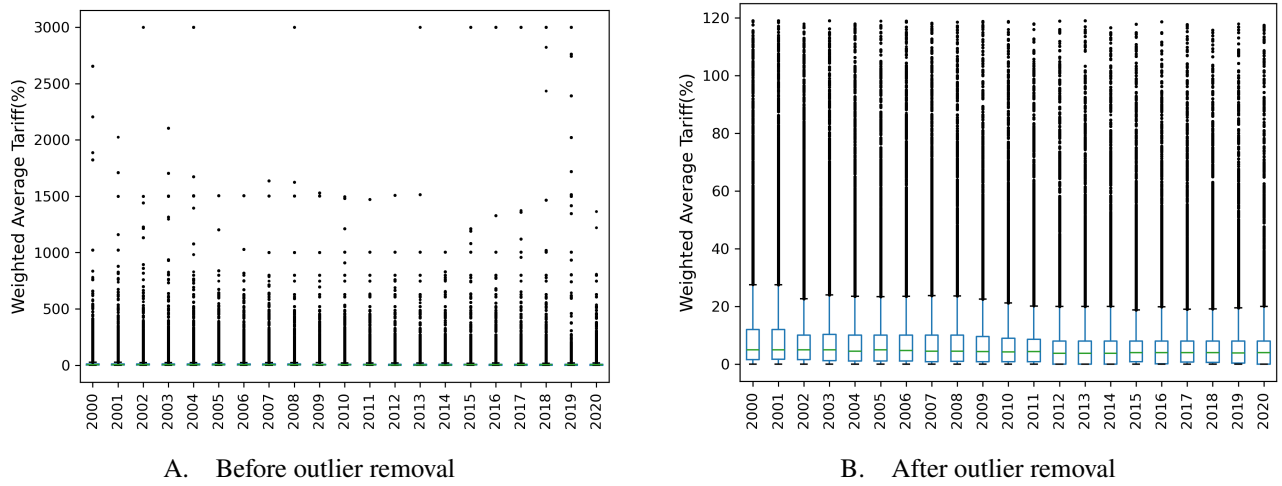


Figure 2: Annual Boxplots of MFN Tariff Rates

Notes. Panel A presents the distribution of MFN tariff rates before removing outliers, based on the merged dataset; panel B shows the distribution after removing outliers, where all tariff rates lie within a 0%–120% range, consistent across all years. Boxplots indicate interquartile ranges, with whiskers representing extreme values. Similar patterns are observed for non-MFN tariff rates.

than 5,000 HS6 products. *BACI* is a harmonized dataset constructed from United Nations *COMTRADE* data, employing a reconciliation methodology that ensures consistency across reporters and partners. By leveraging mirror statistics and correcting for missing or inconsistent entries, it has become one of the most reliable sources for analyzing international trade flows. Its granular classification at the HS6 product level enables detailed examination of trade patterns and the impacts of tariffs on specific goods and varieties.

Second, tariff data, including MFN and bilateral non-MFN rates, is obtained from the *World Integrated Trade Solution* (WITS) database, which also covers over 200 countries and 5,000+ HS6 products. Third, import demand elasticities are sourced from the World Bank’s *Overall Trade Restrictiveness Indices and Import Demand Elasticities* dataset⁷, which provides elasticity estimates for 4,900 HS6 products across 149 countries. Finally, export supply elasticities are drawn from [Nicita, Olarreaga, and Silva \(2018\)](#), covering 3,900 HS6 products and over 100 countries. These estimates capture the responsiveness of export supply to price changes and are based on structural models that account for product- and country-level heterogeneity.

Since trade and tariff data span multiple versions of the Harmonized System (HS), we utilize the World Bank’s *Concordance Table*⁸ to harmonize classifications and unify all data under the HS1988/1992 (H0) version. After merging datasets and excluding observations with missing values, the final dataset contains 132,277,900 observations from 136 countries. We conduct a data validation. Figure 2A illustrates the annual boxplots of MFN tariff rates before removing outliers. While tariffs are expected to exhibit a highly skewed distribution, we observed numerous extreme values, with some tariffs exceeding 1,000% and even reaching 3,000%. Given that our calculation of the TRI includes the square of tariff rates, these extreme values could severely distort our results. To address this, we excluded records with tariff rates above the 99.9th percentile. Figure 2B shows the boxplots after outlier removal. Post-cleaning, tariff rates across all years fall within a range of 0% to 120%, a range deemed more reasonable and suitable for our analysis.

⁷World Bank (2020). Overall Trade Restrictiveness Indices and Import Demand Elasticities. Available at: [World Bank Datacatalog](#).

⁸World Bank. Product Concordance Table. Available at: [Product Concordance](#).

Table 1: Importer Characteristics in 2018

Country	ISO 3	GDP	Imports	Simple Mean		Weighted Mean		Mean Elasticities	
				MFN	Non-MFN	MFN	Non-MFN	Importer	Exporter
USA	USA	20656.52	1254.28	3.50	1.94	2.12	0.93	-3.40	5.18
China	CHN	13894.91	2492.04	9.89	5.71	7.32	3.64	-2.67	23.51
Japan	JPN	5040.88	685.37	2.54	1.48	0.60	0.34	-3.93	12.43
Germany	DEU	3974.44	1287.16	4.99	2.89	4.68	2.91	-1.55	3.06
France	FRA	2790.96	473.22	5.46	3.19	5.28	3.28	-1.53	3.03
India	IND	2702.93	202.21	11.02	6.49	14.78	6.29	-3.77	49.00
Italy	ITA	2091.93	453.71	4.98	2.88	4.71	2.75	-1.60	3.12
Brazil	BRA	1916.93	179.18	15.47	10.41	9.19	6.03	-3.07	35.86
Korea	KOR	1725.37	559.28	7.45	3.39	5.95	2.08	-1.95	23.48
Canada	CAN	1725.33	401.28	3.39	1.94	2.54	1.38	-1.64	10.54
Russian Fed.	RUS	1657.33	288.74	5.94	3.68	3.41	1.73	-2.16	35.92
Australia	AUS	1429.73	117.57	3.37	1.47	0.92	0.36	-2.02	21.83
Spain	ESP	1421.70	277.85	5.38	3.17	6.71	4.07	-1.54	3.07
Mexico	MEX	1256.30	418.42	6.89	3.72	9.64	5.36	-1.53	17.58
Indonesia	IDN	1042.27	150.65	9.90	4.19	8.29	3.33	-2.35	54.36
Netherlands	NLD	914.04	372.89	5.35	3.22	6.17	3.28	-1.54	3.00
Turkey	TUR	778.97	152.28	5.91	3.42	8.17	4.12	-1.91	40.61
Poland	POL	588.78	243.13	5.06	3.02	5.91	3.83	-1.54	25.77
Sweden	SWE	555.46	129.96	4.40	2.42	3.89	2.19	-1.43	3.89
Belgium	BEL	543.30	244.54	5.10	3.03	5.83	3.56	-1.48	2.90
Argentina	ARG	524.82	48.18	14.83	8.53	11.09	3.38	-2.00	45.77
Austria	AUT	454.99	145.29	4.73	2.65	4.60	2.75	-1.44	2.92
South Africa	ZAF	405.26	65.63	8.99	5.15	7.18	3.74	-2.02	40.79
Ireland	IRL	386.69	118.45	5.09	3.44	4.52	3.71	-1.32	2.90
Singapore	SGP	363.11	90.88	6.90	6.00	1.26	1.18	-1.25	4.12
HK SAR	HKG	361.73	90.89	0.00	0.00	0.00	0.00	-1.23	16.54
Denmark	DNK	356.84	80.83	5.09	3.19	6.47	4.82	-1.44	3.00
Philippines	PHL	346.84	67.63	7.06	3.28	3.85	1.77	-1.52	55.95
Colombia	COL	334.20	26.01	8.22	4.38	7.78	5.24	-1.71	56.36
Chile	CHL	295.86	40.24	5.99	2.10	6.00	1.26	-1.48	54.97

Notes. All are in billions of US dollars, in 2018. Elasticities are the mean across all varieties, where the importer is the CES elasticity $-\sigma_I^g$, the exporter is the export supply elasticity $1/\omega_I^g$.

Table 1 presents the summary statistics of the data for the top 30 countries and regions ranked by GDP in 2018. The table includes GDP and total import values, alongside simple and import value-weighted average tariff rates under both MFN and non-MFN tariff structures. Additionally, it provides the mean elasticities of import demand and export supply.

A number of key patterns emerge from the data. The United States, China, and Japan rank as the largest economies, with significant import volumes underscoring their central roles in global trade networks. China, in particular, exhibits relatively high tariff levels under both MFN and non-MFN schedules, reflecting a more protectionist stance compared to other major economies. In contrast, Hong Kong SAR maintains a zero-tariff policy, aligning with its position as a prominent free-trade hub.

The data also reveal substantial variation in tariff levels and elasticities across countries. Brazil and India impose some of the highest simple and value-weighted tariffs, indicative of more restrictive trade policies. Conversely, advanced economies such as Germany and France exhibit relatively lower tariff rates, consistent with their deeper integration into global trade systems. Import demand elasticities are generally negative, as expected under CES demand structures, while export supply elasticities vary widely. Notably, countries like Indonesia and Brazil display high export elasticities, signaling significant responsiveness to changes in global demand conditions.

4. Results

This section applies the data described in Section 3 to the model of Section 2. Appendix A.3 reports the TRIs for every country and year; the cross-country comparisons below use 2018 as the reference year.

Figure 3 plots the global import-value-weighted TRIs against the corresponding value-weighted average tariffs over 2000–2020. Two patterns stand out. First, the TRIs lie well below the average tariffs throughout the period, at roughly half their level, for both MFN and non-MFN tariffs. Second, all four series decline over time, and the non-MFN TRI falls faster than the MFN TRI, in line with the growing share of trade conducted under preferential tariffs over the same period (Figure 1B).

The first pattern needs interpretation, because with a fixed set of weights the welfare-equivalent uniform tariff exceeds the average tariff whenever tariffs are dispersed (Anderson and Neary 1994). Two features of our measure push in the opposite direction. First, the country-level TRIs in Figure 4 are computed under the large-country assumption (they coincide with the vertical axis of Figure 6), so part of the welfare cost of a tariff is offset by terms-of-trade gains; under the small-country assumption the TRIs are considerably higher (Figure 6). Second, the global series is weighted by import value and is therefore dominated by the largest importers, such as the United States, China and Japan, whose large-country TRIs lie far below their average tariffs (Figure 4). For many smaller economies the TRI exceeds the simple average tariff, as the dispersion argument predicts.

Figure 4 compares simple average non-MFN tariffs with non-MFN TRIs across countries in 2018. Countries are widely dispersed around the 45-degree line. The largest economies lie below it. A group of mostly small economies lies well above it: the Maldives (MDV), the Comoros (COM), Guyana (GUY), Namibia (NAM) and Papua New Guinea (PNG) have TRIs far above their average tariffs, which indicates that their tariffs are concentrated on heavily imported products. Ethiopia (ETH) and Togo (TGO) have among the highest simple average tariffs in the sample; in several small and low-income economies, high average tariffs partly reflect reliance on tariffs as a source of government revenue. At the other extreme, Kenya (KEN) has a TRI close to zero

despite a simple average tariff above 10 percent.

Figure 5 repeats the comparison with import-value-weighted average tariffs. Weighting by trade flows does not remove the dispersion. Countries with similar weighted average tariffs, such as Kenya (KEN) and Guyana (GUY), have very different TRIs, because the TRI also depends on the composition of imports and on the import-demand and export-supply elasticities of the products imported.

Figure 6 plots each country's non-MFN TRI under the large-country assumption (vertical axis) against its TRI under the small-country assumption (horizontal axis). Most countries lie below the 45-degree line: ignoring terms-of-trade effects overstates the TRI. The gap is large for big importers. For the United States, the small-country TRI is about 4 percent while the large-country TRI is close to zero. It is also large for several EU members, such as Denmark (DNK), Ireland (IRL) and the Netherlands (NLD). Because EU members apply a common external tariff, their TRIs differ only through the composition of their imports and the associated elasticities.

Finally, Figure 7 compares MFN and non-MFN TRIs across countries in 2018. Most countries lie below the 45-degree line: their non-MFN TRIs are lower than their MFN TRIs because preferential tariffs lower the tariffs actually applied. A few countries, such as Kenya (KEN) and Ghana (GHA), lie above it. Non-MFN TRIs also vary widely among countries with similar MFN TRIs: Kenya (KEN), Suriname (SUR), Ghana (GHA) and Guyana (GUY) have similar MFN TRIs, but their non-MFN TRIs range from 3 percent to over 20 percent. MFN tariffs alone are therefore a poor guide to the restrictiveness of the tariffs countries actually apply.

Taken together, these results show that neither simple nor import-weighted average tariffs track the welfare cost of a country's tariff structure. The TRI accounts for elasticities, trade flows, product varieties and terms-of-trade effects, and in our data these features change both the level of measured restrictiveness and the ranking of countries.

5. Conclusion

This paper extends the trade restrictiveness index to non-MFN tariffs, product varieties and large-country effects. We derive the gradient and Hessian of welfare with respect to tariffs in this setting and compute 2,406 annual TRIs for 136 countries over 2000–2020. The global TRI declined steadily over the period, and the non-MFN TRI declined faster than the MFN TRI as the share of trade under preferential tariffs grew. Across countries, average tariffs track the TRI poorly: the largest importers have TRIs well below their average tariffs once terms-of-trade effects are taken into account, while many small economies have TRIs above their average tariffs because their tariffs are concentrated on heavily imported products. Ignoring terms-of-trade effects, as the small-country assumption does, overstates the TRI, most of all for large importers.

The index has limitations. It covers tariffs only and does not capture non-tariff measures. It relies on externally estimated import-demand and export-supply elasticities. And because it rests on a second-order approximation of welfare, its level depends on the reference point of the approximation, as the comparison between the small- and large-country versions shows. Natural next steps are to compare countries' TRIs before and after specific preferential agreements enter into force, to measure how the 2018–2019 escalation of US–China tariffs changed restrictiveness for both countries and for third countries, and to apply the index to tariff regimes that discriminate by design, such as carbon border adjustments.

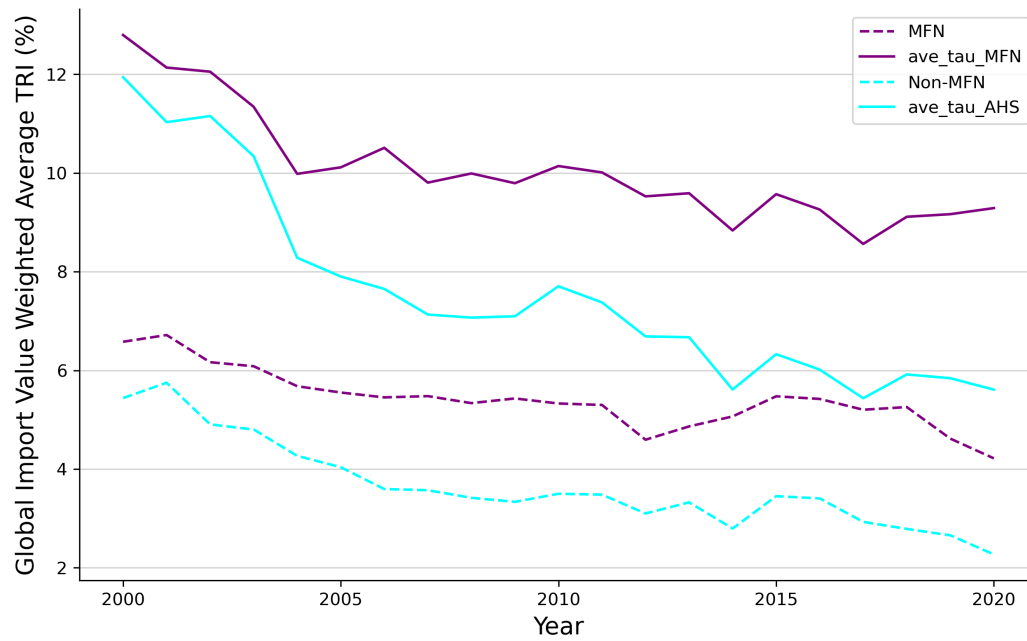


Figure 3: Global Import Value-Weighted Average TRIs and Tariffs (2000–2020)

Notes. Dashed lines indicate value-weighted average TRIs, while solid lines represent value-weighted average tariffs. The purple lines represent MFN tariffs, while the cyan lines denote non-MFN tariffs.

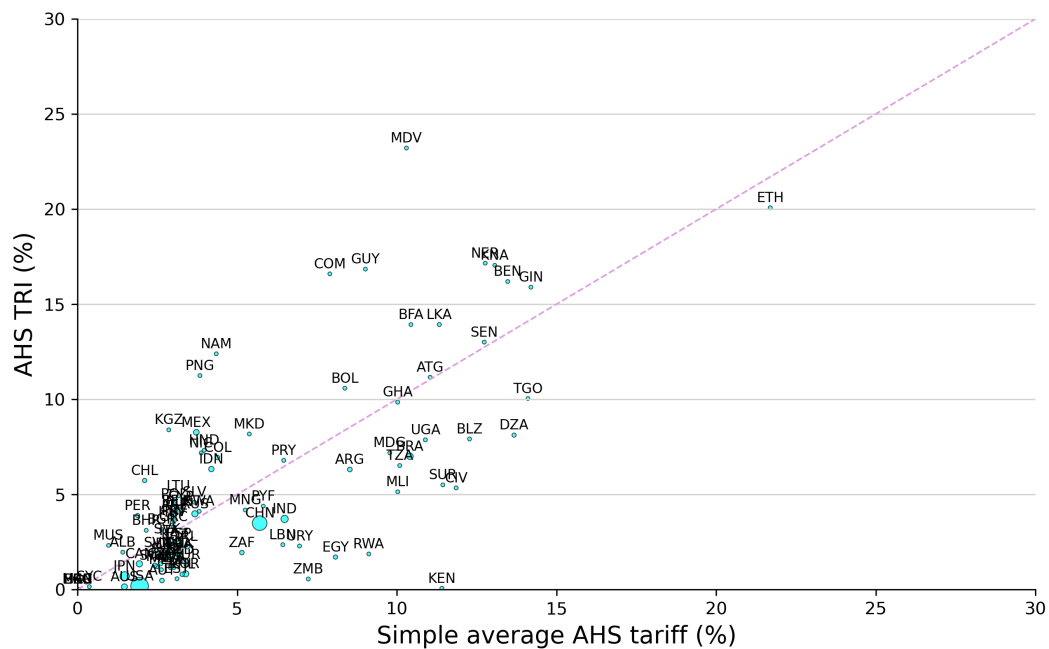


Figure 4: Simple Average Non-MFN Tariffs and Non-MFN TRIs Across Countries in 2018

Notes. The purple dashed line represents the 45-degree line. The radius of each circle corresponds to the country's GDP in 2018.

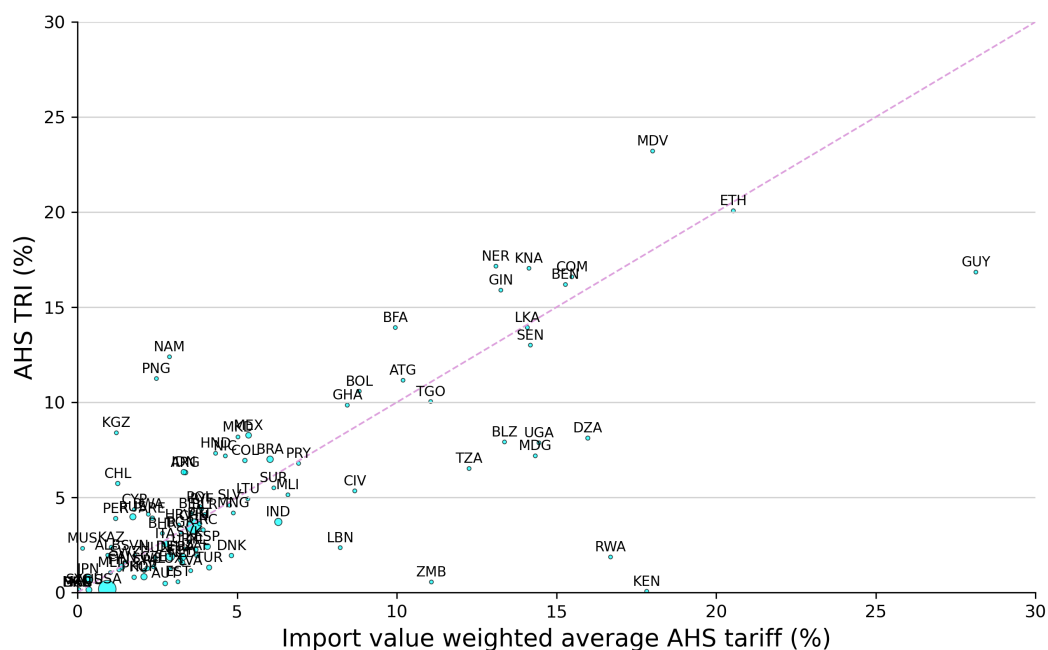


Figure 5: Import Value-Weighted Average Non-MFN Tariffs and Non-MFN TRIs Across Countries in 2018

Notes. The purple dashed line represents the 45-degree line. The radius of each circle corresponds to the country's GDP size in 2018.



Figure 6: Non-MFN TRIs Under Small- and Large-Country Assumptions Across Countries in 2018

Notes. The purple dashed line represents the 45-degree line. The radius of each circle corresponds to the country's GDP in 2018.



20

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Appendix A. Appendix

A.1. Issues in Soderbery’s Methodology

In Soderbery’s analysis, world prices are assumed to satisfy the following equation (note that since MFN tariffs are considered, the tariff subscript for varieties is omitted):

$$(A1) \quad p_{gv}^* = (1 + \tau_g^I)^{\frac{-\omega_{gv}^I(1+\bar{\omega}_{go}^I\sigma_g^I)}{(1+\bar{\omega}_{go}^I)(1+\omega_{gv}^I\sigma_g^I)}} \left(\varphi_{gv} \Phi_g^{\sigma_g^I-1} \right)^{\frac{\omega_{gv}^I}{1+\omega_{gv}^I\sigma_g^I}}.$$

Taking the natural logarithm of Equation (A1) and differentiating with respect to τ_g^I , we have:

$$(A2) \quad \left. \frac{\partial \ln(p_{gv}^*)}{\partial \tau_g^I} \right|_{\tau_g^I=0} = \frac{-\omega_{gv}^I(1 + \bar{\omega}_{go}^I\sigma_g^I)}{(1 + \bar{\omega}_{go}^I)(1 + \omega_{gv}^I\sigma_g^I)}.$$

Consider the extreme case of homogeneous supply elasticities ($\omega_{gv}^I = \bar{\omega}_{go}^I = \omega_g^I$), which simplifies to:

$$(A3) \quad \left. \frac{\partial \ln(p_{gv}^*)}{\partial \tau_g^I} \right|_{\tau_g^I=0} = -\frac{\omega_g^I}{1 + \omega_g^I}.$$

This expression implies that $\partial \ln(p_{gv}^*)/\partial \tau_g^I$ is determined solely by the export elasticities. However, intuitively, it should depend on the balance between export and import elasticities. To further demonstrate the inconsistency, we compare two methods for calculating the elasticity of import quantities with respect to tariffs.

Method 1: Indirect Calculation via Price Elasticities

We first compute the elasticity of delivered prices with respect to tariffs:

$$\begin{aligned}
\left. \frac{\partial \ln(p_{gv}^I)}{\partial \tau_g^I} \right|_{\tau_g^I=0} &= \left. \frac{\partial \ln((1 + \tau_g^I) p_{gv}^*)}{\partial \tau_g^I} \right|_{\tau_g^I=0} \\
&= \left. \frac{\partial \ln(1 + \tau_g^I)}{\partial \tau_g^I} \right|_{\tau_g^I=0} + \left. \frac{\partial \ln(p_{gv}^*)}{\partial \tau_g^I} \right|_{\tau_g^I=0} \\
&= 1 + \frac{-\omega_{gv}^I (1 + \bar{\omega}_{go}^I \sigma_g^I)}{(1 + \bar{\omega}_{go}^I)(1 + \omega_{gv}^I \sigma_g^I)} \\
&= \frac{1 + \bar{\omega}_{go}^I - \omega_{gv}^I + \omega_{gv}^I \sigma_g^I}{(1 + \bar{\omega}_{go}^I)(1 + \omega_{gv}^I \sigma_g^I)}.
\end{aligned}
\tag{A4}$$

Using the elasticity of import quantities with respect to delivered prices, we derive:

$$\begin{aligned}
\left. \frac{\partial \ln(x_{gv}^I)}{\partial \tau_g^I} \right|_{\tau_g^I=0} &= \left. \frac{\partial \ln(x_{gv}^I)}{\partial \ln(p_{gv}^I)} \cdot \frac{\partial \ln(p_{gv}^I)}{\partial \tau_g^I} \right|_{\tau_g^I=0} \\
&= -\frac{\sigma_g^I (1 + \bar{\omega}_{go}^I - \omega_{gv}^I + \omega_{gv}^I \sigma_g^I)}{(1 + \bar{\omega}_{go}^I)(1 + \omega_{gv}^I \sigma_g^I)}.
\end{aligned}
\tag{A5}$$

Method 2: Direct Calculation

Alternatively, substituting Equation (A1) into the demand equation yields:

$$x_{gv} = (1 + \tau_g^I)^{\frac{-(1 + \bar{\omega}_{go}^I \sigma_g^I)}{(1 + \bar{\omega}_{go}^I)(1 + \omega_{gv}^I \sigma_g^I)}} \exp(-\eta_{gv}) \left(\varphi_{gv} \Phi_g^{\sigma_g^I - 1} \right)^{\frac{1}{1 + \omega_{gv}^I \sigma_g^I}}.
\tag{A6}$$

Differentiating with respect to tariffs gives:

$$\left. \frac{\partial \ln(x_{gv}^I)}{\partial \tau_g^I} \right|_{\tau_g^I=0} = -\frac{1 + \bar{\omega}_{go}^I \sigma_g^I}{(1 + \bar{\omega}_{go}^I)(1 + \omega_{gv}^I \sigma_g^I)}.
\tag{A7}$$

Comparison of Results

Equations (A5) and (A7) yield inconsistent results. Combining them, we have:

$$\begin{aligned}
&-\frac{\sigma_g^I (1 + \bar{\omega}_{go}^I - \omega_{gv}^I + \omega_{gv}^I \sigma_g^I)}{(1 + \bar{\omega}_{go}^I)(1 + \omega_{gv}^I \sigma_g^I)} = -\frac{1 + \bar{\omega}_{go}^I \sigma_g^I}{(1 + \bar{\omega}_{go}^I)(1 + \omega_{gv}^I \sigma_g^I)}, \\
&\Rightarrow \sigma_g^I (1 - \omega_{gv}^I + \omega_{gv}^I \sigma_g^I) = 1, \\
&\Rightarrow (1 + \omega_{gv}^I)(\sigma_g^I - 1) = 0, \\
&\Rightarrow \sigma_g^I = 1.
\end{aligned}$$

This implies that Soderbery's conclusion holds only if $\sigma_g^I = 1$, which is an unrealistic assumption.

A.2. The derivatives to tariffs

An important step in calculating the TRI is to determine the marginal effect of tariffs on social welfare. According to Equation 4, the derivative of welfare with respect to tariffs depends on the import tariff level, market-clearing prices, import quantities, import demand elasticity, and the derivative or elasticity of market-clearing prices with respect to the import tariff. In the previous subsection, we examined all components except the last one. In this section, we focus on analyzing the derivative of market clearing prices with respect to tariffs. Taking the natural logarithm of Equation 9 and differentiating both sides with respect to τ_{gk}^I on both sides, we obtain:

$$(A8) \quad \left(\sigma_g^I + \frac{1}{\omega_g^I} \right) \frac{\partial \ln(p_{gv}^{I*})}{\partial \tau_{gk}^I} = -\sigma_g^I \frac{\partial \ln(1 + \tau_{gv}^I)}{\partial \tau_{gk}^I} + (\sigma_g^I - 1) \frac{\partial \ln(\mathcal{P}_g)}{\partial \tau_{gk}^I}.$$

First, consider the term on the right-hand side involving $\ln(1 + \tau_{gv}^I)$. we have

$$(A9) \quad \frac{\partial \ln(1 + \tau_{gv}^I)}{\partial \tau_{gk}^I} = \begin{cases} \frac{1}{1 + \tau_{gk}^I} & \text{if } v = k, \\ 0 & \text{if } v \neq k. \end{cases}$$

Next, substitute Equation 6 into the second term on the right-hand side. This yields:

$$(A10) \quad \begin{aligned} (\sigma_g^I - 1) \frac{\partial \ln(\mathcal{P}_g)}{\partial \tau_{gk}^I} &= - \frac{\partial \ln \left(\sum_m b_{gm}^I (p_{gm}^I)^{1-\sigma_g^I} \right)}{\partial \tau_{gk}^I} \\ &= - (1 - \sigma_g^I) \frac{\sum_m b_{gm}^I (p_{gm}^I)^{-\sigma_g^I} \frac{\partial p_{gm}^I}{\partial \tau_{gk}^I}}{\sum_m b_{gm}^I (p_{gm}^I)^{1-\sigma_g^I}} \\ &= - (1 - \sigma_g^I) \frac{\sum_m b_{gm}^I (p_{gm}^I)^{1-\sigma_g^I} \frac{\partial \ln(p_{gm}^I)}{\partial \tau_{gk}^I}}{\sum_m b_{gm}^I (p_{gm}^I)^{1-\sigma_g^I}} \\ &= - (1 - \sigma_g^I) \sum_m \Psi_{gm} \frac{\partial \ln(p_{gm}^I)}{\partial \tau_{gk}^I}, \end{aligned}$$

where

$$\Psi_{gm} = \frac{b_{gm}^I (p_{gm}^I)^{1-\sigma_g^I}}{\sum_m b_{gm}^I (p_{gm}^I)^{1-\sigma_g^I}}.$$

Now, consider the expression for $\sum_m \Psi_{gm} \partial \ln(p_{gm}^I) / \partial \tau_{gk}^I$:

$$\begin{aligned} \sum_m \Psi_{gm} \frac{\partial \ln(p_{gm}^I)}{\partial \tau_{gk}^I} &= \sum_m \Psi_{gm} \frac{\partial \ln(p_{gm}^{I*}(1 + \tau_{gm}^I))}{\partial \tau_{gk}^I} \\ &= \sum_m \Psi_{gm} \frac{\partial \ln(p_{gm}^{I*})}{\partial \tau_{gk}^I} + \sum_m \Psi_{gm} \frac{\partial \ln(1 + \tau_{gm}^I)}{\partial \tau_{gk}^I} \\ &= \sum_m \Psi_{gm} \frac{\partial \ln(p_{gm}^{I*})}{\partial \tau_{gk}^I} + \Psi_{gk} \frac{1}{1 + \tau_{gk}^I}. \end{aligned}$$

Thus, we have the following system of equations:

$$\begin{aligned} (A11) \quad & \left(\sigma_g + \frac{1}{\omega_g^I} \right) \frac{\partial \ln(p_{gv}^{I*})}{\partial \tau_{gk}^I} + (1 - \sigma_g^I) \sum_m \Psi_{gm} \frac{\partial \ln(p_{gm}^{I*})}{\partial \tau_{gk}^I} \\ &= \begin{cases} -\frac{\sigma_g^I + (1 - \sigma_g^I) \Psi_{gk}}{1 + \tau_{gk}^I} & \text{if } v = k, \\ -\frac{(1 - \sigma_g^I) \Psi_{gk}}{1 + \tau_{gk}^I} & \text{if } v \neq k. \end{cases} \end{aligned}$$

The system can be characterized as a linear arrangement of V^2 equations and V^2 unknowns, specifically $\partial \ln(p_{gv}^{I*}) / \partial \tau_{gk}^I$ for all v and $k \in V$. For analytical simplicity, this system can be expressed in a matrix form as $\mathbf{\Gamma} \mathbf{E}^* = \mathbf{C}$, where $\mathbf{\Gamma}$ denotes the $V^2 \times V^2$ square coefficient matrix. The matrix $\mathbf{\Gamma}$ is composed of V diagonal blocks and can be concisely represented as $\Gamma_1 \oplus \dots \oplus \Gamma_V$:

$$(A12) \quad \mathbf{\Gamma} = \Gamma_1 \oplus \dots \oplus \Gamma_V = \begin{bmatrix} \Gamma_1 & 0 & 0 & \dots & 0 \\ 0 & \Gamma_2 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \Gamma_V \end{bmatrix},$$

where the k -th block Γ_k is defined as:

$$(A13) \quad \Gamma_k = \begin{bmatrix} (1 - \sigma_g^I) \Psi_{g1} + (\sigma_g + \frac{1}{\omega_{g1}^I}) & (1 - \sigma_g^I) \Psi_{g2} & (1 - \sigma_g^I) \Psi_{g3} & \dots & (1 - \sigma_g^I) \Psi_{gV} \\ (1 - \sigma_g^I) \Psi_{g1} & (1 - \sigma_g^I) \Psi_{g2} + (\sigma_g + \frac{1}{\omega_{g2}^I}) & (1 - \sigma_g^I) \Psi_{g3} & \dots & (1 - \sigma_g^I) \Psi_{gV} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ (1 - \sigma_g^I) \Psi_{g1} & (1 - \sigma_g^I) \Psi_{g2} & (1 - \sigma_g^I) \Psi_{g3} & \dots & (1 - \sigma_g^I) \Psi_{gV} + (\sigma_g + \frac{1}{\omega_g^I}) \end{bmatrix}$$

for any $k \in V$.

In parallel, the vectors $\mathbf{E}^*$ and $\mathbf{C}$ can be structured into block form: $\mathbf{E}^* = (\mathbf{E}_1^*, \mathbf{E}_2^*, \dots, \mathbf{E}_V^*)^\top$ and $\mathbf{C} = (\mathbf{C}_1, \mathbf{C}_2, \dots, \mathbf{C}_V)^\top$ where each block $\mathbf{E}_k^*$ and $\mathbf{C}_k$ is defined as:

$$\mathbf{E}_k^* = \left(\frac{\partial \ln(p_{g1}^{I*})}{\partial \tau_{gk}^I}, \frac{\partial \ln(p_{g2}^{I*})}{\partial \tau_{gk}^I}, \dots, \frac{\partial \ln(p_{gV}^{I*})}{\partial \tau_{gk}^I} \right)^\top$$

and

$$\mathbf{C}_k = - \left(\frac{(1 - \sigma_g^I) \Psi_{gk}}{1 + \tau_{gk}^I}, \dots, \frac{\sigma_g^I + (1 - \sigma_g^I) \Psi_{gk}}{1 + \tau_{gk}^I}, \dots, \frac{(1 - \sigma_g^I) \Psi_{gk}}{1 + \tau_{gk}^I} \right)^\top$$

for any $k \in V$. Each block can be solved independently. For the first one,

(A14)

$$\begin{aligned} & \begin{bmatrix} (1 - \sigma_g^I) \Psi_{g1} + (\sigma_g + \frac{1}{\omega_{g1}^I}) & (1 - \sigma_g^I) \Psi_{g2} & (1 - \sigma_g^I) \Psi_{g3} & \dots & (1 - \sigma_g^I) \Psi_{gV} \\ (1 - \sigma_g^I) \Psi_{g1} & (1 - \sigma_g^I) \Psi_{g2} + (\sigma_g + \frac{1}{\omega_{g2}^I}) & (1 - \sigma_g^I) \Psi_{g3} & \dots & (1 - \sigma_g^I) \Psi_{gV} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ (1 - \sigma_g^I) \Psi_{g1} & (1 - \sigma_g^I) \Psi_{g2} & (1 - \sigma_g^I) \Psi_{g3} & \dots & (1 - \sigma_g^I) \Psi_{gV} + (\sigma_g + \frac{1}{\omega_g^I}) \end{bmatrix} \begin{bmatrix} \frac{\partial \ln(p_{g1}^{I*})}{\partial \tau_{g1}^I} \\ \frac{\partial \ln(p_{g2}^{I*})}{\partial \tau_{g1}^I} \\ \vdots \\ \frac{\partial \ln(p_{gV}^{I*})}{\partial \tau_{g1}^I} \end{bmatrix} \\ &= - \begin{bmatrix} \sigma_g^I + (1 - \sigma_g^I) \Psi_{g1} \\ (1 - \sigma_g^I) \Psi_{g1} \\ \vdots \\ (1 - \sigma_g^I) \Psi_{g1} \end{bmatrix} \frac{1}{1 + \tau_{g1}^I}. \end{aligned}$$

If Γ_k is a full rank matrix, applying Cramer's Rule yields $\mathbf{E}_1^* = \mathbf{C}_1 \Gamma_1^{-1}$. To derive an explicit expression for $\mathbf{E}_1^*$, we return to Equation A14, subtracting the first row from subsequent rows:

$$\begin{aligned} & \begin{bmatrix} (1 - \sigma_g^I) \Psi_{g1} + (\sigma_g + \frac{1}{\omega_{g1}^I}) & (1 - \sigma_g^I) \Psi_{g2} & (1 - \sigma_g^I) \Psi_{g3} & \dots & (1 - \sigma_g^I) \Psi_{gV} \\ -(\sigma_g + \frac{1}{\omega_{g1}^I}) & (\sigma_g + \frac{1}{\omega_{g2}^I}) & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -(\sigma_g + \frac{1}{\omega_{g1}^I}) & 0 & 0 & \dots & (\sigma_g + \frac{1}{\omega_g^I}) \end{bmatrix} \begin{bmatrix} \frac{\partial \ln(p_{g1}^{I*})}{\partial \tau_{g1}^I} \\ \frac{\partial \ln(p_{g2}^{I*})}{\partial \tau_{g1}^I} \\ \vdots \\ \frac{\partial \ln(p_{gV}^{I*})}{\partial \tau_{g1}^I} \end{bmatrix} \\ &= \begin{bmatrix} -\sigma_g^I - (1 - \sigma_g^I) \Psi_{g1} \\ \sigma_g^I \\ \vdots \\ \sigma_g^I \end{bmatrix} \frac{1}{1 + \tau_{g1}^I}. \end{aligned} \quad (\text{A15})$$

From the second row onward in Equation A15, we obtain the following expression:

$$(A16) \quad \left(\sigma_g^I + \frac{1}{\omega_g^I} \right) \frac{\partial \ln(p_{gv}^{I*})}{\partial \tau_{g1}^I} = \left(\sigma_g^I + \frac{1}{\omega_{g1}^I} \right) \frac{\partial \ln(p_{g1}^{I*})}{\partial \tau_{g1}^I} + \frac{\sigma_g^I}{1 + \tau_{g1}^I}, \quad \text{for any } v \neq 1.$$

This implies that:

$$(A17) \quad \frac{\partial \ln(p_{gm}^{I*})}{\partial \tau_{g1}^I} = \frac{\partial \ln(p_{gv}^{I*})}{\partial \tau_{g1}^I} \quad \text{for any } v, m \neq 1.$$

By substituting Equation A17 into the first row of A15 and eliminating $\partial \ln(p_{gv}^{I*})/\partial \tau_{g1}^I$ for any $v \neq 1$, we derive:

$$(A18) \quad \frac{\partial \ln(p_{g1}^{I*})}{\partial \tau_{g1}^I} = - \frac{\sigma_g^I + (1 - \sigma_g^I)\Psi_{g1} + \frac{\sigma_g^I(1 - \sigma_g^I)}{\sigma_g^I + \frac{1}{\omega_g^I}}(1 - \Psi_{g1})}{1 + \frac{1}{\omega_g^I}} \cdot \frac{1}{1 + \tau_{g1}^I}.$$

plug equation A18 into (A16), we will have

$$(A19) \quad \frac{\partial \ln(p_{gv}^{I*})}{\partial \tau_{g1}^I} = \left[- \frac{\sigma_g^I + (1 - \sigma_g^I)\Psi_{g1} + \frac{\sigma_g^I(1 - \sigma_g^I)}{\sigma_g^I + \frac{1}{\omega_g^I}}(1 - \Psi_{g1})}{1 + \frac{1}{\omega_g^I}} + \frac{\sigma_g^I}{\sigma_g^I + \frac{1}{\omega_g^I}} \right] \frac{1}{1 + \tau_{g1}^I}, \quad \text{for any } v \neq 1.$$

summarize above 4 equations, we will have

$$(A20) \quad \left(\sigma_g^I + \frac{1}{\omega_g^I} \right) \frac{\partial \ln(p_{gv}^{I*})}{\partial \tau_{gk}^I} = \left(\sigma_g^I + \frac{1}{\omega_g^I} \right) \frac{\partial \ln(p_{gk}^{I*})}{\partial \tau_{gk}^I} + \frac{\sigma_g^I}{1 + \tau_{gk}^I}, \quad \text{for any } v \neq k,$$

$$(A21) \quad \frac{\partial \ln(p_{gm}^{I*})}{\partial \tau_{gk}^I} = \frac{\partial \ln(p_{gv}^{I*})}{\partial \tau_{gk}^I} \quad \text{for any } v, m \neq k,$$

and,

$$(A22) \quad \frac{\partial \ln(p_{gv}^{I*})}{\partial \tau_{gk}^I} = \frac{\Lambda_{gvk}}{1 + \tau_{gk}^I},$$

where

$$(A23) \quad \Lambda_{gvk} = \begin{cases} \Lambda_{gvk}^1 & \text{if } v = k, \\ \Lambda_{gvk}^2 & \text{if } v \neq k, \end{cases}$$

$$\Lambda_{gvk}^1 = - \frac{\sigma_g^I + (1 - \sigma_g^I)\Psi_{gk} + \frac{\sigma_g^I(1 - \sigma_g^I)}{\sigma_g^I + \frac{1}{\omega_g^I}}(1 - \Psi_{gk})}{1 + \frac{1}{\omega_g^I}}$$

and

$$\Lambda_{gvk}^2 = \Lambda_{gvk}^1 + \frac{\sigma_g^I}{\sigma_g^I + \frac{1}{\omega_g^I}}.$$

Rewrite equation A22, we will have

$$(A24) \quad \frac{\partial p_{gv}^{I*}}{\partial \tau_{gk}^I} = \Lambda_{gvk} \frac{p_{gv}^{I*}}{1 + \tau_{gk}^I}.$$

Take the derivative of above equation with respect to τ_{gm}^I , we will have

$$(A25) \quad \frac{\partial^2 p_{gv}^{I*}}{\partial \tau_{gk}^I \partial \tau_{gm}^I} = \begin{cases} \Lambda_{gvv}^1 (\Lambda_{gvv}^1 - 1) \frac{p_{gv}^{I*}}{(1 + \tau_{gv}^I)^2} & \text{if } v = k = m, \\ \Lambda_{gvv}^1 \Lambda_{gvm}^2 \frac{p_{gv}^{I*}}{(1 + \tau_{gv}^I)(1 + \tau_{gm}^I)} & \text{if } v = k \neq m, \\ \Lambda_{gvk}^2 \Lambda_{gvm}^2 \frac{p_{gv}^{I*}}{(1 + \tau_{gv}^I)(1 + \tau_{gm}^I)} & \text{if } v \neq k \text{ and } v \neq m. \end{cases}$$

Define

$$f_k^{gv} = \left. \frac{\partial p_{gv}^{I*}}{\partial \tau_{gk}^I} \right|_{\tau_g^I=0} \quad \text{and} \quad f_{km}^{gv} = \left. \frac{\partial^2 p_{gv}^{I*}}{\partial \tau_{gk}^I \partial \tau_{gm}^I} \right|_{\tau_g^I=0},$$

thus

$$(A26) \quad f_k^{gv} = \begin{cases} \Lambda_{gvv}^1 p_{gv}^{I*} & \text{if } v = k, \\ \Lambda_{gvk}^2 p_{gv}^{I*} & \text{if } v \neq k, \end{cases}$$

and

$$(A27) \quad f_{km}^{gv} = \begin{cases} \Lambda_{gvv}^1 (\Lambda_{gvv}^1 - 1) p_{gv}^{I*} & \text{if } v = k = m, \\ \Lambda_{gvv}^1 \Lambda_{gvm}^2 p_{gv}^{I*} & \text{if } v = k \neq m, \\ \Lambda_{gvk}^2 \Lambda_{gvm}^2 p_{gv}^{I*} & \text{if } v \neq k \text{ and } v \neq m. \end{cases}$$

The first order derivative of imported quantity with respect to tariff

$$\begin{aligned}
\frac{\partial x_{gv}^I}{\partial \tau_{gk}^I} &= \frac{\partial \ln(x_{gv}^I)}{\partial \ln(p_{gv}^I)} \frac{\partial \ln(p_{gv}^I)}{\partial \tau_{gk}^I} \\
&= -\sigma_g^I \left(\frac{\partial \ln(p_{gv}^{I*})}{\partial \tau_{gk}^I} + \frac{\partial \ln(1 + \tau_{gv}^I)}{\partial \tau_{gk}^I} \right) \\
&= -\sigma_g^I \left(\frac{\partial p_{gv}^{I*}}{\partial \tau_{gk}^I} \frac{1}{p_{gv}^{I*}} + \frac{1}{1 + \tau_{gv}^I} \frac{\partial \tau_{gv}^I}{\partial \tau_{gk}^I} \right).
\end{aligned}
\tag{A28}$$

Thus

$$\frac{\partial x_{gv}^I}{\partial \tau_{gk}^I} = \begin{cases} -\sigma_g^I \left(\frac{\partial p_{gv}^{I*}}{\partial \tau_{gv}^I} \frac{1}{p_{gv}^{I*}} + \frac{1}{1 + \tau_{gv}^I} \right) & \text{if } v = k, \\ -\sigma_g^I \frac{\partial p_{gv}^{I*}}{\partial \tau_{gk}^I} \frac{1}{p_{gv}^{I*}} & \text{if } v \neq k. \end{cases}
\tag{A29}$$

Simplify the equation 4

$$\frac{\partial W^I}{\partial \mathbf{T}_g^I} = -\sigma_g^I \frac{\mathbf{T}_g^I}{1 + \mathbf{T}_g^I} \circ \mathbf{P}_g^{I*} \circ \mathbf{X}_g^I - \mathbf{X}_g^I \circ (1 + \sigma_g^I \mathbf{T}_g^I) \frac{\partial \mathbf{P}_g^{I*}}{\partial \mathbf{T}_g^I}
\tag{A30}$$

$$\begin{bmatrix} \frac{\partial W^I}{\partial \tau_{g1}^I} \\ \frac{\partial W^I}{\partial \tau_{g2}^I} \\ \vdots \\ \frac{\partial W^I}{\partial \tau_{gV}^I} \end{bmatrix} = - \begin{bmatrix} \sigma_g^I \frac{\tau_{g1}^I}{1 + \tau_{g1}^I} p_{g1}^{I*} x_{g1}^I \\ \sigma_g^I \frac{\tau_{g2}^I}{1 + \tau_{g2}^I} p_{g2}^{I*} x_{g2}^I \\ \vdots \\ \sigma_g^I \frac{\tau_{gV}^I}{1 + \tau_{gV}^I} p_{gV}^{I*} x_{gV}^I \end{bmatrix} - \begin{bmatrix} \frac{\partial p_{g1}^{I*}}{\partial \tau_{g1}^I} & \frac{\partial p_{g2}^{I*}}{\partial \tau_{g1}^I} & \cdots & \frac{\partial p_{gV}^{I*}}{\partial \tau_{g1}^I} \\ \frac{\partial p_{g1}^{I*}}{\partial \tau_{g2}^I} & \frac{\partial p_{g2}^{I*}}{\partial \tau_{g2}^I} & \cdots & \frac{\partial p_{gV}^{I*}}{\partial \tau_{g2}^I} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial p_{g1}^{I*}}{\partial \tau_{gV}^I} & \frac{\partial p_{g2}^{I*}}{\partial \tau_{gV}^I} & \cdots & \frac{\partial p_{gV}^{I*}}{\partial \tau_{gV}^I} \end{bmatrix} \begin{bmatrix} x_{g1}^I (1 + \sigma_g^I \tau_{g1}^I) \\ x_{g2}^I (1 + \sigma_g^I \tau_{g2}^I) \\ \vdots \\ x_{gV}^I (1 + \sigma_g^I \tau_{gV}^I) \end{bmatrix}
\tag{A31}$$

Specifically, the v -th element is represented as:

$$\frac{\partial W^I}{\partial \tau_{gv}^I} = -\sigma_g^I \frac{\tau_{gv}^I}{1 + \tau_{gv}^I} p_{gv}^{I*} x_{gv}^I - \sum_m \frac{\partial p_{gm}^{I*}}{\partial \tau_{gv}^I} x_{gm}^I (1 + \sigma_g^I \tau_{gm}^I).
\tag{A32}$$

Define

$$F_v^g = \left. \frac{\partial W^I}{\partial \tau_{gv}^I} \right|_{\tau_g^I=0},$$

thus

$$F_v^g = - \sum_m \frac{\partial p_{gm}^{I*}}{\partial \tau_{gv}^I} x_{gm}^I.
\tag{A33}$$

Take the derivative of $\partial W^I / \partial \tau_{gv}^I$ with respect to τ_{gk}^I , we derive

$$(A34) \quad \frac{\partial^2 W^I}{\partial \tau_{gv}^I \partial \tau_{gk}^I} = -\sigma_g^I \left[p_{gv}^{I*} x_{gv}^I \frac{\partial}{\partial \tau_{gk}^I} \left(\frac{\tau_{gv}^I}{1 + \tau_{gv}^I} \right) + \frac{\tau_{gv}^I}{1 + \tau_{gv}^I} \frac{\partial (p_{gv}^{I*} x_{gv}^I)}{\partial \tau_{gk}^I} + \frac{\partial p_{gk}^{I*}}{\partial \tau_{gk}^I} x_{gk}^I \right] \\ - \sum_m \left[\frac{\partial^2 p_{gm}^{I*}}{\partial \tau_{gv}^I \partial \tau_{gk}^I} x_{gm}^I + \frac{\partial p_{gv}^{I*}}{\partial \tau_{gv}^I} \frac{\partial x_{gm}^I}{\partial \tau_{gk}^I} x_{gm}^I \right] (1 + \sigma_g^I \tau_{gm}^I).$$

Define

$$F_{vk}^g = \frac{\partial^2 W^I}{\partial \tau_{gv}^I \partial \tau_{gk}^I} \Big|_{\tau_g^I=0},$$

we derive

$$(A35) \quad F_{vk}^g = \begin{cases} -\sigma_g^I \left[p_{gv}^{I*} x_{gv}^I + \frac{\partial p_{gk}^{I*}}{\partial \tau_{gk}^I} x_{gk}^I \right] - \sum_m \left[\frac{\partial^2 p_{gm}^{I*}}{\partial \tau_{gv}^I \partial \tau_{gk}^I} x_{gm}^I + \frac{\partial p_{gv}^{I*}}{\partial \tau_{gv}^I} \frac{\partial x_{gm}^I}{\partial \tau_{gk}^I} x_{gm}^I \right] & \text{if } v = k, \\ -\sigma_g^I \frac{\partial p_{gk}^{I*}}{\partial \tau_{gk}^I} x_{gk}^I - \sum_m \left[\frac{\partial^2 p_{gm}^{I*}}{\partial \tau_{gv}^I \partial \tau_{gk}^I} x_{gm}^I + \frac{\partial p_{gv}^{I*}}{\partial \tau_{gv}^I} \frac{\partial x_{gm}^I}{\partial \tau_{gk}^I} x_{gm}^I \right] & \text{if } v \neq k. \end{cases}$$

A.3. The Estimated TRIs

Table A1: TRIs

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Albania	ALB	2000	15.31	15.31	15.55	15.55	18.63	18.63
Albania	ALB	2001	15.22	15.22	13.53	13.53	19.61	19.61
Albania	ALB	2002	11.28	11.28	1.41	1.41	14.26	14.26
Albania	ALB	2003	11.37	11.39	21.13	21.04	12.24	12.25
Albania	ALB	2004	11.81	11.81	12.62	12.63	12.72	12.73
Albania	ALB	2005	11.41	9.09	47.41	47.29	10.32	7.21
Albania	ALB	2006	11.12	11.09	49.59	49.59	11.98	11.98
Albania	ALB	2007	11.36	10.49	1.63	1.22	12.31	10.34
Albania	ALB	2008	9.98	4.86	22.00	8.74	13.55	7.80
Albania	ALB	2009	7.54	7.54	1.27	1.27	9.36	9.36
Albania	ALB	2010	10.74	10.74	150.10	150.05	12.76	12.74
Albania	ALB	2011	10.24	5.29	96.32	93.52	12.56	6.10
Albania	ALB	2012	6.25	3.18	—	0.56	5.68	2.37
Albania	ALB	2013	6.10	4.28	9.50	9.84	5.58	5.25
Albania	ALB	2014	6.00	3.32	9.88	7.60	5.27	2.08
Albania	ALB	2015	6.19	3.44	3.74	0.37	5.24	2.12
Albania	ALB	2016	6.20	3.12	3.89	1.41	5.61	2.34
Albania	ALB	2017	6.11	3.62	2.50	58.01	5.50	1.49
Albania	ALB	2018	5.94	3.43	2.26	0.22	5.74	1.96
Albania	ALB	2019	6.36	4.30	—	—	5.53	3.17
Albania	ALB	2020	6.07	4.15	5.01	0.94	5.37	3.41
Algeria	DZA	2001	19.45	19.45	20.21	20.21	17.16	17.16
Algeria	DZA	2002	19.51	19.51	25.09	25.09	17.50	17.50
Algeria	DZA	2003	17.52	17.62	5.89	5.89	10.77	10.78
Algeria	DZA	2005	15.66	11.75	5.66	5.57	5.50	5.37
Algeria	DZA	2006	15.87	12.81	11.27	9.80	11.36	11.11
Algeria	DZA	2007	15.81	10.89	17.23	17.13	9.33	8.36

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Algeria	DZA	2008	16.80	11.92	18.84	18.80	15.66	14.81
Algeria	DZA	2009	19.08	15.06	30.58	31.03	24.95	24.29
Algeria	DZA	2014	22.05	19.98	14.47	14.01	7.72	6.43
Algeria	DZA	2015	18.98	15.10	8.32	5.50	11.86	9.42
Algeria	DZA	2016	20.84	18.83	6.96	1.35	5.27	4.88
Algeria	DZA	2017	21.22	19.05	9.36	5.02	11.79	9.65
Algeria	DZA	2018	19.92	18.95	4.76	3.93	13.40	8.12
Algeria	DZA	2019	20.87	16.67	7.35	0.73	10.40	3.84
Algeria	DZA	2020	21.15	19.38	8.41	7.02	12.57	12.12
Antigua	ATG	2000	22.34	22.34	13.44	13.44	15.35	15.35
Antigua	ATG	2001	19.63	19.63	23.08	23.08	21.64	21.64
Antigua	ATG	2002	18.24	16.98	7.50	7.48	20.27	19.32
Antigua	ATG	2003	19.77	19.77	27.11	27.11	20.38	20.38
Antigua	ATG	2004	18.57	18.57	39.99	39.99	16.46	16.46
Antigua	ATG	2005	18.95	18.95	23.17	23.17	16.97	16.97
Antigua	ATG	2006	27.05	11.88	13.82	14.50	19.30	21.21
Antigua	ATG	2007	18.69	17.12	15.60	75.37	14.06	7.66
Antigua	ATG	2008	21.04	19.60	16.67	16.60	17.14	16.19
Antigua	ATG	2009	14.83	14.93	15.00	15.00	15.07	15.06
Antigua	ATG	2010	18.23	18.16	26.71	26.72	18.34	18.32
Antigua	ATG	2011	20.04	20.04	28.44	28.44	28.40	28.40
Antigua	ATG	2012	18.41	14.25	2.75	5.39	16.22	15.26
Antigua	ATG	2013	23.07	18.61	20.52	16.96	14.74	12.88
Antigua	ATG	2014	15.67	16.96	–	–	20.59	22.05
Antigua	ATG	2015	20.63	12.50	19.98	20.56	18.16	17.47
Antigua	ATG	2016	17.81	14.67	2.11	6.85	19.96	18.97
Antigua	ATG	2017	23.69	14.76	22.73	16.76	21.11	16.94
Antigua	ATG	2018	23.55	15.84	10.38	6.46	25.66	11.16
Antigua	ATG	2019	18.82	17.72	7.69	8.13	17.31	14.28
Antigua	ATG	2020	19.31	15.11	–	–	13.01	9.37
Argentina	ARG	2000	14.14	11.55	13.06	12.03	17.46	11.37
Argentina	ARG	2001	13.79	10.84	14.52	8.70	16.60	11.61
Argentina	ARG	2002	12.07	12.11	11.50	11.47	14.06	14.02
Argentina	ARG	2003	11.44	11.56	9.31	9.26	13.60	13.58
Argentina	ARG	2004	12.60	8.72	16.81	16.81	15.18	13.36
Argentina	ARG	2005	11.92	9.10	15.48	12.56	15.49	12.17
Argentina	ARG	2006	12.53	5.76	14.56	15.91	14.34	6.11
Argentina	ARG	2007	12.80	7.96	3.70	3.67	17.53	11.75
Argentina	ARG	2008	11.18	7.10	2.65	3.08	7.69	5.29
Argentina	ARG	2009	13.04	7.76	433.55	5.65	7.33	3.02
Argentina	ARG	2010	13.14	6.97	32.81	15.39	10.86	5.33
Argentina	ARG	2011	13.43	6.16	8.09	3.85	10.36	5.86
Argentina	ARG	2012	13.97	7.63	35.18	8.21	12.03	4.85
Argentina	ARG	2013	11.34	7.15	0.25	4.89	11.97	5.84
Argentina	ARG	2014	12.30	8.80	22.57	22.69	9.44	8.30
Argentina	ARG	2015	12.31	7.70	2.35	2.81	11.43	10.59
Argentina	ARG	2016	11.89	7.16	22.26	22.92	11.72	11.50
Argentina	ARG	2017	13.82	10.80	3.29	4.34	8.51	5.03
Argentina	ARG	2018	13.06	8.07	9.77	9.44	11.95	6.31
Argentina	ARG	2019	11.96	7.13	4.61	3.93	11.37	8.84
Argentina	ARG	2020	12.05	8.97	4.42	3.00	9.95	6.37
Armenia	ARM	2001	7.90	7.90	0.50	0.50	1.80	1.80
Armenia	ARM	2003	4.63	4.63	130.64	130.64	6.32	6.32
Armenia	ARM	2004	6.95	6.95	88.68	88.59	7.24	7.24
Armenia	ARM	2005	7.88	7.88	209.93	209.93	0.89	0.89

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Armenia	ARM	2006	10.20	5.09	38.42	19.00	9.31	2.07
Armenia	ARM	2007	5.77	5.77	191.49	191.49	3.47	3.47
Armenia	ARM	2008	5.02	4.72	202.40	202.40	3.52	3.39
Armenia	ARM	2009	4.47	4.47	0.00	0.00	5.37	5.37
Armenia	ARM	2010	4.53	4.54	0.72	0.72	5.98	6.01
Armenia	ARM	2011	5.46	5.43	0.40	0.40	6.57	6.54
Armenia	ARM	2012	6.58	3.54	0.12	0.11	6.99	4.15
Armenia	ARM	2013	7.31	6.85	1.02	0.27	7.75	7.47
Armenia	ARM	2014	7.77	2.29	6.50	6.32	8.49	6.04
Armenia	ARM	2015	14.65	5.84	1.70	1.34	5.31	2.50
Armenia	ARM	2016	18.55	17.57	0.34	0.26	2.98	2.76
Armenia	ARM	2017	13.11	3.89	0.13	0.10	2.65	0.60
Armenia	ARM	2019	5.64	4.91	4.52	0.31	3.64	1.93
Armenia	ARM	2020	13.13	12.65	2.84	2.83	3.37	3.35
Australia	AUS	2000	6.19	3.93	30.11	30.03	1.50	1.37
Australia	AUS	2001	4.13	3.57	0.03	0.03	0.40	0.34
Australia	AUS	2002	3.61	3.03	33.02	33.02	1.11	0.80
Australia	AUS	2003	3.64	2.48	0.02	0.00	1.09	0.89
Australia	AUS	2004	3.54	2.83	30.31	0.02	1.02	0.79
Australia	AUS	2005	3.13	2.45	38.59	0.01	0.61	0.46
Australia	AUS	2006	3.51	2.45	42.22	0.01	0.73	0.55
Australia	AUS	2007	3.48	2.35	0.02	0.01	0.50	0.33
Australia	AUS	2008	3.53	2.94	0.00	–	0.58	0.46
Australia	AUS	2009	3.29	2.58	35.11	0.00	0.38	0.28
Australia	AUS	2010	2.98	2.12	37.30	37.08	0.40	0.34
Australia	AUS	2011	2.66	1.49	39.30	39.25	0.44	0.29
Australia	AUS	2012	2.77	1.63	38.19	0.02	0.28	0.35
Australia	AUS	2013	2.63	2.27	0.01	0.00	0.27	0.15
Australia	AUS	2014	2.46	1.49	33.80	33.80	0.25	0.19
Australia	AUS	2015	2.47	1.49	30.78	30.77	0.29	0.22
Australia	AUS	2016	2.78	2.17	0.01	29.98	0.26	0.12
Australia	AUS	2017	2.53	1.12	24.38	24.26	0.31	0.13
Australia	AUS	2018	2.64	1.30	34.15	34.13	0.38	0.14
Australia	AUS	2019	2.57	1.41	0.00	0.01	0.26	0.21
Australia	AUS	2020	2.31	1.47	39.17	39.15	0.22	0.21
Austria	AUT	2000	10.57	8.39	1.70	1.15	3.04	2.09
Austria	AUT	2001	13.28	11.67	0.56	0.38	1.93	1.63
Austria	AUT	2002	11.82	10.08	0.29	0.11	1.25	0.70
Austria	AUT	2003	12.13	9.65	47.82	46.27	2.67	1.68
Austria	AUT	2004	12.25	10.89	1.92	1.35	2.48	1.95
Austria	AUT	2005	12.37	10.20	0.27	0.17	1.81	1.34
Austria	AUT	2006	12.13	9.51	0.19	0.12	1.28	0.96
Austria	AUT	2007	11.16	8.94	0.09	0.04	0.86	0.58
Austria	AUT	2008	11.10	7.71	0.10	0.04	1.01	0.75
Austria	AUT	2009	11.04	9.37	0.06	0.03	0.72	0.52
Austria	AUT	2010	11.54	8.44	0.02	63.45	0.82	0.63
Austria	AUT	2011	11.45	8.88	0.06	0.03	0.76	0.56
Austria	AUT	2012	10.60	8.30	64.29	64.30	0.61	0.39
Austria	AUT	2013	10.51	8.04	0.04	0.02	0.72	0.50
Austria	AUT	2014	10.87	8.82	0.09	0.07	0.72	0.52
Austria	AUT	2015	10.35	8.80	63.03	0.01	0.75	0.57
Austria	AUT	2016	10.25	8.11	0.03	0.03	0.58	0.36
Austria	AUT	2017	10.59	8.40	0.14	0.09	0.97	0.70
Austria	AUT	2018	9.95	8.28	0.05	0.03	0.70	0.47
Austria	AUT	2019	8.90	7.57	0.03	0.01	0.59	0.44

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Austria	AUT	2020	4.84	3.45	0.03	0.01	0.46	0.34
Azerbaijan	AZE	2002	12.55	12.53	18.11	11.36	11.84	9.67
Azerbaijan	AZE	2005	14.00	14.52	15.00	5.00	14.84	4.99
Azerbaijan	AZE	2007	14.07	2.75	6.76	6.14	4.16	3.83
Azerbaijan	AZE	2008	9.24	8.34	1.01	6.05	7.68	3.66
Azerbaijan	AZE	2009	13.22	4.03	3.75	4.46	5.36	5.06
Azerbaijan	AZE	2011	9.61	4.70	5.68	1.00	6.01	5.52
Azerbaijan	AZE	2012	13.33	4.35	2.77	1.97	8.32	8.23
Azerbaijan	AZE	2013	12.98	13.01	8.01	4.84	6.31	5.81
Azerbaijan	AZE	2014	13.66	5.69	13.23	12.71	7.19	6.34
Azerbaijan	AZE	2015	12.81	4.75	8.40	4.44	2.25	3.07
Azerbaijan	AZE	2019	12.90	6.74	4.56	2.28	6.71	6.02
Azerbaijan	AZE	2020	5.22	3.85	5.27	8.43	5.66	13.93
Bahamas	BHS	2002	22.68	22.67	23.30	23.30	21.40	21.39
Bahamas	BHS	2006	28.27	28.29	39.27	39.30	30.01	30.02
Bahamas	BHS	2010	30.12	30.06	45.63	45.63	33.05	32.91
Bahamas	BHS	2011	35.97	35.97	28.03	28.25	27.58	27.58
Bahamas	BHS	2013	37.52	35.74	29.63	28.33	38.32	36.46
Bahamas	BHS	2014	38.26	34.93	37.79	19.91	41.10	24.34
Bahamas	BHS	2015	33.58	31.47	2.32	2.46	44.70	44.51
Bahamas	BHS	2016	30.36	28.82	43.56	43.38	31.44	29.00
Bahamas	BHS	2018	33.68	31.22	41.04	33.54	40.78	38.74
Bahrain	BHR	2000	7.12	7.12	10.03	10.03	19.41	19.41
Bahrain	BHR	2001	7.57	7.57	9.68	9.68	12.67	12.67
Bahrain	BHR	2002	6.80	6.80	5.00	5.00	7.37	7.37
Bahrain	BHR	2003	6.92	6.96	5.00	5.01	7.20	7.50
Bahrain	BHR	2004	6.65	6.65	5.26	5.26	13.95	13.95
Bahrain	BHR	2005	5.00	1.39	5.93	55.26	4.90	4.62
Bahrain	BHR	2006	5.34	1.33	4.94	5.18	5.33	5.78
Bahrain	BHR	2007	4.99	1.27	4.62	6.24	5.00	1.22
Bahrain	BHR	2008	4.99	4.50	19.33	18.62	4.77	1.77
Bahrain	BHR	2009	5.00	3.86	4.64	4.12	4.48	–
Bahrain	BHR	2010	6.72	6.71	4.60	4.61	7.12	7.08
Bahrain	BHR	2011	14.16	2.79	14.94	10.03	11.77	3.60
Bahrain	BHR	2012	13.80	12.71	4.53	0.68	10.22	9.07
Bahrain	BHR	2013	4.92	3.04	6.93	–	4.58	2.79
Bahrain	BHR	2014	4.85	3.60	5.75	1.83	3.73	3.38
Bahrain	BHR	2015	4.55	3.12	4.09	2.71	4.29	3.18
Bahrain	BHR	2016	5.00	2.10	2.87	1.03	3.91	2.77
Bahrain	BHR	2017	5.09	2.07	4.14	4.69	4.30	3.61
Bahrain	BHR	2018	4.89	4.07	41.43	41.44	3.59	3.11
Bahrain	BHR	2019	4.96	1.34	1.99	50.29	4.57	2.97
Bahrain	BHR	2020	4.97	4.81	6.13	3.60	4.77	3.42
Bangladesh	BGD	2000	34.92	34.92	36.65	36.65	37.09	37.09
Bangladesh	BGD	2001	33.30	33.30	5.89	5.89	37.25	37.25
Bangladesh	BGD	2002	36.08	36.08	36.95	36.95	37.15	37.15
Bangladesh	BGD	2003	31.21	31.21	32.49	32.49	32.34	32.34
Bangladesh	BGD	2004	28.69	28.69	29.84	29.84	29.81	29.81
Bangladesh	BGD	2005	23.83	23.83	25.61	25.62	25.52	25.52
Bangladesh	BGD	2006	24.24	24.24	22.52	22.52	25.72	25.72
Bangladesh	BGD	2007	24.00	23.99	18.04	18.04	24.28	24.25
Bangladesh	BGD	2008	23.21	23.21	6.54	6.54	24.81	24.81
Bangladesh	BGD	2009	24.34	24.34	5.73	5.71	24.79	24.79
Bangladesh	BGD	2010	24.52	24.52	5.04	5.03	24.87	24.86
Bangladesh	BGD	2011	24.57	24.57	24.43	24.43	24.82	24.82

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Bangladesh	BGD	2012	24.61	24.61	19.57	19.56	24.77	24.77
Bangladesh	BGD	2013	24.65	24.65	13.12	13.12	24.76	24.76
Bangladesh	BGD	2015	24.69	24.43	0.22	23.17	24.82	23.97
Bangladesh	BGD	2016	24.83	24.75	24.99	25.79	24.88	23.86
Bangladesh	BGD	2019	24.85	24.80	24.14	25.25	24.90	24.30
Bangladesh	BGD	2020	24.87	24.74	5.22	0.79	24.68	22.87
Barbados	BRB	2000	24.05	22.43	9.03	9.03	15.97	15.97
Barbados	BRB	2001	21.38	20.16	5.98	6.07	15.02	14.99
Barbados	BRB	2002	17.22	16.33	10.25	10.26	7.74	7.73
Barbados	BRB	2003	26.20	16.97	31.60	31.83	10.77	9.74
Barbados	BRB	2004	40.78	25.55	5.69	5.35	18.88	10.68
Barbados	BRB	2006	33.16	23.09	19.98	15.30	22.00	16.56
Barbados	BRB	2007	34.85	28.85	17.81	4.11	17.57	13.21
Barbados	BRB	2010	21.12	21.16	15.48	15.48	31.02	31.01
Barbados	BRB	2011	19.88	19.68	3.70	3.69	9.28	9.27
Barbados	BRB	2012	28.91	18.00	4.82	11.53	23.74	7.14
Barbados	BRB	2013	26.17	20.81	20.91	22.25	20.77	20.16
Barbados	BRB	2020	23.68	14.31	5.87	4.46	19.95	9.03
Belarus	BLR	2000	10.37	10.37	9.49	9.49	10.16	10.16
Belarus	BLR	2001	11.15	11.16	12.49	12.49	11.64	11.64
Belarus	BLR	2002	13.22	13.22	8.28	8.28	15.73	15.73
Belarus	BLR	2008	11.02	7.25	6.32	1.06	13.89	9.70
Belarus	BLR	2009	10.22	7.70	5.52	0.61	13.06	9.00
Belarus	BLR	2010	12.48	7.51	2.12	0.11	9.32	5.56
Belarus	BLR	2011	11.21	6.26	2.97	0.96	4.67	3.83
Belarus	BLR	2012	11.34	6.22	7.33	7.75	7.89	4.70
Belarus	BLR	2013	11.30	7.41	8.90	6.34	7.92	3.89
Belarus	BLR	2014	9.78	6.31	3.87	3.03	6.24	4.17
Belarus	BLR	2015	9.22	5.75	0.87	0.90	5.06	3.45
Belarus	BLR	2016	10.03	7.26	9.72	7.71	5.14	3.90
Belarus	BLR	2017	10.07	6.94	2.97	0.60	3.72	3.58
Belarus	BLR	2018	11.37	7.92	1.46	0.35	5.59	4.10
Belarus	BLR	2019	10.97	7.43	5.26	4.22	5.86	4.81
Belarus	BLR	2020	10.75	6.98	2.75	2.46	5.54	4.25
Belgium	BEL	2000	16.15	13.82	2.82	2.51	3.97	3.52
Belgium	BEL	2001	15.05	14.05	1.93	1.56	3.47	3.08
Belgium	BEL	2002	12.75	11.07	1.41	0.14	3.81	1.51
Belgium	BEL	2003	13.38	11.39	1.00	0.23	3.32	1.81
Belgium	BEL	2004	13.62	11.45	2.05	0.45	3.29	1.85
Belgium	BEL	2005	12.92	11.48	1.70	1.24	3.30	2.32
Belgium	BEL	2006	13.46	11.47	1.44	1.22	2.93	2.29
Belgium	BEL	2007	12.36	9.97	91.28	91.12	3.61	2.05
Belgium	BEL	2008	12.19	9.67	1.75	0.69	3.32	1.90
Belgium	BEL	2009	12.87	10.37	0.43	0.34	1.82	1.45
Belgium	BEL	2010	13.74	10.35	0.71	0.61	2.60	2.05
Belgium	BEL	2011	14.37	11.05	96.59	96.59	2.80	2.34
Belgium	BEL	2012	13.74	11.08	0.18	0.02	2.05	1.64
Belgium	BEL	2013	13.43	10.12	0.26	0.16	2.29	1.79
Belgium	BEL	2014	13.57	10.54	0.37	0.25	2.22	1.91
Belgium	BEL	2015	13.81	10.77	91.18	90.79	2.54	1.80
Belgium	BEL	2016	14.39	12.15	1.84	1.42	2.83	2.18
Belgium	BEL	2017	14.39	10.43	0.64	0.48	2.44	2.12
Belgium	BEL	2018	12.48	10.60	2.81	2.62	4.40	4.14
Belgium	BEL	2019	13.29	11.96	0.55	0.39	2.22	1.89
Belgium	BEL	2020	7.68	3.76	0.24	0.19	1.39	0.98

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Belize	BLZ	2001	19.36	19.36	19.93	19.93	19.96	19.96
Belize	BLZ	2002	30.41	30.41	0.15	0.15	12.72	12.72
Belize	BLZ	2003	16.59	16.59	0.30	0.30	12.48	12.48
Belize	BLZ	2006	22.39	21.84	1.15	1.08	4.05	3.99
Belize	BLZ	2007	20.53	19.03	4.61	4.61	14.10	13.65
Belize	BLZ	2008	12.39	11.65	9.96	559.00	11.01	9.78
Belize	BLZ	2009	12.79	10.62	5.02	4.97	14.09	13.04
Belize	BLZ	2010	18.97	15.69	10.03	13.57	18.99	18.54
Belize	BLZ	2011	16.30	15.01	5.66	5.66	9.09	9.05
Belize	BLZ	2012	14.50	12.83	0.23	0.62	15.36	11.68
Belize	BLZ	2013	15.77	14.95	135.03	135.14	15.79	14.49
Belize	BLZ	2014	15.99	13.57	20.79	20.94	21.71	20.41
Belize	BLZ	2015	19.94	16.83	7.27	7.26	20.67	19.76
Belize	BLZ	2016	17.62	14.02	1.10	0.65	19.27	13.43
Belize	BLZ	2017	18.88	15.44	89.96	100.25	20.16	19.00
Belize	BLZ	2018	20.40	16.86	5.00	84.65	9.38	7.92
Belize	BLZ	2019	33.46	32.69	0.57	0.66	10.78	10.49
Belize	BLZ	2020	34.43	20.43	20.01	8.00	20.04	8.05
Benin	BEN	2001	14.48	14.52	–	–	14.97	15.00
Benin	BEN	2002	15.86	15.86	–	–	19.28	19.28
Benin	BEN	2003	10.71	10.71	22.93	22.93	11.78	11.78
Benin	BEN	2004	12.68	11.63	24.61	24.97	17.47	17.25
Benin	BEN	2005	12.27	12.27	19.30	19.30	16.77	16.77
Benin	BEN	2006	12.40	8.56	49.68	–	19.18	4.08
Benin	BEN	2007	13.15	8.10	20.56	104.80	19.24	0.40
Benin	BEN	2008	11.68	12.32	19.22	18.76	17.00	17.17
Benin	BEN	2009	14.79	10.61	–	–	16.34	15.25
Benin	BEN	2010	11.70	10.60	7.80	7.48	8.15	8.22
Benin	BEN	2011	11.32	7.54	10.52	8.00	10.52	7.03
Benin	BEN	2012	11.68	7.83	7.94	5.68	8.10	6.55
Benin	BEN	2013	14.57	9.99	18.87	17.96	0.79	21.73
Benin	BEN	2014	10.90	9.11	9.73	7.62	10.02	7.91
Benin	BEN	2015	15.45	9.03	10.10	4.55	10.04	0.90
Benin	BEN	2016	16.59	12.85	10.46	8.72	10.37	8.51
Benin	BEN	2017	22.40	22.44	21.48	21.58	14.51	14.55
Benin	BEN	2018	24.03	22.86	12.16	10.93	14.91	16.19
Benin	BEN	2019	21.50	19.96	–	0.66	19.42	20.10
Benin	BEN	2020	13.18	7.41	11.15	8.73	12.45	13.23
Bolivia	BOL	2000	9.32	9.32	10.05	10.05	9.91	9.91
Bolivia	BOL	2001	9.13	9.13	10.01	10.01	9.80	9.80
Bolivia	BOL	2002	10.24	10.24	9.99	9.99	9.82	9.82
Bolivia	BOL	2003	9.04	9.04	10.00	10.00	9.85	9.85
Bolivia	BOL	2004	9.54	8.41	10.00	9.12	9.97	8.31
Bolivia	BOL	2005	9.77	9.14	10.00	10.00	9.93	9.43
Bolivia	BOL	2006	9.54	4.27	7.03	4.04	9.71	8.38
Bolivia	BOL	2007	9.73	8.95	9.49	0.45	9.93	5.93
Bolivia	BOL	2008	9.68	7.89	9.95	4.84	9.84	6.98
Bolivia	BOL	2009	12.12	8.93	–	10.37	6.75	4.27
Bolivia	BOL	2010	14.31	11.22	5.17	4.38	9.16	8.24
Bolivia	BOL	2011	12.63	7.80	8.35	8.69	13.72	9.27
Bolivia	BOL	2012	11.75	8.67	6.70	3.73	12.06	10.30
Bolivia	BOL	2013	14.19	10.07	9.46	8.66	12.51	7.08
Bolivia	BOL	2014	14.03	12.09	–	0.89	12.08	10.22
Bolivia	BOL	2015	13.87	10.97	10.26	3.71	12.01	10.82
Bolivia	BOL	2016	12.86	11.89	6.81	4.27	10.98	10.41

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Bolivia	BOL	2017	12.73	7.96	9.27	6.16	10.46	8.32
Bolivia	BOL	2018	13.02	10.92	9.01	4.44	11.68	10.58
Bolivia	BOL	2019	12.41	5.62	0.87	0.80	8.15	7.72
Bolivia	BOL	2020	12.55	7.62	5.71	4.77	9.02	8.68
Botswana	BWA	2001	33.26	23.94	8.10	0.46	20.40	8.45
Botswana	BWA	2002	29.73	18.73	3.43	2.61	13.22	9.56
Botswana	BWA	2004	28.32	19.91	8.20	79.01	33.21	28.28
Botswana	BWA	2005	30.18	21.48	245.91	210.02	29.58	23.84
Botswana	BWA	2006	30.69	17.50	–	–	32.73	18.14
Botswana	BWA	2007	31.80	23.18	0.05	0.00	19.02	15.04
Botswana	BWA	2008	27.83	16.34	193.36	0.08	36.84	17.77
Botswana	BWA	2009	26.10	15.79	304.94	276.52	23.18	11.23
Botswana	BWA	2010	25.50	17.62	58.36	57.49	23.49	17.00
Botswana	BWA	2011	30.67	10.45	1.05	0.03	26.31	10.67
Botswana	BWA	2012	21.84	15.13	0.01	0.01	2.29	1.03
Botswana	BWA	2013	19.33	12.34	0.71	0.07	5.65	2.93
Botswana	BWA	2014	18.47	8.13	0.90	0.07	3.21	1.34
Botswana	BWA	2015	20.71	17.18	6.55	3.96	14.37	10.62
Botswana	BWA	2016	16.14	10.32	6.56	0.83	8.36	6.80
Botswana	BWA	2017	15.38	11.01	22.18	12.33	13.95	7.89
Botswana	BWA	2018	15.28	7.59	5.37	0.06	7.95	4.11
Botswana	BWA	2019	15.19	7.73	3.07	0.52	12.08	7.27
Botswana	BWA	2020	13.29	6.93	2.45	2.13	12.80	8.45
Brazil	BRA	2000	16.11	16.11	48.98	49.02	17.36	17.32
Brazil	BRA	2001	15.28	15.28	–	–	6.80	6.84
Brazil	BRA	2002	14.35	14.33	6.94	6.83	9.73	9.58
Brazil	BRA	2003	13.93	13.93	6.09	5.95	10.42	10.31
Brazil	BRA	2004	14.71	10.77	7.57	6.82	10.25	9.27
Brazil	BRA	2005	14.24	9.76	8.80	0.33	10.62	4.77
Brazil	BRA	2006	12.84	9.24	10.19	2.21	10.28	5.06
Brazil	BRA	2007	12.86	8.27	4.53	3.84	8.96	6.84
Brazil	BRA	2008	13.65	11.06	18.94	27.94	10.27	6.96
Brazil	BRA	2009	12.76	9.88	4.21	1.98	10.25	7.78
Brazil	BRA	2010	12.32	10.46	23.94	19.86	6.84	5.93
Brazil	BRA	2011	12.36	10.29	9.36	10.55	11.21	9.03
Brazil	BRA	2012	12.11	10.45	0.03	67.08	9.85	7.01
Brazil	BRA	2013	11.97	12.06	–	–	10.20	10.18
Brazil	BRA	2014	11.74	10.05	24.78	25.05	9.59	7.83
Brazil	BRA	2015	11.55	9.93	12.35	155.16	8.34	4.91
Brazil	BRA	2016	11.80	8.94	–	–	6.23	4.34
Brazil	BRA	2017	12.03	8.62	1.42	17.20	6.44	4.47
Brazil	BRA	2018	11.15	8.96	16.73	17.84	9.29	7.00
Brazil	BRA	2019	10.83	9.11	2.24	2.28	10.42	8.92
Brazil	BRA	2020	10.84	8.96	13.60	13.14	9.69	7.94
Brunei	BRN	2000	1.12	1.12	0.08	0.08	1.46	1.46
Brunei	BRN	2001	2.31	2.27	0.00	0.02	4.26	4.25
Brunei	BRN	2002	2.53	2.37	2.81	2.77	3.34	3.13
Brunei	BRN	2003	1.42	1.12	1575.68	931.83	3.77	2.14
Brunei	BRN	2004	4.57	4.42	27.06	27.06	1.42	1.33
Brunei	BRN	2005	2.54	2.54	4.89	0.00	–	–
Brunei	BRN	2006	2.86	2.30	3.75	1.56	3.65	2.83
Brunei	BRN	2007	3.86	3.46	22.37	22.43	1.64	1.27
Brunei	BRN	2008	2.61	1.87	0.01	0.01	2.10	1.66
Brunei	BRN	2009	3.34	2.21	0.19	0.26	1.75	1.36
Brunei	BRN	2010	4.91	5.59	0.67	0.67	5.58	5.60

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Brunei	BRN	2011	4.84	3.24	0.00	0.01	1.33	0.35
Brunei	BRN	2014	5.53	3.44	0.32	0.07	3.50	0.92
Brunei	BRN	2016	5.74	2.06	3.48	1.03	3.10	0.98
Brunei	BRN	2017	0.41	0.11	0.03	0.00	0.31	0.31
Brunei	BRN	2018	0.44	0.04	0.00	0.00	0.04	0.00
Brunei	BRN	2019	0.19	0.07	0.00	0.00	0.06	0.01
Brunei	BRN	2020	0.21	0.19	0.01	0.00	0.22	0.21
Bulgaria	BGR	2000	21.76	21.76	1.70	1.73	20.64	20.74
Bulgaria	BGR	2001	21.39	17.15	0.78	0.95	22.20	14.73
Bulgaria	BGR	2002	19.64	19.63	40.31	38.81	16.39	16.37
Bulgaria	BGR	2003	19.90	19.89	107.39	107.41	17.73	17.74
Bulgaria	BGR	2004	20.27	20.27	6.41	6.37	18.20	18.18
Bulgaria	BGR	2005	18.30	18.85	1.67	0.52	7.49	7.47
Bulgaria	BGR	2006	16.82	12.19	27.98	13.12	15.80	9.20
Bulgaria	BGR	2007	11.80	9.55	22.65	21.06	7.93	4.09
Bulgaria	BGR	2008	12.57	10.06	—	5.01	6.08	3.82
Bulgaria	BGR	2009	15.10	13.06	0.24	0.52	5.99	2.73
Bulgaria	BGR	2010	14.64	13.92	1.21	1.41	5.43	4.28
Bulgaria	BGR	2011	13.64	12.77	11.83	12.54	6.16	4.72
Bulgaria	BGR	2012	15.26	11.13	1.63	36.27	6.90	4.49
Bulgaria	BGR	2013	14.67	11.21	6.36	4.36	6.40	4.00
Bulgaria	BGR	2014	15.27	9.05	2.82	1.10	6.61	3.75
Bulgaria	BGR	2015	15.32	10.33	4.57	4.38	5.65	3.31
Bulgaria	BGR	2016	14.60	11.38	24.22	23.87	5.70	3.92
Bulgaria	BGR	2017	14.66	13.42	1.45	0.87	5.08	3.74
Bulgaria	BGR	2018	14.37	12.45	2.74	1.17	4.69	3.17
Bulgaria	BGR	2019	12.77	6.93	54.09	53.34	2.92	1.58
Burkina Faso	BFA	2001	15.52	15.52	5.24	5.24	13.62	13.62
Burkina Faso	BFA	2002	15.52	15.52	4.92	4.92	16.49	16.49
Burkina Faso	BFA	2003	18.53	18.53	5.07	5.07	19.30	19.30
Burkina Faso	BFA	2004	17.77	17.77	5.05	5.05	18.71	18.71
Burkina Faso	BFA	2005	17.38	17.22	4.94	4.94	16.01	16.01
Burkina Faso	BFA	2006	13.76	10.78	7.19	6.95	17.60	13.38
Burkina Faso	BFA	2007	14.17	12.57	5.55	5.43	11.84	11.26
Burkina Faso	BFA	2008	13.88	11.78	5.71	5.59	7.60	7.27
Burkina Faso	BFA	2009	13.38	9.13	5.40	5.38	16.72	7.97
Burkina Faso	BFA	2010	13.58	8.75	12.05	12.21	16.35	11.70
Burkina Faso	BFA	2011	14.60	9.21	5.43	3.73	11.33	8.04
Burkina Faso	BFA	2012	12.69	10.10	9.84	5.98	9.74	8.94
Burkina Faso	BFA	2013	11.78	10.49	19.53	0.43	18.03	4.03
Burkina Faso	BFA	2014	10.87	10.58	4.97	5.24	11.21	10.72
Burkina Faso	BFA	2015	15.00	11.08	4.99	4.29	9.78	9.14
Burkina Faso	BFA	2016	17.81	11.09	4.99	4.86	7.94	7.57
Burkina Faso	BFA	2017	18.08	18.08	11.22	11.22	6.68	6.68
Burkina Faso	BFA	2018	17.67	16.84	20.61	20.57	13.80	13.93
Burkina Faso	BFA	2019	16.80	12.06	5.26	4.61	6.47	17.67
Burkina Faso	BFA	2020	18.06	13.21	11.91	8.81	8.33	7.64
Burundi	BDI	2002	18.34	11.72	16.31	15.51	23.36	20.30
Burundi	BDI	2003	17.24	12.27	14.37	14.64	19.96	17.12
Burundi	BDI	2005	21.57	21.57	30.00	30.00	29.54	29.54
Burundi	BDI	2006	14.47	15.35	10.02	10.03	13.96	14.28
Burundi	BDI	2007	14.59	17.91	10.34	9.61	14.81	14.76
Burundi	BDI	2008	19.52	14.25	10.00	178.76	10.35	3.09
Burundi	BDI	2009	22.47	16.12	10.00	10.03	17.37	13.79
Burundi	BDI	2010	40.67	35.95	10.45	10.45	20.56	20.75

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Burundi	BDI	2011	17.61	12.41	11.32	11.02	14.77	11.13
Burundi	BDI	2012	15.53	11.73	111.65	108.78	14.01	12.77
Burundi	BDI	2013	19.27	4.28	0.33	0.26	22.24	19.63
Burundi	BDI	2014	32.21	20.95	21.63	22.45	26.76	18.36
Burundi	BDI	2015	29.27	18.94	0.38	0.02	26.20	20.24
Burundi	BDI	2016	26.04	18.87	0.42	0.26	21.31	15.78
Burundi	BDI	2018	30.67	28.66	29.58	1.56	–	–
Burundi	BDI	2019	36.14	34.82	0.44	0.26	28.19	23.67
Burundi	BDI	2020	30.21	26.48	0.07	0.01	–	–
CIV	CIV	2001	18.07	18.07	5.60	5.59	7.61	7.61
CIV	CIV	2002	18.68	18.54	5.17	5.17	16.35	16.25
CIV	CIV	2003	17.96	17.81	–	–	16.88	16.72
CIV	CIV	2004	17.30	17.37	6.20	6.25	12.04	12.12
CIV	CIV	2005	16.26	16.36	8.80	8.91	11.28	11.50
CIV	CIV	2006	14.66	13.49	4.99	5.08	9.36	8.73
CIV	CIV	2007	16.13	14.94	5.23	5.12	11.72	10.27
CIV	CIV	2008	15.09	13.82	5.31	5.36	9.83	9.42
CIV	CIV	2009	15.10	14.16	4.84	5.22	9.24	8.92
CIV	CIV	2010	15.80	14.22	5.01	5.00	9.65	7.67
CIV	CIV	2011	14.76	14.08	6.74	6.88	9.57	9.31
CIV	CIV	2012	15.65	15.50	7.08	–	10.31	9.81
CIV	CIV	2013	15.77	13.64	4.24	4.80	7.61	7.78
CIV	CIV	2014	16.10	15.65	3.57	6.61	10.45	10.35
CIV	CIV	2015	18.70	14.99	5.74	3.04	8.83	5.71
CIV	CIV	2016	18.30	17.04	11.35	10.94	9.31	5.47
CIV	CIV	2017	19.58	19.47	7.30	7.26	12.64	12.46
CIV	CIV	2018	18.77	13.42	5.02	5.07	5.04	5.34
CIV	CIV	2019	18.04	12.00	5.25	4.89	7.03	6.51
CIV	CIV	2020	19.28	17.01	11.77	11.92	14.06	14.00
Cabo Verde	CPV	2005	6.29	6.29	4.59	4.59	8.66	8.66
Cabo Verde	CPV	2006	11.21	11.21	4.65	4.65	19.92	19.92
Cabo Verde	CPV	2007	12.06	12.06	0.02	0.01	2.84	2.83
Cabo Verde	CPV	2008	19.76	19.76	1.11	0.14	8.55	7.66
Cabo Verde	CPV	2009	4.92	6.66	9.45	6.76	3.59	3.79
Cabo Verde	CPV	2010	3.45	4.89	1.45	1.41	3.36	3.48
Cabo Verde	CPV	2011	4.81	5.56	0.30	0.30	5.65	5.65
Cabo Verde	CPV	2012	7.12	4.89	0.03	0.04	2.80	2.80
Cabo Verde	CPV	2013	19.27	19.27	0.14	0.14	4.04	4.05
Cabo Verde	CPV	2015	17.52	17.68	2.88	2.87	2.77	2.78
Cabo Verde	CPV	2019	26.58	26.71	0.14	0.14	15.41	15.26
Cabo Verde	CPV	2020	26.64	24.90	0.77	0.08	12.20	5.75
Cameroon	CMR	2001	12.09	12.09	6.43	6.43	12.55	12.55
Cameroon	CMR	2002	13.49	13.49	12.21	12.20	11.64	11.64
Cameroon	CMR	2003	12.48	12.48	11.37	11.37	12.33	12.33
Cameroon	CMR	2004	11.60	11.60	10.72	10.72	12.01	12.01
Cameroon	CMR	2005	11.70	11.69	11.25	11.24	13.12	13.12
Cameroon	CMR	2006	11.20	11.20	6.99	6.89	11.53	11.57
Cameroon	CMR	2007	11.81	9.85	11.39	18.89	10.36	18.38
Cameroon	CMR	2008	12.29	9.52	27.30	35.35	12.51	10.55
Cameroon	CMR	2009	13.52	11.53	20.82	19.28	14.16	12.97
Cameroon	CMR	2010	12.07	12.01	4.04	5.29	10.13	9.90
Cameroon	CMR	2011	13.33	10.92	1.37	21.51	13.60	13.43
Cameroon	CMR	2012	12.09	10.11	9.51	712.78	10.91	6.60
Cameroon	CMR	2013	12.17	9.84	11.23	9.87	11.77	10.56
Cameroon	CMR	2014	10.83	10.65	29.70	29.69	11.31	11.17

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Cameroon	CMR	2019	23.10	22.37	19.31	14.23	17.42	15.10
Canada	CAN	2000	18.21	4.56	0.11	0.30	4.26	0.74
Canada	CAN	2001	17.95	5.21	–	–	3.06	0.73
Canada	CAN	2002	16.17	14.15	–	29.95	3.26	2.03
Canada	CAN	2003	16.29	15.75	4.83	1.85	12.51	1.60
Canada	CAN	2004	17.70	17.31	1.92	1.38	2.75	2.24
Canada	CAN	2005	15.70	15.23	1.48	35.82	2.57	1.36
Canada	CAN	2006	17.26	5.38	0.93	38.20	2.09	1.37
Canada	CAN	2007	18.82	18.56	0.36	10.48	1.96	1.18
Canada	CAN	2008	18.57	4.49	–	–	1.28	0.95
Canada	CAN	2009	20.29	4.70	38.21	0.17	1.36	0.91
Canada	CAN	2010	17.98	4.62	30.57	1.07	0.99	0.79
Canada	CAN	2011	17.96	4.32	0.28	0.15	0.88	0.78
Canada	CAN	2012	18.37	4.37	3.16	–	1.53	1.01
Canada	CAN	2013	18.77	4.35	1.54	–	1.30	0.86
Canada	CAN	2014	19.42	4.39	–	0.72	1.03	0.88
Canada	CAN	2015	21.59	5.41	6.18	–	1.31	0.78
Canada	CAN	2016	19.89	5.75	1.90	0.26	1.67	0.92
Canada	CAN	2017	20.02	19.66	–	–	1.17	0.81
Canada	CAN	2018	20.53	5.51	12.76	12.28	1.49	1.35
Canada	CAN	2019	20.39	20.14	2.75	2.03	1.32	1.00
Canada	CAN	2020	22.21	5.90	0.40	0.17	1.56	0.90
Central Afri.	CAF	2001	26.64	26.64	8.67	8.67	24.98	24.98
Central Afri.	CAF	2002	27.89	27.89	21.53	21.53	28.82	28.82
Central Afri.	CAF	2003	27.14	27.14	27.19	27.19	27.17	27.17
Central Afri.	CAF	2004	25.89	25.89	25.49	25.49	25.68	25.68
Central Afri.	CAF	2005	25.74	25.74	25.05	25.05	25.32	25.32
Central Afri.	CAF	2006	27.92	27.92	27.88	27.88	27.94	27.94
Central Afri.	CAF	2007	22.26	18.98	10.16	10.06	11.80	14.06
Central Afri.	CAF	2008	18.96	17.56	19.15	18.50	19.75	19.47
Central Afri.	CAF	2009	27.44	14.60	–	2.43	25.29	17.35
Central Afri.	CAF	2010	14.32	12.18	–	–	12.08	10.59
Central Afri.	CAF	2011	23.34	23.04	–	–	23.36	23.16
Central Afri.	CAF	2012	20.75	20.73	2.48	2.47	17.43	17.41
Central Afri.	CAF	2013	21.40	15.78	–	9.36	21.15	16.98
Central Afri.	CAF	2015	20.02	20.01	10.00	10.00	14.74	14.74
Central Afri.	CAF	2016	19.01	18.63	–	–	19.22	18.50
Central Afri.	CAF	2017	20.21	19.11	7.45	3.98	19.33	19.24
Chile	CHL	2000	9.00	9.00	9.00	9.00	9.00	9.00
Chile	CHL	2001	8.00	8.00	8.00	8.00	8.00	8.00
Chile	CHL	2002	6.98	6.98	8.12	8.12	6.41	6.41
Chile	CHL	2003	5.99	5.99	5.99	5.99	5.96	5.96
Chile	CHL	2004	6.07	3.76	6.00	0.70	6.00	3.62
Chile	CHL	2005	6.08	4.70	6.00	5.95	6.00	5.13
Chile	CHL	2006	5.99	3.92	6.00	4.32	6.00	4.45
Chile	CHL	2007	5.99	4.36	5.84	2.92	5.99	5.28
Chile	CHL	2008	6.03	3.88	6.00	3.00	6.00	3.62
Chile	CHL	2009	6.01	4.46	6.00	5.46	6.00	4.37
Chile	CHL	2010	6.03	3.51	6.00	2.56	6.00	3.42
Chile	CHL	2011	6.00	4.70	5.76	1.85	6.00	2.64
Chile	CHL	2012	6.00	3.50	5.99	2.72	6.00	3.66
Chile	CHL	2013	6.00	6.00	6.00	6.00	6.00	6.00
Chile	CHL	2015	6.00	2.48	6.00	5.72	6.00	4.69
Chile	CHL	2016	6.00	6.00	6.00	6.00	6.00	6.00
Chile	CHL	2017	6.00	3.50	6.00	0.17	6.00	4.15

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Chile	CHL	2018	6.00	2.13	6.00	5.71	6.00	5.73
Chile	CHL	2019	6.00	2.58	5.99	1.84	6.00	4.62
Chile	CHL	2020	6.00	2.67	5.88	0.14	6.00	2.89
China	CHN	2000	27.06	23.54	16.44	16.45	18.95	18.94
China	CHN	2001	23.73	21.63	15.60	15.59	17.52	17.52
China	CHN	2002	19.25	19.20	7.65	7.65	12.82	12.81
China	CHN	2003	16.73	16.71	4.92	4.88	10.69	10.69
China	CHN	2004	15.13	15.12	4.44	4.44	9.45	9.44
China	CHN	2005	14.36	13.92	4.13	4.03	9.74	9.61
China	CHN	2006	14.43	12.48	3.20	2.50	8.05	6.53
China	CHN	2007	14.66	12.96	4.33	2.95	9.91	8.57
China	CHN	2008	13.28	10.90	2.96	1.90	7.75	6.48
China	CHN	2009	13.80	10.41	1.08	2.72	11.56	5.81
China	CHN	2010	13.73	11.10	3.43	2.21	7.85	5.06
China	CHN	2011	13.68	10.26	4.59	2.41	7.86	5.61
China	CHN	2014	13.30	9.06	0.86	0.61	4.17	2.98
China	CHN	2015	13.38	10.10	0.16	0.10	3.83	2.24
China	CHN	2016	13.74	9.60	1.12	0.66	5.52	3.26
China	CHN	2017	13.02	10.05	3.06	1.62	6.54	4.16
China	CHN	2018	12.97	8.27	2.00	1.04	5.20	3.49
China	CHN	2019	7.81	6.02	1.79	1.11	3.32	2.72
China	CHN	2020	7.50	5.29	1.33	0.59	3.29	2.47
Hong Kong	HKG	2000	0.00	0.00	51.89	51.89	0.04	0.04
Hong Kong	HKG	2001	0.00	0.00	61.55	61.55	0.04	0.04
Hong Kong	HKG	2002	0.00	0.00	49.76	49.76	0.03	0.03
Hong Kong	HKG	2003	0.00	0.00	29.20	29.20	0.02	0.02
Hong Kong	HKG	2004	0.00	0.00	41.95	41.95	0.02	0.02
Hong Kong	HKG	2005	0.00	0.00	52.64	52.64	0.02	0.02
Hong Kong	HKG	2006	0.00	0.00	39.30	39.30	0.02	0.02
Hong Kong	HKG	2007	0.00	0.00	27.35	27.35	0.01	0.01
Hong Kong	HKG	2008	0.00	0.00	32.15	32.15	0.01	0.01
Hong Kong	HKG	2009	0.00	0.00	33.31	33.31	0.01	0.01
Hong Kong	HKG	2010	0.00	0.00	31.80	31.80	0.00	0.00
Hong Kong	HKG	2011	0.00	0.00	26.70	26.70	0.00	0.00
Hong Kong	HKG	2012	0.00	0.00	31.87	31.87	0.00	0.00
Hong Kong	HKG	2013	0.00	0.00	6.19	6.19	0.00	0.00
Hong Kong	HKG	2014	0.00	0.00	24.85	24.85	0.00	0.00
Hong Kong	HKG	2015	0.00	0.00	21.81	21.81	0.00	0.00
Hong Kong	HKG	2016	0.00	0.00	30.96	30.96	0.00	0.00
Hong Kong	HKG	2017	0.00	0.00	27.18	27.18	0.00	0.00
Hong Kong	HKG	2018	0.00	0.00	33.22	33.22	0.00	0.00
Hong Kong	HKG	2019	0.00	0.00	30.04	30.04	0.00	0.00
Hong Kong	HKG	2020	0.00	0.00	25.78	25.78	0.00	0.00
Macao	MAC	2000	0.00	0.00	–	–	0.26	0.26
Macao	MAC	2001	0.00	0.00	38.77	38.77	0.27	0.27
Macao	MAC	2002	0.00	0.00	29.32	29.32	0.29	0.29
Macao	MAC	2003	0.00	0.00	40.95	40.95	0.24	0.24
Macao	MAC	2004	0.00	0.00	–	–	0.26	0.26
Macao	MAC	2005	0.00	0.00	–	–	0.31	0.31
Macao	MAC	2006	0.00	0.00	45.34	45.34	0.21	0.21
Macao	MAC	2007	0.00	0.00	39.42	39.42	0.19	0.19
Macao	MAC	2008	0.00	0.00	51.19	51.19	0.21	0.21
Macao	MAC	2009	0.00	0.00	0.47	0.47	–	–
Macao	MAC	2010	0.00	0.00	10.75	10.75	0.03	0.03
Macao	MAC	2011	0.00	0.00	–	–	0.21	0.21

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Macao	MAC	2012	0.00	0.00	–	–	0.10	0.10
Macao	MAC	2013	0.00	0.00	105.79	105.79	0.15	0.15
Macao	MAC	2014	0.00	0.00	47.93	47.93	0.14	0.14
Macao	MAC	2015	0.00	0.00	–	–	0.00	0.00
Macao	MAC	2016	0.00	0.00	35.41	35.41	0.00	0.00
Macao	MAC	2017	0.00	0.00	527.45	527.45	0.01	0.01
Macao	MAC	2018	0.00	0.00	–	–	0.02	0.02
Macao	MAC	2019	0.00	0.00	8.77	8.77	0.00	0.00
Macao	MAC	2020	0.00	0.00	30.04	30.04	0.02	0.02
Colombia	COL	2000	13.03	12.94	18.05	19.81	12.69	12.65
Colombia	COL	2001	13.95	13.95	15.02	15.04	12.26	12.26
Colombia	COL	2002	13.83	13.83	9.20	9.22	12.63	12.60
Colombia	COL	2003	13.76	13.76	18.63	18.62	15.08	15.08
Colombia	COL	2004	14.46	11.26	4.33	5.67	14.49	11.56
Colombia	COL	2005	14.26	11.17	10.86	8.44	15.36	12.97
Colombia	COL	2006	14.32	9.42	8.19	6.93	13.12	10.52
Colombia	COL	2007	14.47	10.67	4.43	3.67	16.52	11.82
Colombia	COL	2008	13.60	8.08	9.30	8.87	11.76	8.20
Colombia	COL	2009	12.81	8.50	8.81	4.21	13.33	9.22
Colombia	COL	2010	12.01	10.22	13.63	14.75	12.50	12.28
Colombia	COL	2011	7.77	5.26	–	–	7.52	2.67
Colombia	COL	2012	8.03	6.28	10.07	0.74	8.47	3.23
Colombia	COL	2013	9.50	6.05	12.60	12.11	9.87	8.61
Colombia	COL	2014	7.92	4.87	1.38	112.69	12.30	10.09
Colombia	COL	2015	9.10	9.11	11.42	11.42	9.22	9.22
Colombia	COL	2016	9.44	9.45	3.94	4.32	9.96	9.99
Colombia	COL	2017	9.70	6.39	7.45	4.33	10.56	6.53
Colombia	COL	2018	9.27	6.51	5.84	5.76	10.87	6.94
Colombia	COL	2019	8.61	5.88	7.29	4.25	11.34	7.87
Colombia	COL	2020	9.33	4.53	4.56	–	4.49	1.99
Comoros	COM	2008	9.87	9.75	14.97	15.12	23.28	23.31
Comoros	COM	2010	9.75	3.95	15.02	14.58	17.36	16.71
Comoros	COM	2011	9.95	9.92	13.39	13.06	49.23	49.27
Comoros	COM	2012	9.91	9.80	3.45	2.48	9.89	9.73
Comoros	COM	2013	18.35	18.15	12.32	8.46	20.35	19.97
Comoros	COM	2014	19.76	19.76	20.03	19.87	19.76	19.76
Comoros	COM	2015	17.67	16.20	19.99	19.99	18.90	18.45
Comoros	COM	2017	19.62	18.96	20.12	20.02	19.80	19.08
Comoros	COM	2018	18.40	17.51	20.02	11.67	18.80	16.60
Comoros	COM	2019	19.96	0.61	20.21	10.19	20.34	6.26
Comoros	COM	2020	17.98	4.73	17.89	4.23	18.17	8.56
Costa Rica	CRI	2000	13.17	6.72	42.04	42.02	2.32	1.53
Costa Rica	CRI	2001	12.59	12.59	0.03	0.06	1.63	1.63
Costa Rica	CRI	2002	13.38	13.43	–	–	1.40	1.43
Costa Rica	CRI	2003	13.39	5.99	74.75	74.89	1.42	1.46
Costa Rica	CRI	2004	13.42	8.42	–	–	2.98	2.74
Costa Rica	CRI	2005	13.36	13.40	–	–	2.24	2.24
Costa Rica	CRI	2006	13.61	7.05	2.23	2.25	2.64	2.05
Costa Rica	CRI	2007	13.57	12.96	12.32	–	2.47	1.81
Costa Rica	CRI	2008	13.52	5.08	0.17	0.21	2.35	1.28
Costa Rica	CRI	2009	12.59	5.66	13.83	8.81	8.26	5.24
Costa Rica	CRI	2010	12.59	6.30	0.03	0.03	1.64	1.38
Costa Rica	CRI	2011	12.54	4.53	8.09	3.77	2.14	1.72
Costa Rica	CRI	2012	12.25	12.20	–	–	3.22	2.06
Costa Rica	CRI	2013	14.60	12.05	410.38	409.75	1.99	0.97

Continued on next page...

Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Costa Rica	CRI	2014	14.28	6.74	1.08	0.77	2.14	0.92
Costa Rica	CRI	2016	13.80	12.06	1.11	37.69	2.11	0.43
Costa Rica	CRI	2018	12.15	4.74	25.39	24.37	1.15	1.15
Costa Rica	CRI	2019	11.92	11.21	3.87	0.48	0.80	0.39
Costa Rica	CRI	2020	12.06	3.71	0.43	0.29	1.27	0.95
Croatia	HRV	2001	19.75	17.48	5.45	4.29	16.18	11.54
Croatia	HRV	2002	29.37	29.41	7.25	7.25	11.19	11.19
Croatia	HRV	2003	27.03	21.36	71.66	71.59	9.12	6.77
Croatia	HRV	2004	24.04	23.65	4.81	0.85	9.12	6.49
Croatia	HRV	2005	22.59	22.02	6.68	2.95	8.12	5.96
Croatia	HRV	2006	20.52	19.91	206.84	196.49	5.99	4.52
Croatia	HRV	2007	18.77	9.40	103.44	103.30	6.07	3.80
Croatia	HRV	2008	17.32	9.23	243.79	243.26	5.52	3.59
Croatia	HRV	2009	18.66	8.68	201.85	198.74	6.81	4.35
Croatia	HRV	2010	17.64	7.89	3.41	0.69	6.57	3.89
Croatia	HRV	2011	16.68	8.30	139.33	137.71	6.10	3.15
Croatia	HRV	2012	16.63	8.28	24.14	–	7.78	3.08
Croatia	HRV	2013	15.26	14.03	3.11	0.04	4.94	2.98
Croatia	HRV	2014	14.97	11.38	1.36	0.71	4.33	3.05
Croatia	HRV	2015	16.20	11.07	2.12	1.30	3.60	2.58
Croatia	HRV	2016	16.70	11.25	82.67	0.02	3.20	1.64
Croatia	HRV	2017	21.31	20.31	0.58	0.19	5.23	3.33
Croatia	HRV	2018	21.03	9.16	0.92	0.74	3.77	3.55
Croatia	HRV	2019	20.32	18.20	11.75	–	4.92	2.91
Croatia	HRV	2020	17.04	16.66	1.30	1.63	3.91	3.00
Cyprus	CYP	2000	32.29	31.49	–	–	29.27	29.18
Cyprus	CYP	2001	31.64	22.13	2.47	0.60	18.08	3.66
Cyprus	CYP	2002	32.86	32.11	0.66	2.46	14.17	14.12
Cyprus	CYP	2004	16.20	14.35	4.55	4.41	3.92	2.85
Cyprus	CYP	2005	17.39	16.02	2.78	1.57	7.31	3.48
Cyprus	CYP	2006	13.41	12.51	7.00	6.36	0.77	0.41
Cyprus	CYP	2007	13.59	12.28	1.16	1.04	6.49	3.30
Cyprus	CYP	2008	12.21	9.82	2.54	1.84	24.64	24.64
Cyprus	CYP	2009	16.52	15.02	3.54	0.48	6.87	3.79
Cyprus	CYP	2010	13.91	12.98	1.57	1.55	5.99	3.12
Cyprus	CYP	2011	15.33	13.79	5.72	5.48	7.34	6.98
Cyprus	CYP	2012	10.07	8.47	5.26	1.48	4.98	2.31
Cyprus	CYP	2013	11.03	9.83	3.31	3.00	5.72	2.89
Cyprus	CYP	2014	13.35	8.04	5.69	5.34	6.07	1.71
Cyprus	CYP	2015	12.12	6.84	2.12	1.71	3.34	2.87
Cyprus	CYP	2016	14.63	8.49	21.81	21.83	2.42	2.11
Cyprus	CYP	2017	14.43	6.00	0.11	0.14	4.35	1.69
Cyprus	CYP	2018	14.11	6.67	16.01	16.03	4.97	4.37
Cyprus	CYP	2019	10.54	5.69	4.28	5.64	6.81	1.43
Cyprus	CYP	2020	7.67	3.25	0.21	0.22	1.49	0.34
Czechia	CZE	2000	12.65	12.45	1.80	1.80	4.48	4.47
Czechia	CZE	2001	11.62	11.64	7.43	7.43	2.82	2.82
Czechia	CZE	2002	13.02	13.23	0.12	0.24	3.84	3.85
Czechia	CZE	2003	11.20	11.37	8.94	8.66	3.22	3.24
Czechia	CZE	2004	9.80	8.71	–	–	3.15	2.08
Czechia	CZE	2005	11.29	10.05	0.94	1.85	2.53	2.19
Czechia	CZE	2006	10.01	8.22	112.59	113.79	3.33	2.42
Czechia	CZE	2007	9.12	8.19	4.05	4.14	3.25	2.41
Czechia	CZE	2008	10.30	8.74	5.92	4.53	3.93	2.70
Czechia	CZE	2009	10.76	7.99	7.34	4.15	3.13	2.05

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Czechia	CZE	2010	10.44	8.92	0.04	0.79	3.13	2.43
Czechia	CZE	2011	10.69	8.79	2.35	0.79	2.92	2.24
Czechia	CZE	2012	11.15	8.88	0.37	0.40	2.74	1.57
Czechia	CZE	2013	10.89	8.32	88.87	88.99	2.51	1.65
Czechia	CZE	2014	11.00	9.16	0.24	0.22	2.52	1.99
Czechia	CZE	2015	11.25	7.66	81.12	82.58	2.78	1.73
Czechia	CZE	2016	12.20	10.15	27.66	27.67	2.57	1.75
Czechia	CZE	2017	11.53	8.47	0.61	0.29	2.45	1.55
Czechia	CZE	2018	10.83	8.32	0.13	0.04	2.29	1.35
Czechia	CZE	2019	10.11	8.07	0.02	74.83	2.10	1.36
Czechia	CZE	2020	7.56	6.23	70.01	70.03	1.74	1.30
Denmark	DNK	2000	16.97	14.74	2.33	1.88	3.54	2.27
Denmark	DNK	2001	17.03	16.59	95.83	95.92	3.53	3.10
Denmark	DNK	2002	17.67	16.83	204.41	205.48	3.14	2.82
Denmark	DNK	2003	17.81	16.47	103.95	104.89	3.36	2.25
Denmark	DNK	2004	17.48	16.78	69.24	68.50	3.13	2.90
Denmark	DNK	2005	17.62	15.93	0.00	112.42	2.80	2.07
Denmark	DNK	2006	18.14	16.58	130.69	130.68	2.09	1.67
Denmark	DNK	2007	16.81	14.90	0.23	0.35	1.99	1.65
Denmark	DNK	2008	14.88	12.48	0.26	0.23	2.29	1.78
Denmark	DNK	2009	17.10	14.54	95.47	95.41	1.89	1.45
Denmark	DNK	2010	17.47	15.55	0.30	0.25	2.05	1.60
Denmark	DNK	2011	17.69	15.99	0.21	0.11	2.03	1.62
Denmark	DNK	2012	16.86	14.69	123.88	0.02	1.76	1.27
Denmark	DNK	2013	17.44	15.72	0.07	0.03	1.73	1.45
Denmark	DNK	2014	18.04	14.29	129.89	129.96	1.88	0.92
Denmark	DNK	2015	17.04	13.36	0.04	0.03	1.65	0.87
Denmark	DNK	2016	19.39	17.45	0.08	0.04	2.11	1.81
Denmark	DNK	2017	21.77	20.16	0.19	94.64	3.26	2.63
Denmark	DNK	2018	20.24	19.39	0.14	0.09	2.17	1.94
Denmark	DNK	2019	20.71	18.33	92.11	91.91	3.11	2.16
Denmark	DNK	2020	9.27	4.85	84.11	84.22	2.20	1.10
Dominica	DMA	2000	27.25	27.25	35.22	35.22	27.60	27.60
Dominica	DMA	2001	45.91	45.91	93.42	93.42	26.84	26.84
Dominica	DMA	2002	46.63	46.63	–	–	33.98	33.98
Dominica	DMA	2003	46.57	46.57	–	–	30.38	30.38
Dominica	DMA	2006	46.45	39.10	–	5.57	10.51	9.30
Dominica	DMA	2007	44.27	18.41	–	–	40.97	17.26
Dominica	DMA	2010	19.31	19.31	5.00	5.00	5.49	5.49
Dominica	DMA	2011	51.56	10.31	–	0.79	59.79	6.58
Dominica	DMA	2012	31.99	10.31	1.47	0.76	34.34	5.55
Dominica	DMA	2013	53.60	28.79	19.72	19.91	38.18	24.66
Dominica	DMA	2015	54.44	54.48	17.02	17.02	61.27	61.27
Dominica	DMA	2016	29.65	19.58	10.00	10.00	10.94	10.61
Dominica	DMA	2017	25.16	15.09	–	9.61	28.85	14.64
Dominica	DMA	2020	39.62	39.13	–	–	42.92	42.60
Egypt	EGY	2000	24.56	24.53	7.00	7.00	28.76	28.82
Egypt	EGY	2001	26.15	26.15	25.44	25.44	27.40	27.40
Egypt	EGY	2002	26.50	26.50	32.35	32.35	26.63	26.62
Egypt	EGY	2003	24.93	24.93	5.05	5.05	9.58	9.58
Egypt	EGY	2004	18.67	18.68	7.58	7.58	17.42	17.43
Egypt	EGY	2005	22.94	20.75	8.53	2.50	18.13	15.43
Egypt	EGY	2006	18.73	18.74	3.03	3.03	35.80	35.80
Egypt	EGY	2007	18.56	18.55	15.98	15.46	16.07	16.08
Egypt	EGY	2008	18.97	10.34	7.56	74.89	9.05	5.86

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Egypt	EGY	2009	18.50	13.76	4.26	4.30	13.51	8.53
Egypt	EGY	2010	19.99	17.05	30.15	26.56	20.89	18.48
Egypt	EGY	2011	17.27	10.97	14.21	10.51	17.23	6.82
Egypt	EGY	2012	17.01	12.94	4.21	1.12	11.05	8.19
Egypt	EGY	2013	16.79	12.13	–	–	29.11	27.04
Egypt	EGY	2014	18.09	10.63	3.34	1.45	11.88	6.69
Egypt	EGY	2015	16.08	11.00	4.85	1.04	11.96	8.87
Egypt	EGY	2016	20.48	10.45	4.81	2.72	8.05	4.98
Egypt	EGY	2017	32.41	17.18	4.92	4.35	9.00	5.60
Egypt	EGY	2018	28.75	12.70	4.03	1.43	4.07	1.71
Egypt	EGY	2019	29.95	17.14	6.02	0.89	11.67	6.69
El Salvador	SLV	2000	18.20	18.20	–	–	21.79	21.79
El Salvador	SLV	2001	20.50	20.51	–	–	22.87	22.87
El Salvador	SLV	2002	21.08	14.30	–	–	23.34	18.99
El Salvador	SLV	2003	20.84	16.98	61.09	58.24	23.37	20.33
El Salvador	SLV	2004	20.06	15.52	–	–	22.96	20.91
El Salvador	SLV	2005	14.50	14.48	10.63	10.63	14.59	14.59
El Salvador	SLV	2006	20.64	14.58	280.99	219.16	22.71	10.10
El Salvador	SLV	2007	15.89	13.58	1306.03	0.44	8.93	6.42
El Salvador	SLV	2008	16.51	11.70	–	23.29	13.92	7.89
El Salvador	SLV	2009	14.90	9.13	3.09	73.29	13.91	9.73
El Salvador	SLV	2010	13.03	10.99	1.32	72.60	13.56	9.96
El Salvador	SLV	2011	13.49	10.21	–	–	14.21	11.33
El Salvador	SLV	2012	13.79	10.66	145.37	114.66	12.33	9.90
El Salvador	SLV	2013	13.29	8.68	21.22	6.35	13.87	8.26
El Salvador	SLV	2014	13.21	7.61	–	–	14.16	7.12
El Salvador	SLV	2015	13.29	8.41	29.58	7.85	14.22	6.57
El Salvador	SLV	2017	11.20	6.65	–	–	13.88	6.12
El Salvador	SLV	2018	10.84	6.20	50.82	46.09	9.96	4.59
El Salvador	SLV	2019	11.60	9.01	–	8.10	13.72	9.99
El Salvador	SLV	2020	11.96	7.83	352.27	343.16	12.68	6.69
Estonia	EST	2000	5.92	5.92	0.04	0.04	0.41	0.41
Estonia	EST	2001	7.06	7.06	–	–	0.51	0.51
Estonia	EST	2002	6.82	6.82	0.20	0.20	0.43	0.43
Estonia	EST	2003	6.82	6.82	0.01	0.01	0.54	0.54
Estonia	EST	2004	11.30	10.65	5.16	4.11	3.76	3.09
Estonia	EST	2005	9.69	8.66	1.23	0.57	2.38	1.41
Estonia	EST	2006	9.65	8.78	3.57	1888.99	2.63	1.17
Estonia	EST	2007	11.15	10.37	480.82	0.21	4.53	3.05
Estonia	EST	2008	10.52	9.43	3.55	1.36	4.37	2.99
Estonia	EST	2009	9.25	7.38	13.22	–	4.16	3.24
Estonia	EST	2010	9.68	7.26	2.81	1.62	3.83	2.78
Estonia	EST	2011	10.33	9.10	–	–	3.31	2.27
Estonia	EST	2012	8.77	6.83	3.22	0.01	3.05	1.67
Estonia	EST	2013	10.80	9.00	4.69	–	3.25	2.64
Estonia	EST	2014	10.04	7.45	0.90	3.53	1.04	3.41
Estonia	EST	2015	11.09	10.51	0.88	0.01	0.95	0.28
Estonia	EST	2016	11.44	10.90	1.22	0.69	2.15	1.68
Estonia	EST	2017	12.37	9.81	0.06	0.04	1.83	0.99
Estonia	EST	2018	11.68	10.64	78.91	78.90	0.77	0.57
Estonia	EST	2019	12.23	10.94	77.67	77.66	0.68	0.44
Estonia	EST	2020	5.20	3.59	0.00	0.00	0.63	0.46
Eswatini	SWZ	2001	24.19	8.85	98.92	0.70	9.90	6.52
Eswatini	SWZ	2002	16.69	16.69	0.27	0.27	4.87	4.87
Eswatini	SWZ	2004	22.43	21.32	73.65	73.66	6.15	5.84

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Eswatini	SWZ	2005	19.69	18.75	75.89	75.89	1.70	1.65
Eswatini	SWZ	2006	19.35	18.69	70.73	70.75	5.88	5.76
Eswatini	SWZ	2007	14.34	14.33	0.43	0.43	3.16	3.16
Eswatini	SWZ	2008	21.98	19.17	76.31	76.34	5.23	4.34
Eswatini	SWZ	2009	19.39	15.54	75.70	75.75	6.34	5.54
Eswatini	SWZ	2010	20.89	19.43	45.06	45.04	4.17	4.00
Eswatini	SWZ	2011	21.54	17.58	65.82	0.01	4.25	4.11
Eswatini	SWZ	2012	16.78	9.64	80.11	80.04	3.43	2.94
Eswatini	SWZ	2013	19.98	10.62	0.12	137.48	4.10	1.73
Eswatini	SWZ	2014	18.87	12.77	2.40	2.25	5.47	4.10
Eswatini	SWZ	2015	16.73	6.48	90.08	0.02	2.54	1.31
Eswatini	SWZ	2016	17.53	7.58	0.12	0.01	3.01	0.96
Eswatini	SWZ	2017	17.09	9.53	0.10	0.01	3.27	1.92
Eswatini	SWZ	2018	17.60	8.64	0.20	76.61	4.20	1.62
Eswatini	SWZ	2019	20.31	13.01	0.36	0.09	4.88	2.69
Eswatini	SWZ	2020	18.78	12.45	76.43	76.54	5.45	3.88
Ethiopia	ETH	2001	16.78	16.79	8.44	8.44	12.54	12.54
Ethiopia	ETH	2002	15.60	15.60	–	–	18.39	18.39
Ethiopia	ETH	2006	19.63	19.63	2.35	2.35	13.60	13.60
Ethiopia	ETH	2008	22.26	21.85	–	10.20	17.13	17.00
Ethiopia	ETH	2009	21.00	20.31	14.86	14.67	19.84	19.85
Ethiopia	ETH	2010	23.49	22.85	0.70	1.74	11.83	11.01
Ethiopia	ETH	2011	22.08	21.63	2.64	1.92	10.93	10.57
Ethiopia	ETH	2012	20.96	20.61	5.01	6.13	17.21	16.56
Ethiopia	ETH	2015	24.75	24.38	–	–	12.63	12.59
Ethiopia	ETH	2018	26.48	26.06	25.75	25.48	20.40	20.07
Finland	FIN	2000	6.67	5.62	6.69	4.71	4.24	3.34
Finland	FIN	2001	6.57	6.24	1.35	1.18	2.37	2.24
Finland	FIN	2002	6.41	4.83	1.62	1.13	1.32	0.84
Finland	FIN	2003	7.03	6.60	40.59	40.46	1.33	0.90
Finland	FIN	2004	7.53	6.21	0.78	0.58	1.77	0.92
Finland	FIN	2005	7.39	5.78	6.67	4.62	1.49	0.82
Finland	FIN	2006	7.24	5.77	24.39	24.04	1.36	1.09
Finland	FIN	2007	6.88	5.03	23.81	24.01	2.04	1.80
Finland	FIN	2008	7.30	5.97	44.44	0.15	1.62	0.91
Finland	FIN	2009	5.96	4.50	59.40	0.17	1.43	1.40
Finland	FIN	2010	8.10	6.52	97.06	96.72	1.99	1.57
Finland	FIN	2011	6.23	5.04	1.47	0.78	1.57	1.04
Finland	FIN	2012	7.38	5.69	47.21	46.62	1.67	1.17
Finland	FIN	2013	6.48	5.35	0.29	0.26	1.99	1.01
Finland	FIN	2014	10.70	9.97	110.66	110.69	2.16	1.71
Finland	FIN	2015	9.62	8.00	0.70	0.54	1.79	1.41
Finland	FIN	2016	7.24	5.68	4.90	4.89	1.12	0.80
Finland	FIN	2017	6.41	4.23	72.33	72.56	1.80	1.03
Finland	FIN	2018	6.37	4.79	92.02	0.15	1.83	1.20
Finland	FIN	2019	11.34	9.88	0.63	0.50	1.80	1.63
Finland	FIN	2020	3.89	2.55	1.47	1.43	2.36	2.24
France	FRA	2000	18.39	14.72	1.55	0.49	2.55	1.59
France	FRA	2001	24.88	22.94	1.61	1.25	2.37	1.95
France	FRA	2002	14.97	13.27	1.54	0.72	2.66	1.93
France	FRA	2003	14.93	12.86	1.75	1.02	2.76	1.99
France	FRA	2004	18.09	16.66	2.37	0.67	3.20	1.88
France	FRA	2005	16.17	13.31	1.82	1.34	3.14	2.43
France	FRA	2006	15.98	12.11	1.41	0.69	2.75	2.10
France	FRA	2007	14.38	11.64	1.60	0.51	2.98	1.66

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
France	FRA	2008	14.32	11.27	1.74	1.04	2.99	2.07
France	FRA	2009	15.02	10.45	1.23	0.51	2.52	1.57
France	FRA	2010	15.04	9.46	1.50	1.27	2.76	2.45
France	FRA	2011	15.92	12.89	1.30	0.31	2.51	1.49
France	FRA	2012	15.10	12.48	0.76	0.46	1.85	1.33
France	FRA	2013	15.21	11.09	0.98	0.43	2.20	1.49
France	FRA	2014	15.18	13.12	1.40	0.76	2.68	1.76
France	FRA	2015	14.22	12.78	0.98	0.81	2.10	1.73
France	FRA	2016	15.96	12.06	0.83	0.64	1.95	1.50
France	FRA	2017	15.28	12.43	1.05	0.96	2.39	1.80
France	FRA	2018	14.66	12.73	3.33	0.18	2.34	1.87
France	FRA	2019	14.22	10.56	0.71	0.28	1.77	1.15
France	FRA	2020	5.49	3.19	0.59	0.43	1.46	1.00
Polynesia	PYF	2008	6.06	3.65	1.99	1.31	7.05	6.79
Polynesia	PYF	2009	8.19	5.75	3.60	1.28	8.62	6.37
Polynesia	PYF	2011	7.23	4.90	0.68	0.54	4.83	2.42
Polynesia	PYF	2012	7.94	5.15	2.08	1.75	6.46	5.78
Polynesia	PYF	2013	8.21	4.89	0.24	0.23	2.66	2.60
Polynesia	PYF	2014	6.57	4.14	1.24	1.10	7.01	6.94
Polynesia	PYF	2015	7.02	5.65	8.98	9.42	10.10	8.51
Polynesia	PYF	2016	7.14	4.86	1.81	–	6.20	4.19
Polynesia	PYF	2017	6.13	4.41	9.21	4.15	7.94	6.14
Polynesia	PYF	2018	7.89	5.12	6.00	4.01	5.99	4.38
Polynesia	PYF	2019	8.84	6.12	6.40	3.57	10.56	5.17
Polynesia	PYF	2020	8.25	6.65	8.02	4.49	9.92	6.40
Gabon	GAB	2000	19.37	19.37	11.90	11.90	12.98	12.98
Gabon	GAB	2001	22.25	22.24	10.65	10.66	11.03	11.02
Gabon	GAB	2002	15.45	15.45	10.80	10.80	10.79	10.79
Gabon	GAB	2003	12.72	12.73	2.70	2.70	8.78	8.78
Gabon	GAB	2004	18.04	18.04	4.94	4.94	25.58	25.58
Gabon	GAB	2005	18.40	18.41	10.01	10.01	11.40	11.40
Gabon	GAB	2007	16.09	16.09	9.58	9.58	10.01	10.01
Gabon	GAB	2008	14.28	14.28	1.83	1.76	11.08	11.10
Gabon	GAB	2009	16.55	16.46	9.80	9.80	10.22	10.22
Gabon	GAB	2010	12.95	12.95	14.17	14.17	10.29	10.29
Gabon	GAB	2011	11.71	11.71	12.45	12.45	13.32	13.32
Gabon	GAB	2012	11.56	11.55	29.50	29.49	24.48	24.49
Gabon	GAB	2013	11.10	10.96	10.97	10.98	11.86	11.76
Gabon	GAB	2015	11.80	11.36	–	7.62	26.53	26.47
Gabon	GAB	2016	11.61	12.35	10.13	10.13	14.34	14.45
Gabon	GAB	2019	10.67	8.02	10.19	9.48	11.76	10.43
Gambia	GMB	2003	12.40	12.40	18.62	18.62	1.05	1.05
Gambia	GMB	2007	17.88	17.88	18.53	18.53	5.63	5.63
Gambia	GMB	2008	14.72	14.72	0.15	0.15	5.42	5.42
Gambia	GMB	2009	16.04	16.04	20.01	20.01	20.04	20.04
Gambia	GMB	2010	11.58	11.58	–	–	11.46	11.46
Gambia	GMB	2011	11.68	11.68	12.77	12.77	12.81	12.81
Gambia	GMB	2012	12.44	12.44	6.76	6.76	13.97	13.97
Gambia	GMB	2013	11.49	11.52	12.08	12.12	12.01	12.06
Gambia	GMB	2018	19.36	18.33	–	–	–	–
Gambia	GMB	2020	18.59	11.71	–	–	–	3.08
Georgia	GEO	2001	11.76	11.76	9.65	9.65	8.31	8.31
Georgia	GEO	2002	11.40	11.40	5.32	5.32	10.23	10.23
Georgia	GEO	2003	11.81	11.81	0.67	0.67	6.02	6.02
Georgia	GEO	2004	11.53	11.52	0.90	0.90	4.62	4.62

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Georgia	GEO	2005	11.63	11.63	4.37	4.37	5.09	5.09
Georgia	GEO	2006	12.15	6.26	0.47	0.39	3.82	2.48
Georgia	GEO	2007	8.42	3.15	0.02	0.01	–	–
Georgia	GEO	2008	7.42	2.81	0.00	0.00	–	–
Georgia	GEO	2009	7.13	0.99	19.58	19.58	–	–
Georgia	GEO	2010	6.03	5.29	266.49	0.00	0.06	–
Georgia	GEO	2011	2.74	1.92	0.00	0.00	0.08	0.08
Georgia	GEO	2012	3.10	2.48	392.35	392.35	0.04	–
Georgia	GEO	2013	3.37	1.23	0.04	0.00	0.10	0.03
Georgia	GEO	2015	3.40	2.16	93.26	93.58	1.61	1.32
Georgia	GEO	2016	4.48	3.29	116.34	0.00	1.08	0.92
Georgia	GEO	2020	5.26	2.23	0.01	0.01	1.55	0.42
Germany	DEU	2000	13.29	11.12	1.62	0.94	3.44	2.56
Germany	DEU	2001	14.22	12.72	1.44	1.04	3.16	2.67
Germany	DEU	2002	11.98	10.22	1.66	0.87	3.57	2.07
Germany	DEU	2003	12.06	10.59	–	2.14	3.57	2.77
Germany	DEU	2004	12.13	11.05	1.28	1.03	2.83	2.39
Germany	DEU	2005	11.94	10.49	1.06	0.52	3.03	2.07
Germany	DEU	2006	11.58	9.28	1.00	0.62	2.77	1.84
Germany	DEU	2007	10.95	7.52	0.73	0.29	2.52	1.32
Germany	DEU	2008	11.37	8.38	0.57	0.25	2.26	1.26
Germany	DEU	2009	11.68	9.03	0.51	0.27	2.07	1.45
Germany	DEU	2010	11.61	9.44	0.44	0.25	1.84	1.44
Germany	DEU	2011	12.08	9.37	0.91	0.46	2.53	1.63
Germany	DEU	2012	11.66	7.92	0.51	0.28	1.92	1.32
Germany	DEU	2013	11.79	8.65	0.94	0.42	2.52	1.41
Germany	DEU	2014	11.60	8.08	0.77	0.44	2.36	1.74
Germany	DEU	2015	11.57	7.53	0.92	0.49	2.53	1.59
Germany	DEU	2016	11.65	9.27	0.67	0.42	2.27	1.78
Germany	DEU	2017	11.54	9.57	1.15	0.75	2.75	2.05
Germany	DEU	2018	10.96	8.28	1.24	0.69	2.76	1.89
Germany	DEU	2019	11.57	6.80	1.12	0.73	2.68	1.86
Germany	DEU	2020	7.58	4.06	1.16	0.78	2.46	1.83
Ghana	GHA	2000	56.92	56.92	14.31	14.31	23.85	23.85
Ghana	GHA	2001	77.73	16.83	3.35	1.14	31.14	16.60
Ghana	GHA	2002	69.06	17.94	9.97	9.75	17.56	10.53
Ghana	GHA	2003	63.63	17.45	4.62	3.33	27.17	12.88
Ghana	GHA	2007	15.09	14.92	2.27	2.26	6.34	6.33
Ghana	GHA	2008	14.51	14.49	0.62	0.62	14.45	14.41
Ghana	GHA	2009	16.89	16.86	5.78	5.76	14.02	14.01
Ghana	GHA	2010	15.87	15.36	7.81	130.55	8.27	4.81
Ghana	GHA	2013	14.96	14.41	6.14	6.13	9.84	9.83
Ghana	GHA	2016	15.75	14.02	0.26	0.24	4.05	3.98
Ghana	GHA	2017	17.77	15.91	8.00	5.01	13.53	9.90
Ghana	GHA	2018	18.38	11.53	11.95	9.39	10.63	9.85
Ghana	GHA	2019	18.76	13.08	20.71	4.93	12.67	7.21
Ghana	GHA	2020	18.57	15.98	6.15	5.01	11.88	11.45
Greece	GRC	2000	17.49	15.24	889.89	796.18	6.07	4.33
Greece	GRC	2001	19.85	18.10	0.07	0.00	6.70	5.65
Greece	GRC	2002	16.00	12.54	0.72	0.12	4.68	3.63
Greece	GRC	2003	16.50	15.61	1.67	0.59	5.24	4.38
Greece	GRC	2004	14.46	11.61	3.84	1.49	3.90	2.53
Greece	GRC	2005	13.62	9.02	4.71	2.90	4.59	3.58
Greece	GRC	2006	13.60	11.35	–	–	6.78	5.34
Greece	GRC	2007	14.91	11.52	1.83	1.25	5.03	3.99

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Greece	GRC	2008	12.85	11.59	8.17	6.60	2.02	1.52
Greece	GRC	2009	15.02	12.47	11.87	11.72	1.80	1.25
Greece	GRC	2010	14.95	9.52	2.62	–	1.69	3.50
Greece	GRC	2011	17.13	15.21	2.83	1.62	4.68	4.15
Greece	GRC	2012	16.49	12.28	6.12	0.34	3.56	2.59
Greece	GRC	2013	17.54	15.92	3.60	2.34	5.22	4.20
Greece	GRC	2014	15.28	12.48	6.97	3.92	4.69	3.91
Greece	GRC	2015	16.01	11.44	0.72	0.26	3.50	2.45
Greece	GRC	2016	16.83	15.75	6.20	3.02	4.79	4.60
Greece	GRC	2017	18.62	12.23	396.18	0.05	6.28	2.66
Greece	GRC	2018	16.29	11.34	486.50	476.61	4.50	3.29
Greece	GRC	2019	15.59	12.28	1.63	1.70	4.18	3.59
Greece	GRC	2020	11.42	9.16	182.16	0.02	1.46	1.15
Grenada	GRD	2000	9.79	9.79	20.80	20.80	5.74	5.74
Grenada	GRD	2001	22.67	22.67	17.53	17.53	23.23	23.23
Grenada	GRD	2002	15.52	15.52	–	–	9.35	9.35
Grenada	GRD	2003	15.13	15.15	16.77	16.79	16.06	16.07
Grenada	GRD	2006	17.27	9.92	–	5.29	19.08	16.98
Grenada	GRD	2007	16.19	11.92	21.25	20.85	17.23	13.83
Grenada	GRD	2008	15.28	3.80	24.39	24.24	8.86	7.09
Grenada	GRD	2010	15.97	10.69	36.99	42.89	19.59	18.58
Grenada	GRD	2011	14.88	11.30	0.12	15.24	14.83	14.24
Grenada	GRD	2012	14.34	10.08	–	3.06	16.05	13.14
Grenada	GRD	2013	15.74	12.44	–	2.88	8.07	7.11
Grenada	GRD	2015	13.76	13.73	15.69	10.44	12.44	12.26
Grenada	GRD	2016	14.66	14.56	5.68	5.65	15.55	15.46
Grenada	GRD	2019	21.43	21.01	5.01	5.01	7.65	7.53
Guatemala	GTM	2000	13.90	13.91	–	–	20.89	20.89
Guatemala	GTM	2001	14.96	13.74	23.12	23.04	19.86	19.87
Guatemala	GTM	2002	14.91	14.96	17.92	18.01	15.67	15.68
Guatemala	GTM	2003	14.67	14.05	–	–	15.69	15.17
Guatemala	GTM	2004	13.41	10.03	4.44	130.31	13.03	6.80
Guatemala	GTM	2005	13.61	13.61	44.87	44.88	12.17	12.17
Guatemala	GTM	2006	14.64	14.63	40.54	40.55	13.22	13.21
Guatemala	GTM	2007	13.72	11.02	0.92	1.03	10.28	8.63
Guatemala	GTM	2008	15.52	6.96	0.30	75.98	10.87	8.89
Guatemala	GTM	2009	15.45	9.51	26.82	26.51	8.47	5.78
Guatemala	GTM	2010	14.79	12.58	4.48	4.50	5.13	5.04
Guatemala	GTM	2011	14.57	7.25	0.00	35.21	1.95	1.28
Guatemala	GTM	2012	16.37	5.91	2.31	0.17	3.87	2.41
Guatemala	GTM	2013	16.68	6.24	1.94	1.96	5.36	3.77
Guatemala	GTM	2014	16.25	14.89	1.21	1.23	4.00	3.28
Guatemala	GTM	2015	15.33	12.87	1.79	0.69	4.77	2.38
Guatemala	GTM	2020	15.26	10.69	0.38	22.78	4.73	3.25
Guinea	GIN	2004	6.81	6.81	6.40	6.40	6.64	6.64
Guinea	GIN	2005	14.84	15.60	10.24	10.27	17.89	17.98
Guinea	GIN	2008	14.78	12.25	1.33	1.24	11.65	10.95
Guinea	GIN	2009	11.88	11.88	5.02	5.02	9.29	9.29
Guinea	GIN	2010	9.21	9.31	9.36	9.39	12.62	12.65
Guinea	GIN	2012	11.89	11.89	4.62	4.62	8.10	8.10
Guinea	GIN	2017	16.60	16.61	12.47	12.47	14.17	14.18
Guinea	GIN	2018	19.25	18.92	5.06	5.05	16.53	15.90
Guinea	GIN	2019	14.93	14.57	6.31	6.28	5.62	5.25
Guinea	GIN	2020	13.83	10.20	9.19	2.98	11.16	9.98
Guyana	GUY	2000	39.07	39.07	26.17	26.17	17.73	17.73

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Guyana	GUY	2001	38.34	38.34	16.65	16.65	21.52	21.52
Guyana	GUY	2002	37.90	37.90	13.49	13.49	23.01	23.01
Guyana	GUY	2003	38.02	38.02	9.17	9.17	12.31	12.31
Guyana	GUY	2006	39.03	38.99	13.35	13.18	32.28	32.05
Guyana	GUY	2007	38.24	38.24	–	–	32.94	32.94
Guyana	GUY	2008	39.06	39.03	52.20	51.84	26.55	26.49
Guyana	GUY	2009	17.81	17.81	10.48	10.48	22.18	22.18
Guyana	GUY	2010	38.54	38.46	5.44	0.93	11.11	10.37
Guyana	GUY	2011	38.69	38.62	0.49	2.57	19.39	17.99
Guyana	GUY	2012	12.23	8.66	69.41	65.63	10.91	9.28
Guyana	GUY	2013	38.75	37.29	–	–	30.93	29.06
Guyana	GUY	2015	37.64	4.36	6.02	4.75	15.35	7.77
Guyana	GUY	2016	33.54	33.58	51.04	51.08	25.63	25.62
Guyana	GUY	2017	35.70	35.70	–	–	25.34	24.55
Guyana	GUY	2018	35.99	34.96	12.36	12.04	17.24	16.84
Guyana	GUY	2019	31.93	31.80	4.72	5.00	5.95	6.81
Guyana	GUY	2020	32.41	32.15	2.62	5.67	9.66	7.87
Honduras	HND	2000	26.01	26.01	–	–	19.69	19.69
Honduras	HND	2001	28.29	28.26	–	–	18.61	18.61
Honduras	HND	2002	26.06	26.06	16.52	16.52	14.98	14.98
Honduras	HND	2003	27.69	27.70	–	–	14.80	14.80
Honduras	HND	2004	28.07	9.20	–	12.24	15.00	10.85
Honduras	HND	2005	28.94	28.94	17.18	17.18	15.21	15.21
Honduras	HND	2006	19.91	19.91	7.48	7.30	14.89	14.88
Honduras	HND	2007	26.87	26.29	–	–	14.82	13.87
Honduras	HND	2008	25.54	7.70	–	11.11	14.99	10.98
Honduras	HND	2009	26.32	26.32	11.87	11.86	15.02	15.01
Honduras	HND	2010	27.11	27.11	7.80	7.80	14.48	14.48
Honduras	HND	2011	26.28	26.52	–	–	14.96	15.09
Honduras	HND	2012	25.89	25.74	–	–	15.07	14.98
Honduras	HND	2013	29.11	29.02	9.01	8.80	16.52	16.39
Honduras	HND	2015	26.75	12.85	14.30	4.68	14.82	5.24
Honduras	HND	2018	24.66	6.72	–	–	11.02	7.32
Honduras	HND	2020	21.98	7.03	4.89	3.93	9.43	6.08
Hungary	HUN	2002	18.51	18.52	12.86	12.88	13.11	13.11
Hungary	HUN	2004	20.90	20.52	0.28	0.13	1.74	1.02
Hungary	HUN	2005	16.27	15.69	61.73	61.79	1.82	1.37
Hungary	HUN	2006	17.36	16.88	63.89	64.12	2.22	1.42
Hungary	HUN	2007	11.31	10.99	0.10	0.20	2.16	1.62
Hungary	HUN	2008	10.15	9.31	0.26	0.14	1.76	1.47
Hungary	HUN	2009	10.66	9.02	0.13	72.39	1.80	1.17
Hungary	HUN	2010	9.71	8.71	77.90	77.93	1.62	1.15
Hungary	HUN	2011	15.52	13.59	71.26	71.31	1.54	0.93
Hungary	HUN	2012	13.44	13.03	0.39	0.13	2.09	1.43
Hungary	HUN	2013	15.81	14.05	22.28	18.86	2.85	2.34
Hungary	HUN	2014	15.11	14.02	0.33	0.37	2.69	1.58
Hungary	HUN	2015	16.21	14.92	0.25	28.90	3.05	1.57
Hungary	HUN	2016	17.37	16.20	3.99	4.45	2.89	1.82
Hungary	HUN	2017	17.27	16.22	12.15	12.43	3.64	2.84
Hungary	HUN	2018	16.79	15.16	1.75	0.72	3.25	1.83
Hungary	HUN	2019	15.75	11.15	1.08	1.13	3.16	2.04
Hungary	HUN	2020	9.07	7.28	1.13	0.89	2.80	1.81
Iceland	ISL	2000	3.99	3.86	1.80	1.51	2.84	0.83
Iceland	ISL	2001	4.01	3.94	1.07	0.16	3.18	3.16
Iceland	ISL	2003	3.31	3.28	0.32	0.19	2.78	2.60

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Iceland	ISL	2004	3.50	3.43	2.07	1.96	3.55	3.38
Iceland	ISL	2005	2.82	2.66	0.05	0.05	2.96	2.69
Iceland	ISL	2006	3.36	1.96	7.62	0.01	7.68	1.22
Iceland	ISL	2007	2.36	1.92	2.79	0.20	4.87	1.30
Iceland	ISL	2008	3.74	3.17	0.90	0.60	2.61	2.10
Iceland	ISL	2009	2.87	2.58	2.96	0.11	3.97	–
Iceland	ISL	2010	6.25	5.55	0.61	0.59	3.35	2.93
Iceland	ISL	2011	5.16	4.32	3.15	2.53	2.37	1.60
Iceland	ISL	2012	2.06	1.39	0.16	0.06	2.20	0.89
Iceland	ISL	2013	2.10	1.68	0.08	0.05	4.03	3.86
Iceland	ISL	2014	2.26	1.25	1.15	0.26	2.18	0.71
Iceland	ISL	2015	2.34	2.14	2.09	0.28	2.42	1.32
Iceland	ISL	2016	2.55	1.54	0.77	0.49	1.92	0.70
Iceland	ISL	2017	1.74	1.41	62.91	62.84	1.35	0.44
Iceland	ISL	2019	1.97	3.28	106.62	1.30	1.14	9.79
India	IND	2000	42.15	41.76	40.66	40.95	37.73	37.89
India	IND	2001	41.98	41.05	53.07	52.57	31.09	30.30
India	IND	2002	38.32	38.35	30.52	29.04	30.50	30.96
India	IND	2003	35.48	35.52	27.12	27.09	27.44	27.46
India	IND	2004	38.58	38.00	30.70	31.39	30.78	30.59
India	IND	2005	31.07	28.77	16.48	15.99	21.84	17.37
India	IND	2006	31.23	31.36	10.48	10.95	14.85	14.93
India	IND	2007	30.19	29.19	20.17	15.09	17.82	15.07
India	IND	2008	26.84	24.72	7.90	7.86	12.45	12.02
India	IND	2009	26.40	24.73	3.24	0.32	10.57	8.69
India	IND	2010	27.07	24.08	8.03	6.01	9.48	6.01
India	IND	2011	28.16	28.11	–	0.20	11.22	11.83
India	IND	2012	30.21	30.24	2.98	2.13	12.83	12.98
India	IND	2013	31.49	30.38	–	7.48	11.07	11.07
India	IND	2015	28.74	28.36	2664.19	2619.77	10.01	9.84
India	IND	2016	32.96	29.82	588.10	7.54	9.37	6.05
India	IND	2017	30.30	20.98	5.69	2.23	11.11	7.03
India	IND	2018	28.68	16.94	16.58	14.93	10.08	3.71
India	IND	2019	29.58	26.15	1.52	2.08	10.81	8.81
India	IND	2020	29.54	26.78	5.67	2.24	13.31	9.50
Indonesia	IDN	2000	12.99	12.99	5.15	5.15	11.00	10.99
Indonesia	IDN	2001	10.40	8.77	238.26	237.12	7.77	6.69
Indonesia	IDN	2002	9.79	8.21	6.49	4.44	8.92	6.79
Indonesia	IDN	2003	9.54	7.95	129.31	127.07	6.44	4.81
Indonesia	IDN	2004	9.63	8.09	112.95	110.01	7.96	6.27
Indonesia	IDN	2005	9.72	7.31	2.13	1.23	9.11	7.02
Indonesia	IDN	2006	9.35	7.87	–	–	7.92	6.64
Indonesia	IDN	2007	9.40	7.37	0.91	1.44	6.39	4.16
Indonesia	IDN	2008	9.55	9.54	–	–	8.16	8.16
Indonesia	IDN	2009	9.89	8.31	85.12	84.89	6.92	5.33
Indonesia	IDN	2010	11.14	9.15	4.05	2.19	5.02	3.33
Indonesia	IDN	2011	10.95	9.66	19.09	19.38	4.97	3.36
Indonesia	IDN	2012	12.37	6.96	5.88	0.88	7.71	4.27
Indonesia	IDN	2013	13.07	7.24	2.74	2.90	7.73	5.66
Indonesia	IDN	2016	16.83	8.80	12.62	10.91	12.73	10.08
Indonesia	IDN	2017	16.38	7.32	1.91	3.02	10.97	6.23
Indonesia	IDN	2018	16.64	8.47	18.13	19.56	10.97	6.33
Indonesia	IDN	2019	17.80	12.17	6.14	5.72	11.31	6.20
Indonesia	IDN	2020	17.14	13.20	1.87	1.48	10.94	7.65
Iran	IRN	2000	34.85	33.19	49.81	49.94	52.45	51.70

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Iran	IRN	2003	37.81	37.75	9.43	9.39	20.87	20.83
Iran	IRN	2004	36.56	36.52	16.03	16.05	16.55	16.51
Iran	IRN	2007	33.32	31.72	6.53	6.02	24.93	24.86
Iran	IRN	2008	19.77	21.78	19.80	19.82	24.04	23.99
Iran	IRN	2011	46.26	46.29	42.20	27.24	35.15	27.56
Iran	IRN	2019	31.09	31.10	8.55	8.59	17.17	17.36
Iran	IRN	2020	25.79	25.80	5.47	4.96	13.35	10.35
Ireland	IRL	2000	22.87	22.46	2.33	1.57	2.89	2.62
Ireland	IRL	2001	20.80	20.67	142.01	0.27	3.86	1.98
Ireland	IRL	2002	16.39	14.70	2.30	1.96	3.92	3.69
Ireland	IRL	2003	19.28	17.18	1.90	1.08	4.21	3.61
Ireland	IRL	2004	19.97	18.10	3.36	0.72	4.79	2.39
Ireland	IRL	2005	19.25	16.66	3.41	0.17	4.86	1.25
Ireland	IRL	2006	22.47	20.78	2.82	92.33	3.72	1.77
Ireland	IRL	2007	20.21	18.27	1.76	0.86	3.16	1.88
Ireland	IRL	2008	20.12	18.04	2.45	0.69	3.84	1.21
Ireland	IRL	2009	19.76	18.12	1.97	1.88	3.30	3.23
Ireland	IRL	2010	20.98	15.61	2.07	1.87	3.92	3.69
Ireland	IRL	2011	21.23	20.70	2.69	2.58	4.37	4.31
Ireland	IRL	2012	18.25	17.77	4.43	0.06	5.02	0.64
Ireland	IRL	2013	20.00	18.68	0.30	0.09	1.20	0.78
Ireland	IRL	2014	19.24	18.00	4.85	0.04	5.54	0.60
Ireland	IRL	2015	16.76	14.98	3.45	0.09	5.07	0.80
Ireland	IRL	2016	18.02	16.34	3.70	0.87	5.21	1.62
Ireland	IRL	2017	18.84	17.27	4.49	0.06	5.28	0.68
Ireland	IRL	2018	17.14	15.66	1.15	1.89	2.11	2.25
Ireland	IRL	2019	14.62	11.88	2.35	4.86	3.31	5.66
Ireland	IRL	2020	4.15	2.81	4.87	0.57	5.72	1.50
Israel	ISR	2000	19.71	17.45	29.03	29.03	2.30	2.24
Israel	ISR	2001	9.61	9.64	–	–	3.98	3.98
Israel	ISR	2002	10.02	10.02	0.14	0.15	3.95	3.95
Israel	ISR	2003	9.83	9.86	–	–	3.90	3.91
Israel	ISR	2004	8.24	5.02	–	–	3.97	3.87
Israel	ISR	2005	9.71	14.48	–	–	3.99	2.05
Israel	ISR	2006	9.51	8.55	1.03	0.43	4.03	1.31
Israel	ISR	2007	9.19	12.95	7.70	3.64	5.09	2.61
Israel	ISR	2008	8.66	11.39	4.91	3.88	4.69	3.59
Israel	ISR	2009	9.24	9.12	7.78	2.90	8.87	4.20
Israel	ISR	2011	7.81	7.84	0.09	0.16	1.58	3.36
Israel	ISR	2012	7.77	5.55	0.13	0.07	1.62	1.20
Israel	ISR	2013	7.74	6.12	9.36	2.15	3.08	0.98
Israel	ISR	2014	8.25	6.97	1.47	0.19	2.11	1.35
Israel	ISR	2015	8.27	7.05	34.71	3.33	1.46	3.18
Israel	ISR	2016	8.83	7.61	591.09	2.18	6.26	1.53
Israel	ISR	2017	7.84	4.85	1.17	0.98	2.75	2.36
Italy	ITA	2000	9.53	7.32	4.38	3.46	3.95	2.73
Italy	ITA	2001	10.13	9.47	1.07	0.81	3.22	2.72
Italy	ITA	2002	9.45	7.69	1.04	0.55	4.18	3.15
Italy	ITA	2003	9.30	7.96	1.79	1.01	4.14	2.99
Italy	ITA	2004	9.68	8.17	1.31	0.77	4.26	3.49
Italy	ITA	2005	9.42	7.99	1.60	0.91	4.02	2.92
Italy	ITA	2006	9.42	7.65	1.43	0.67	3.78	2.94
Italy	ITA	2007	8.79	7.17	1.64	0.90	3.88	2.99
Italy	ITA	2008	8.62	6.25	1.35	0.74	3.56	2.42
Italy	ITA	2009	8.98	6.44	0.80	0.20	2.87	1.86

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Italy	ITA	2010	9.01	7.14	0.81	0.44	2.39	1.71
Italy	ITA	2011	9.40	7.24	0.46	0.33	2.03	1.55
Italy	ITA	2012	9.44	7.05	0.59	0.14	1.84	1.00
Italy	ITA	2013	9.30	7.00	35.73	36.80	2.32	1.26
Italy	ITA	2014	8.73	6.86	1.20	0.95	2.87	2.28
Italy	ITA	2015	8.74	6.39	1.23	0.47	2.92	1.98
Italy	ITA	2016	10.46	8.34	0.98	0.66	3.09	2.32
Italy	ITA	2017	8.59	5.61	1.60	0.75	3.51	2.09
Italy	ITA	2018	8.46	6.11	1.43	0.96	3.55	2.55
Italy	ITA	2019	8.55	5.86	0.87	0.65	2.93	1.99
Italy	ITA	2020	5.78	4.00	0.56	0.33	2.26	1.72
Jamaica	JAM	2000	20.73	20.73	0.12	0.12	17.35	17.35
Jamaica	JAM	2001	21.39	21.39	0.19	0.16	18.24	18.27
Jamaica	JAM	2002	20.77	20.77	1.36	1.36	15.84	15.85
Jamaica	JAM	2003	23.08	23.08	150.95	150.95	18.33	18.33
Jamaica	JAM	2004	24.78	24.78	0.05	0.05	14.18	14.17
Jamaica	JAM	2006	25.07	25.00	0.16	0.12	14.73	14.61
Jamaica	JAM	2007	26.39	26.39	1.97	1.97	14.95	14.94
Jamaica	JAM	2010	22.47	18.51	1.91	4.10	14.88	10.82
Jamaica	JAM	2011	19.11	16.62	3.26	2.98	14.86	14.14
Jamaica	JAM	2012	25.09	24.17	9.71	9.35	15.61	13.27
Jamaica	JAM	2013	24.65	22.65	5.23	0.37	12.89	10.71
Jamaica	JAM	2015	21.83	21.90	4.55	4.56	14.46	14.43
Jamaica	JAM	2016	20.80	20.83	5.13	5.13	11.11	11.11
Jamaica	JAM	2020	21.04	16.01	4.78	3.94	4.97	3.06
Japan	JPN	2000	2.86	2.07	0.16	0.05	0.87	0.72
Japan	JPN	2001	2.86	2.55	0.13	0.05	0.84	0.73
Japan	JPN	2002	6.68	2.21	0.02	0.00	0.79	0.69
Japan	JPN	2003	5.45	2.05	0.07	0.12	0.77	0.76
Japan	JPN	2004	4.59	4.46	0.08	0.04	0.81	0.60
Japan	JPN	2005	4.30	4.10	0.10	0.10	0.89	0.81
Japan	JPN	2006	4.01	3.84	0.12	0.10	0.81	0.70
Japan	JPN	2007	3.68	3.20	–	–	0.70	0.54
Japan	JPN	2008	3.52	3.09	0.11	0.04	1.25	0.60
Japan	JPN	2009	3.73	1.94	0.12	0.31	1.92	0.73
Japan	JPN	2010	2.76	2.02	0.07	39.66	1.57	0.56
Japan	JPN	2011	3.82	3.50	30.64	0.01	0.66	0.49
Japan	JPN	2012	2.69	2.33	0.01	0.01	0.30	0.22
Japan	JPN	2013	2.06	1.65	28.88	0.01	0.33	0.18
Japan	JPN	2014	3.42	3.07	0.07	0.10	0.84	0.66
Japan	JPN	2015	2.56	1.98	0.03	0.02	0.45	0.35
Japan	JPN	2016	3.65	3.28	0.02	0.01	0.46	0.34
Japan	JPN	2017	3.52	3.09	0.06	0.03	0.70	0.55
Japan	JPN	2018	3.51	1.76	0.09	0.07	0.93	0.69
Japan	JPN	2019	3.68	2.54	0.13	0.06	0.91	0.64
Japan	JPN	2020	4.39	2.13	0.25	0.14	1.08	0.81
Jordan	JOR	2000	24.20	24.20	9.01	9.01	16.20	16.20
Jordan	JOR	2001	25.35	25.34	0.27	0.27	18.87	18.86
Jordan	JOR	2002	26.62	26.62	101.39	101.47	13.26	13.27
Jordan	JOR	2003	27.43	27.48	7.02	7.02	14.25	14.25
Jordan	JOR	2004	25.54	25.63	18.78	18.77	12.77	12.72
Jordan	JOR	2005	25.31	21.34	200.96	196.27	10.68	8.65
Jordan	JOR	2006	22.37	20.23	5.13	5.21	13.46	11.01
Jordan	JOR	2007	18.82	12.40	–	–	15.32	11.39
Jordan	JOR	2008	19.44	16.34	3.77	3.57	12.73	10.50

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Jordan	JOR	2009	11.99	8.04	0.13	0.14	1.62	1.17
Jordan	JOR	2010	13.98	12.66	30.01	0.01	29.09	2.18
Jordan	JOR	2011	14.87	13.47	5.03	5.14	10.58	13.30
Jordan	JOR	2012	13.72	13.70	3.26	3.14	9.71	8.89
Jordan	JOR	2013	14.28	14.44	6.16	16.93	10.02	18.32
Jordan	JOR	2014	19.12	10.32	8.56	3.72	13.02	5.13
Jordan	JOR	2015	15.41	9.62	0.28	1.91	7.59	6.28
Jordan	JOR	2016	15.13	10.42	–	15.62	20.33	9.73
Jordan	JOR	2017	26.28	12.16	30.04	0.01	27.65	3.64
Jordan	JOR	2020	24.45	10.12	–	0.48	2.23	0.79
Kazakhstan	KAZ	2004	4.85	4.85	0.02	0.02	1.92	1.92
Kazakhstan	KAZ	2008	3.67	1.29	3.21	2.18	5.45	4.20
Kazakhstan	KAZ	2010	5.88	4.81	3.52	2.10	7.09	4.92
Kazakhstan	KAZ	2011	5.96	1.89	5.77	1.99	5.36	4.21
Kazakhstan	KAZ	2012	6.13	1.76	4.19	1.98	6.03	3.33
Kazakhstan	KAZ	2013	6.37	4.50	3.26	2.80	5.73	5.07
Kazakhstan	KAZ	2014	4.88	1.76	5.66	3.93	3.88	3.32
Kazakhstan	KAZ	2015	4.69	4.47	1.97	0.40	3.84	2.24
Kazakhstan	KAZ	2016	4.68	4.22	2.40	0.73	3.39	2.11
Kazakhstan	KAZ	2017	4.48	1.76	1.94	1.10	3.32	2.89
Kazakhstan	KAZ	2018	4.61	1.66	1.96	1.00	2.80	2.38
Kazakhstan	KAZ	2019	4.76	1.85	2.92	1.79	2.99	2.52
Kazakhstan	KAZ	2020	4.83	1.20	2.99	1.96	3.36	1.96
Kenya	KEN	2000	23.23	23.23	6.08	6.08	6.25	6.22
Kenya	KEN	2001	26.27	26.27	11.92	11.91	17.13	17.29
Kenya	KEN	2002	25.18	25.18	7.32	7.32	8.21	8.21
Kenya	KEN	2004	22.94	22.94	9.15	9.16	16.60	16.60
Kenya	KEN	2005	23.38	20.77	27.20	27.49	29.11	29.03
Kenya	KEN	2006	26.95	22.21	39.56	35.47	41.37	42.08
Kenya	KEN	2007	29.93	29.01	31.67	31.86	–	–
Kenya	KEN	2008	29.61	26.08	1.53	1.63	–	9.72
Kenya	KEN	2009	28.69	26.25	0.46	1.97	19.69	20.42
Kenya	KEN	2010	28.15	24.99	49.93	3.51	3.12	4.85
Kenya	KEN	2011	28.68	24.82	3.34	3.85	3.82	4.88
Kenya	KEN	2012	25.90	20.25	6.79	7.88	12.20	12.37
Kenya	KEN	2013	27.71	24.76	1.14	3.11	7.78	8.90
Kenya	KEN	2014	25.85	9.72	1.91	0.72	4.89	5.58
Kenya	KEN	2015	26.59	26.06	47.34	48.00	2.58	2.08
Kenya	KEN	2016	26.71	22.65	8.41	1.21	–	2.63
Kenya	KEN	2018	27.45	18.33	1.38	3.34	2.19	0.06
Kenya	KEN	2019	26.72	12.04	9.45	9.69	0.74	1.97
Kenya	KEN	2020	28.40	25.37	0.17	0.18	–	3.16
Kiribati	KIR	2019	0.00	0.00	0.96	0.96	0.81	0.81
Kiribati	KIR	2020	0.00	0.00	1.82	1.82	1.47	1.47
Kyrgyzstan	KGZ	2000	9.14	9.14	10.06	10.06	12.63	12.63
Kyrgyzstan	KGZ	2001	6.41	6.42	8.72	–	7.55	7.20
Kyrgyzstan	KGZ	2002	8.70	8.69	10.54	10.51	–	–
Kyrgyzstan	KGZ	2003	10.35	10.35	4.77	4.77	17.30	17.30
Kyrgyzstan	KGZ	2006	6.48	6.08	11.36	11.56	–	–
Kyrgyzstan	KGZ	2007	5.91	1.64	8.74	3.73	7.71	7.27
Kyrgyzstan	KGZ	2008	4.59	2.52	4.55	0.00	0.01	6.21
Kyrgyzstan	KGZ	2009	6.79	4.77	–	–	9.00	7.28
Kyrgyzstan	KGZ	2010	5.54	3.46	–	–	9.38	8.11
Kyrgyzstan	KGZ	2011	6.78	5.87	7.65	0.27	8.07	7.12
Kyrgyzstan	KGZ	2012	4.89	2.21	–	–	7.81	5.53

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Kyrgyzstan	KGZ	2013	5.00	1.90	–	–	8.54	5.69
Kyrgyzstan	KGZ	2014	6.08	5.09	1.62	9.98	8.76	8.32
Kyrgyzstan	KGZ	2015	4.94	3.34	11.68	–	9.04	4.51
Kyrgyzstan	KGZ	2016	6.69	3.79	–	–	8.83	0.70
Kyrgyzstan	KGZ	2017	5.03	2.90	2.15	0.53	7.48	5.38
Kyrgyzstan	KGZ	2018	4.55	3.31	5.74	4.53	8.23	8.40
Kyrgyzstan	KGZ	2019	5.11	2.84	3.34	0.56	7.24	6.03
Kyrgyzstan	KGZ	2020	6.30	1.44	0.22	–	10.18	0.05
Latvia	LVA	2001	6.99	6.99	1806.58	1808.02	11.22	11.22
Latvia	LVA	2004	9.49	8.21	1.51	0.51	3.21	2.03
Latvia	LVA	2005	9.52	8.98	29.71	0.59	4.32	2.36
Latvia	LVA	2006	10.07	8.72	11.70	11.15	5.41	2.77
Latvia	LVA	2007	8.84	6.29	0.85	0.68	5.53	3.04
Latvia	LVA	2008	8.93	6.95	1.65	0.78	4.29	2.15
Latvia	LVA	2009	13.67	13.09	1.32	3.06	3.35	2.57
Latvia	LVA	2010	8.04	6.65	1.41	0.56	3.79	3.09
Latvia	LVA	2011	9.90	8.52	0.96	0.75	4.44	3.42
Latvia	LVA	2012	12.06	10.98	2.03	0.71	2.65	1.27
Latvia	LVA	2013	11.57	10.09	3.61	2.92	2.10	1.21
Latvia	LVA	2014	11.48	10.21	3.63	0.65	1.80	1.37
Latvia	LVA	2015	10.81	9.47	43.90	0.13	1.40	0.60
Latvia	LVA	2016	9.56	7.77	24.22	0.34	1.48	1.26
Latvia	LVA	2017	12.61	12.43	0.01	29.99	1.76	1.51
Latvia	LVA	2018	12.89	12.14	15.85	0.11	1.92	1.15
Latvia	LVA	2019	13.48	12.61	0.15	22.67	1.92	1.19
Latvia	LVA	2020	5.47	3.58	9.10	5.15	1.95	1.51
Lebanon	LBN	2000	29.83	30.02	3.30	3.30	12.63	12.63
Lebanon	LBN	2001	23.87	23.98	21.43	21.43	18.66	18.66
Lebanon	LBN	2002	21.95	19.14	9.52	9.50	16.19	16.16
Lebanon	LBN	2004	19.68	19.67	9.70	9.70	5.29	5.29
Lebanon	LBN	2005	18.18	18.18	3.59	3.59	5.56	5.56
Lebanon	LBN	2006	18.33	14.82	–	0.97	3.93	2.56
Lebanon	LBN	2007	18.07	15.84	3.49	3.18	4.13	3.54
Lebanon	LBN	2014	19.89	13.87	6.47	5.90	3.94	2.63
Lebanon	LBN	2015	20.97	13.83	0.83	0.68	4.60	3.17
Lebanon	LBN	2016	22.35	14.16	6.41	5.78	3.37	2.41
Lebanon	LBN	2017	22.20	14.56	0.45	0.54	3.51	2.19
Lebanon	LBN	2018	24.66	20.40	5.59	4.28	3.62	2.36
Lebanon	LBN	2019	22.44	11.30	5.55	5.40	2.32	1.55
Lebanon	LBN	2020	24.96	10.27	5.68	5.55	2.40	0.95
Lithuania	LTU	2002	17.99	8.11	87.19	86.71	8.95	5.01
Lithuania	LTU	2003	19.14	15.80	101.36	99.59	9.81	6.43
Lithuania	LTU	2004	14.51	13.65	1.22	65.42	8.28	5.86
Lithuania	LTU	2005	16.57	16.04	1.00	54.13	7.26	5.60
Lithuania	LTU	2006	16.25	15.74	1.55	1.22	6.75	5.45
Lithuania	LTU	2007	13.60	11.54	2.23	2.25	6.16	3.05
Lithuania	LTU	2008	14.67	13.31	1.66	2.17	6.57	4.00
Lithuania	LTU	2009	15.23	12.12	–	–	6.62	4.26
Lithuania	LTU	2010	14.18	11.81	0.50	–	7.06	3.33
Lithuania	LTU	2011	14.27	13.05	2.02	0.82	6.78	5.75
Lithuania	LTU	2012	13.39	12.29	0.69	0.49	6.26	5.35
Lithuania	LTU	2013	15.62	14.88	6.40	6.56	5.46	4.78
Lithuania	LTU	2014	15.53	15.03	4.21	4.13	5.32	4.37
Lithuania	LTU	2015	16.65	18.29	2.97	245.27	5.04	4.81
Lithuania	LTU	2016	20.86	18.34	0.03	0.25	4.78	2.50

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Lithuania	LTU	2017	19.72	19.16	1.17	0.32	5.30	4.75
Lithuania	LTU	2018	18.86	17.86	0.97	0.49	5.51	4.92
Lithuania	LTU	2019	19.09	18.08	–	0.98	5.07	4.69
Lithuania	LTU	2020	10.43	3.30	0.39	–	4.38	1.75
Luxembourg	LUX	2000	14.04	13.62	0.54	0.54	0.99	0.88
Luxembourg	LUX	2001	20.30	18.82	–	–	2.52	2.29
Luxembourg	LUX	2002	18.41	13.99	560.02	547.15	2.17	1.65
Luxembourg	LUX	2003	14.66	12.30	48.92	48.75	1.77	1.27
Luxembourg	LUX	2004	13.23	7.48	2.56	1.26	1.97	1.33
Luxembourg	LUX	2005	20.20	13.09	4.24	2.34	2.84	2.11
Luxembourg	LUX	2006	15.58	6.64	1.39	1.36	2.03	1.21
Luxembourg	LUX	2007	14.52	6.47	3.05	0.97	1.90	1.19
Luxembourg	LUX	2008	15.30	14.27	26.58	23.87	2.76	1.90
Luxembourg	LUX	2009	18.22	16.92	0.06	0.05	2.64	2.34
Luxembourg	LUX	2010	16.13	12.14	38.24	33.32	1.92	1.29
Luxembourg	LUX	2011	16.56	12.07	41.82	41.72	0.66	0.44
Luxembourg	LUX	2012	15.85	11.29	2.67	2.25	1.69	1.22
Luxembourg	LUX	2013	15.91	10.48	71.92	70.79	2.12	1.43
Luxembourg	LUX	2014	16.31	14.82	–	–	2.01	1.81
Luxembourg	LUX	2015	17.19	14.50	9.43	9.87	2.38	1.97
Luxembourg	LUX	2016	15.64	12.63	157.72	156.01	2.32	1.64
Luxembourg	LUX	2017	15.82	9.82	5.73	0.09	1.67	0.89
Luxembourg	LUX	2018	15.95	10.40	–	–	2.58	1.26
Luxembourg	LUX	2020	12.01	3.74	1.27	1.06	2.05	1.24
Madagascar	MDG	2000	12.98	12.98	0.05	0.05	1.93	1.93
Madagascar	MDG	2001	14.35	14.35	3.79	3.79	6.02	6.02
Madagascar	MDG	2002	11.86	11.86	0.00	0.00	0.50	0.50
Madagascar	MDG	2003	5.04	5.04	4.77	4.77	5.00	5.00
Madagascar	MDG	2004	6.28	6.28	1.52	1.52	4.86	4.86
Madagascar	MDG	2005	13.59	12.43	9.97	9.90	10.03	10.00
Madagascar	MDG	2006	14.78	11.28	21.54	57.56	19.97	0.78
Madagascar	MDG	2007	18.27	18.16	10.67	10.21	14.69	14.49
Madagascar	MDG	2008	14.27	10.54	11.43	3.72	13.15	11.86
Madagascar	MDG	2009	10.23	8.40	5.66	4.45	11.13	9.36
Madagascar	MDG	2010	11.83	9.88	19.16	0.21	19.80	1.29
Madagascar	MDG	2011	10.21	6.55	–	34.77	19.88	0.93
Madagascar	MDG	2012	13.07	11.43	4.81	4.75	11.90	6.39
Madagascar	MDG	2013	14.50	12.39	11.18	10.18	13.40	13.32
Madagascar	MDG	2014	14.79	9.11	5.39	5.05	13.47	11.20
Madagascar	MDG	2015	13.50	15.27	9.48	9.21	12.80	12.77
Madagascar	MDG	2016	15.90	15.75	5.39	0.83	15.07	14.17
Madagascar	MDG	2017	17.24	16.01	8.79	4.93	15.97	10.52
Madagascar	MDG	2018	13.99	10.61	7.05	4.71	10.10	7.18
Madagascar	MDG	2019	14.92	11.56	7.21	1.63	13.20	10.18
Madagascar	MDG	2020	16.52	7.29	7.37	2.90	17.06	14.89
Malawi	MWI	2000	22.63	22.63	23.46	23.46	22.42	22.42
Malawi	MWI	2001	22.47	13.39	–	–	22.37	12.18
Malawi	MWI	2003	22.08	22.08	24.63	24.63	21.28	21.28
Malawi	MWI	2006	22.35	16.86	23.79	16.58	22.16	19.74
Malawi	MWI	2008	22.27	22.16	22.16	22.25	22.12	21.97
Malawi	MWI	2009	22.08	22.00	22.14	24.00	22.36	22.33
Malawi	MWI	2010	22.15	6.17	210.68	207.93	11.18	1.16
Malawi	MWI	2011	24.54	12.71	25.29	5.05	24.93	12.38
Malawi	MWI	2012	22.39	8.20	–	0.19	22.67	8.13
Malawi	MWI	2013	22.28	19.70	–	–	22.26	20.67

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Malawi	MWI	2014	22.08	19.87	20.78	10.70	22.49	21.11
Malawi	MWI	2015	22.00	9.51	–	24.60	21.36	6.25
Malawi	MWI	2016	22.23	21.09	18.43	13.84	20.94	18.73
Malawi	MWI	2019	22.02	5.45	–	30.65	22.33	2.76
Malawi	MWI	2020	22.17	3.17	22.38	1.17	21.77	2.78
Malaysia	MYS	2000	13.65	13.62	4.25	4.35	6.33	6.39
Malaysia	MYS	2001	15.36	12.38	7.00	2.10	9.57	4.41
Malaysia	MYS	2002	14.43	10.07	3.54	3.43	7.99	7.30
Malaysia	MYS	2003	15.41	12.76	0.17	0.44	12.42	6.54
Malaysia	MYS	2004	13.78	13.51	0.75	0.72	2.99	2.89
Malaysia	MYS	2005	12.92	12.50	0.76	0.76	2.51	2.48
Malaysia	MYS	2006	12.83	8.85	0.06	0.20	2.28	1.60
Malaysia	MYS	2007	13.36	9.06	1.95	1.77	5.06	4.82
Malaysia	MYS	2008	12.52	8.75	54.22	0.24	2.83	1.31
Malaysia	MYS	2009	13.03	9.02	0.33	0.15	2.42	1.26
Malaysia	MYS	2010	11.90	8.67	35.20	0.05	1.72	1.06
Malaysia	MYS	2011	11.72	11.71	0.15	0.15	1.54	1.54
Malaysia	MYS	2012	11.88	12.07	0.05	0.05	0.75	0.75
Malaysia	MYS	2013	10.32	10.14	0.01	0.01	0.37	0.36
Malaysia	MYS	2014	10.11	5.35	0.27	0.09	1.59	0.82
Malaysia	MYS	2016	10.33	10.41	0.00	0.00	0.25	0.25
Malaysia	MYS	2020	9.58	9.68	0.08	0.08	1.79	1.79
Maldives	MDV	2000	24.86	24.85	–	–	24.94	24.94
Maldives	MDV	2001	23.16	22.56	5.74	5.73	24.74	24.63
Maldives	MDV	2002	23.65	23.64	25.03	24.99	24.71	24.71
Maldives	MDV	2003	20.98	20.98	41.23	41.25	26.13	26.11
Maldives	MDV	2004	22.23	22.23	0.62	0.62	12.38	12.38
Maldives	MDV	2005	23.48	23.48	7.46	7.46	35.81	35.81
Maldives	MDV	2006	22.55	22.55	24.61	24.61	15.76	15.76
Maldives	MDV	2008	23.36	23.18	20.43	18.73	21.97	21.56
Maldives	MDV	2009	25.39	24.26	8.49	7.37	19.56	14.85
Maldives	MDV	2010	15.65	15.65	16.09	16.17	21.76	21.75
Maldives	MDV	2011	17.13	16.92	24.83	24.74	21.21	19.69
Maldives	MDV	2012	18.97	18.50	4.33	4.40	7.57	7.20
Maldives	MDV	2013	14.45	13.64	4.07	4.00	5.95	5.89
Maldives	MDV	2014	14.46	13.24	3.82	3.99	13.47	12.61
Maldives	MDV	2015	11.37	10.58	10.86	7.99	13.68	12.76
Maldives	MDV	2016	3.92	4.05	3.52	3.23	79.19	80.67
Maldives	MDV	2017	3.27	3.17	5.84	2.04	32.91	32.89
Maldives	MDV	2018	23.08	22.88	5.88	6.69	23.14	23.21
Maldives	MDV	2019	6.81	6.28	8.99	7.03	20.52	20.93
Maldives	MDV	2020	25.64	18.76	5.38	5.73	20.79	20.10
Mali	MLI	2001	11.08	11.13	18.72	18.72	7.38	7.39
Mali	MLI	2002	13.44	13.44	5.29	5.29	4.59	4.59
Mali	MLI	2003	12.80	13.01	8.26	8.27	8.80	8.89
Mali	MLI	2004	14.55	14.55	8.71	8.71	6.08	6.08
Mali	MLI	2005	14.76	14.75	5.05	5.05	6.77	6.76
Mali	MLI	2006	12.65	10.36	81.44	81.12	5.39	5.29
Mali	MLI	2007	13.26	11.24	20.80	20.58	8.17	8.10
Mali	MLI	2008	11.02	9.82	15.62	15.62	10.77	10.70
Mali	MLI	2009	9.15	7.65	6.32	6.55	6.63	6.57
Mali	MLI	2010	10.05	9.99	6.05	6.04	9.74	9.70
Mali	MLI	2011	9.85	8.74	14.63	14.59	7.66	7.59
Mali	MLI	2012	9.00	7.05	5.73	6.28	7.68	5.91
Mali	MLI	2013	6.96	5.22	11.31	11.22	5.80	5.82

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Mali	MLI	2014	8.49	6.31	5.10	5.08	5.65	4.68
Mali	MLI	2015	8.04	5.01	6.35	5.28	7.35	6.34
Mali	MLI	2016	9.21	7.32	5.57	3.67	7.33	6.46
Mali	MLI	2017	10.12	10.12	8.37	5.81	9.32	7.54
Mali	MLI	2018	10.26	8.31	–	–	5.36	5.14
Mali	MLI	2019	10.12	8.00	–	9.54	8.86	7.07
Mali	MLI	2020	7.37	6.56	5.26	5.00	6.71	6.61
Malta	MLT	2000	12.07	11.60	3.95	3.95	5.65	5.66
Malta	MLT	2002	7.20	7.21	3.14	3.14	4.81	4.81
Malta	MLT	2003	6.82	6.83	4.37	4.37	4.48	4.49
Malta	MLT	2004	7.40	6.58	1.44	1.43	3.27	2.67
Malta	MLT	2005	8.32	6.71	1.50	0.06	3.46	1.42
Malta	MLT	2006	6.79	5.69	0.24	0.18	3.80	2.60
Malta	MLT	2007	4.87	3.38	2.07	0.82	2.65	1.49
Malta	MLT	2008	4.32	3.40	1.25	1.50	3.80	3.08
Malta	MLT	2009	5.76	4.58	0.37	1.55	2.19	1.90
Malta	MLT	2010	4.88	4.01	3.90	0.66	2.81	2.66
Malta	MLT	2011	4.40	3.07	2.58	0.88	2.58	1.46
Malta	MLT	2012	7.66	6.36	2.08	1.35	2.24	1.52
Malta	MLT	2013	5.29	4.32	0.28	0.03	2.11	1.41
Malta	MLT	2014	6.20	5.12	0.93	0.78	2.04	1.44
Malta	MLT	2015	6.32	3.73	3.35	3.26	2.82	2.07
Malta	MLT	2016	6.42	3.64	2.60	0.84	1.06	0.86
Malta	MLT	2017	7.14	3.46	3.33	1.65	3.69	2.68
Malta	MLT	2018	7.31	4.83	1.37	0.51	1.86	1.04
Malta	MLT	2019	7.36	6.04	4.50	4.73	2.76	1.89
Mauritius	MUS	2000	73.07	66.21	82.04	76.40	78.87	73.47
Mauritius	MUS	2001	69.70	69.70	101.66	101.67	78.62	78.62
Mauritius	MUS	2002	68.59	68.58	76.46	76.45	77.90	77.90
Mauritius	MUS	2004	68.75	44.03	–	33.95	77.66	59.81
Mauritius	MUS	2005	10.69	8.17	–	–	27.42	27.37
Mauritius	MUS	2006	16.57	9.57	5.10	8.25	20.55	10.97
Mauritius	MUS	2007	13.56	5.51	–	–	15.96	11.80
Mauritius	MUS	2008	12.05	8.88	–	–	15.87	8.47
Mauritius	MUS	2009	6.68	4.75	375.01	283.98	8.29	4.83
Mauritius	MUS	2010	5.84	4.37	127.66	0.06	8.06	3.48
Mauritius	MUS	2011	6.24	4.96	51.95	51.50	9.69	8.82
Mauritius	MUS	2012	5.97	5.26	269.89	269.47	9.57	8.29
Mauritius	MUS	2013	5.23	3.88	38.75	38.62	8.02	5.91
Mauritius	MUS	2014	5.26	3.34	13.09	12.89	6.33	3.64
Mauritius	MUS	2015	4.24	3.09	–	0.14	6.96	4.28
Mauritius	MUS	2016	4.12	3.69	103.86	103.85	3.85	2.82
Mauritius	MUS	2017	5.72	4.71	6.79	6.10	–	–
Mauritius	MUS	2018	3.67	1.47	0.03	273.23	1.49	2.32
Mauritius	MUS	2019	24.55	24.51	2.43	2.17	–	–
Mauritius	MUS	2020	27.26	1.81	1.11	0.11	–	1.20
Mexico	MEX	2000	20.83	20.84	11.78	11.71	14.58	14.56
Mexico	MEX	2001	20.23	20.25	1.80	1.68	13.33	13.35
Mexico	MEX	2002	20.21	12.47	15.21	310.96	15.37	5.02
Mexico	MEX	2003	20.17	20.22	10.43	10.44	14.26	14.30
Mexico	MEX	2004	20.38	11.47	8.63	3.39	13.19	8.08
Mexico	MEX	2005	25.14	10.38	7.19	4.06	14.51	7.25
Mexico	MEX	2006	25.81	9.32	7.29	0.54	13.53	4.14
Mexico	MEX	2007	25.55	19.47	5.77	1.40	14.13	5.20
Mexico	MEX	2008	24.01	10.22	24.08	21.33	14.35	9.07

Continued on next page...

Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Mexico	MEX	2009	21.41	7.43	–	35.17	12.65	4.12
Mexico	MEX	2010	19.50	15.95	13.18	9.45	7.66	5.97
Mexico	MEX	2011	19.61	19.56	8.77	8.76	10.97	10.90
Mexico	MEX	2012	16.91	16.90	0.91	0.91	3.31	3.31
Mexico	MEX	2013	17.82	17.83	0.82	1.19	3.01	3.12
Mexico	MEX	2014	17.44	5.30	4.97	1.17	8.99	2.34
Mexico	MEX	2015	17.24	17.27	2.50	2.60	4.86	5.00
Mexico	MEX	2016	17.59	17.59	1.79	1.96	4.99	4.99
Mexico	MEX	2017	17.17	9.92	0.84	0.70	1.93	1.33
Mexico	MEX	2018	16.66	13.84	6.54	6.43	8.43	8.27
Mongolia	MNG	2000	5.03	5.03	4.94	4.94	4.97	4.97
Mongolia	MNG	2001	6.76	6.76	11.86	11.86	–	–
Mongolia	MNG	2002	6.86	6.86	6.16	6.16	6.74	6.74
Mongolia	MNG	2003	4.99	4.99	2.33	2.33	4.27	4.27
Mongolia	MNG	2004	5.00	5.00	1.56	1.56	3.84	3.84
Mongolia	MNG	2005	4.87	4.87	3.82	3.81	2.50	2.46
Mongolia	MNG	2006	4.88	4.88	1.81	1.81	4.13	4.13
Mongolia	MNG	2007	5.01	5.01	0.04	5.00	0.98	5.00
Mongolia	MNG	2008	5.00	5.00	35.90	35.96	4.75	4.75
Mongolia	MNG	2009	5.02	5.02	3.37	2.48	4.54	–
Mongolia	MNG	2010	5.05	5.03	5.00	0.54	5.00	4.39
Mongolia	MNG	2011	4.95	4.96	2.19	3.01	3.64	4.19
Mongolia	MNG	2013	5.20	5.20	4.97	4.97	4.88	4.88
Mongolia	MNG	2015	5.06	5.18	4.39	4.19	4.78	4.77
Mongolia	MNG	2017	5.64	5.43	4.97	1.72	5.42	3.06
Mongolia	MNG	2018	7.68	7.08	2.96	1.05	5.32	4.18
Mongolia	MNG	2019	5.64	3.79	0.47	0.47	4.75	4.74
Mongolia	MNG	2020	5.47	5.36	5.00	2.63	5.48	4.67
Morocco	MAR	2000	46.18	46.18	43.06	43.06	43.40	43.40
Morocco	MAR	2001	46.14	46.14	10.20	10.18	40.88	40.92
Morocco	MAR	2002	45.46	45.46	41.17	42.68	42.90	43.01
Morocco	MAR	2003	45.10	45.10	40.95	40.98	43.51	43.52
Morocco	MAR	2005	43.78	34.00	1.03	53.70	42.52	25.89
Morocco	MAR	2006	42.44	30.46	9.08	72.67	39.20	31.68
Morocco	MAR	2007	41.88	29.25	2.89	–	28.59	19.70
Morocco	MAR	2008	36.91	24.47	2.43	0.30	3.52	1.92
Morocco	MAR	2009	29.98	19.67	3.96	0.15	23.57	15.64
Morocco	MAR	2010	28.03	28.03	104.21	104.22	24.99	24.99
Morocco	MAR	2011	26.13	26.13	1.26	1.26	16.82	16.82
Morocco	MAR	2012	21.72	18.34	0.71	64.56	19.32	11.29
Morocco	MAR	2014	17.60	11.40	0.92	63.02	19.36	11.36
Morocco	MAR	2015	18.21	11.39	57.56	56.79	19.33	11.17
Morocco	MAR	2016	18.06	12.56	–	11.62	18.70	10.16
Morocco	MAR	2017	17.39	10.79	1.41	19.24	7.80	4.45
Morocco	MAR	2019	18.26	12.50	17.36	17.31	5.56	2.94
Morocco	MAR	2020	17.38	10.41	2.18	19.73	6.32	3.70
Namibia	NAM	2001	22.47	16.09	18.40	15.43	15.92	11.28
Namibia	NAM	2002	22.28	14.58	7.14	1.78	20.54	13.35
Namibia	NAM	2004	21.30	15.17	–	8.33	28.04	20.32
Namibia	NAM	2005	17.75	8.71	79.86	91.13	19.85	9.21
Namibia	NAM	2006	19.48	8.66	30.97	0.24	28.61	16.33
Namibia	NAM	2007	17.28	10.21	18.01	2.16	12.74	6.44
Namibia	NAM	2008	15.78	9.68	14.59	13.22	17.78	12.07
Namibia	NAM	2009	19.01	13.63	5.75	3.99	18.79	11.90
Namibia	NAM	2010	19.09	17.40	17.81	0.20	13.72	11.25

Continued on next page...

Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Namibia	NAM	2011	16.27	11.49	–	–	12.72	7.80
Namibia	NAM	2012	15.75	10.13	12.24	1.71	13.09	7.41
Namibia	NAM	2013	16.77	9.78	0.23	18.58	11.43	6.58
Namibia	NAM	2014	15.65	10.78	29.87	73.36	15.46	12.87
Namibia	NAM	2015	13.73	8.90	10.79	0.34	9.10	4.78
Namibia	NAM	2016	15.74	9.97	3.58	0.93	10.67	6.18
Namibia	NAM	2017	14.64	12.21	7.61	5.07	16.31	15.29
Namibia	NAM	2018	11.26	8.31	24.62	15.06	28.84	12.39
Namibia	NAM	2019	14.04	6.54	12.31	143.51	7.34	3.47
Namibia	NAM	2020	14.89	9.44	6.45	0.10	8.97	5.03
Netherlands	NLD	2000	22.20	18.32	0.50	0.33	4.50	3.09
Netherlands	NLD	2001	21.72	18.73	59.55	59.15	4.80	4.02
Netherlands	NLD	2002	20.34	19.02	1.33	0.69	5.43	4.84
Netherlands	NLD	2003	20.20	15.64	1.69	1.71	4.87	3.15
Netherlands	NLD	2004	19.77	15.79	0.49	0.16	4.51	3.11
Netherlands	NLD	2005	19.54	16.72	0.27	0.19	3.76	2.65
Netherlands	NLD	2006	19.54	18.16	0.38	0.26	3.65	3.11
Netherlands	NLD	2007	18.05	15.67	0.59	0.31	3.28	2.68
Netherlands	NLD	2008	17.07	14.60	0.53	0.03	3.57	2.68
Netherlands	NLD	2009	18.83	13.58	2.48	0.01	2.40	1.32
Netherlands	NLD	2010	17.83	15.11	0.68	0.37	3.24	2.44
Netherlands	NLD	2011	18.83	16.71	0.44	0.16	2.92	2.28
Netherlands	NLD	2012	18.53	14.59	0.18	0.10	2.32	1.53
Netherlands	NLD	2013	18.87	14.90	0.38	0.27	2.89	2.15
Netherlands	NLD	2014	17.75	12.53	113.61	113.91	3.22	2.46
Netherlands	NLD	2015	16.40	11.79	0.48	0.41	4.14	3.27
Netherlands	NLD	2016	18.98	16.99	0.84	0.68	5.09	4.73
Netherlands	NLD	2017	18.84	13.14	0.70	0.46	3.43	2.07
Netherlands	NLD	2018	17.62	13.48	0.59	0.32	3.19	1.57
Netherlands	NLD	2019	15.52	13.26	0.64	0.37	3.10	2.68
Netherlands	NLD	2020	9.97	8.52	0.53	0.32	3.06	2.52
New Zealand	NZL	2000	3.25	2.10	0.02	0.01	2.52	1.81
New Zealand	NZL	2001	10.45	9.80	0.05	0.05	4.22	3.58
New Zealand	NZL	2002	6.19	5.22	0.00	60.08	1.12	0.96
New Zealand	NZL	2003	6.84	5.23	0.01	0.00	2.96	1.92
New Zealand	NZL	2004	7.02	4.54	0.24	0.23	4.69	3.78
New Zealand	NZL	2005	16.32	3.29	0.00	67.38	6.02	1.54
New Zealand	NZL	2006	17.50	4.18	0.00	0.00	9.42	1.63
New Zealand	NZL	2007	3.37	2.47	46.45	46.45	2.48	1.51
New Zealand	NZL	2008	2.75	2.12	0.00	49.62	1.47	1.24
New Zealand	NZL	2009	2.73	2.19	50.61	50.60	3.16	1.29
New Zealand	NZL	2010	2.75	2.04	0.30	0.25	3.33	1.96
New Zealand	NZL	2011	2.76	2.68	0.57	0.46	3.54	3.19
New Zealand	NZL	2012	2.81	2.67	0.96	0.72	2.35	1.93
New Zealand	NZL	2013	2.90	2.06	0.43	0.27	2.13	1.03
New Zealand	NZL	2014	2.98	2.30	0.62	0.08	1.52	0.52
New Zealand	NZL	2015	2.87	1.70	0.52	0.20	1.80	1.38
New Zealand	NZL	2016	2.81	2.33	0.11	0.00	3.33	1.87
New Zealand	NZL	2017	3.19	2.30	53.42	53.41	1.69	1.22
New Zealand	NZL	2019	2.70	1.84	0.75	38.77	2.15	1.06
New Zealand	NZL	2020	2.18	1.38	0.13	0.13	0.59	0.45
Nicaragua	NIC	2000	9.08	7.09	8.83	7.83	9.60	8.69
Nicaragua	NIC	2001	12.88	9.59	–	–	13.72	12.15
Nicaragua	NIC	2002	14.78	11.21	–	–	14.95	11.81
Nicaragua	NIC	2003	14.30	8.76	9.95	51.19	14.70	6.79

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Nicaragua	NIC	2004	16.15	12.73	496.40	1.95	14.79	6.61
Nicaragua	NIC	2005	17.39	17.39	3.54	3.54	14.62	14.62
Nicaragua	NIC	2006	16.24	14.60	0.43	0.30	13.28	5.40
Nicaragua	NIC	2007	23.78	21.71	13.98	15.54	14.27	12.71
Nicaragua	NIC	2008	20.53	19.31	3.26	0.93	8.99	8.23
Nicaragua	NIC	2009	22.80	12.43	15.54	12.53	14.26	11.53
Nicaragua	NIC	2010	22.46	16.49	8.72	0.40	9.33	5.31
Nicaragua	NIC	2011	21.15	16.93	15.29	14.12	13.37	12.70
Nicaragua	NIC	2012	18.82	15.30	–	–	11.75	10.16
Nicaragua	NIC	2013	18.93	9.25	15.20	3.86	14.41	2.62
Nicaragua	NIC	2014	19.71	13.70	63.30	54.99	12.97	8.76
Nicaragua	NIC	2015	17.41	12.53	–	9.30	12.88	9.47
Nicaragua	NIC	2016	17.34	7.94	–	–	14.31	10.32
Nicaragua	NIC	2017	18.46	11.44	–	23.35	14.44	5.68
Nicaragua	NIC	2018	16.80	7.31	0.31	104.74	12.59	7.18
Nicaragua	NIC	2019	18.84	12.07	–	–	14.17	9.91
Nicaragua	NIC	2020	19.63	10.92	–	–	13.17	8.88
Niger	NER	2001	16.41	16.40	21.39	21.38	18.77	18.76
Niger	NER	2002	15.71	15.70	20.11	20.11	19.74	19.74
Niger	NER	2003	16.74	16.69	22.25	22.25	19.43	19.41
Niger	NER	2004	14.47	15.01	8.59	8.59	15.73	15.80
Niger	NER	2005	11.94	11.92	12.67	12.67	11.52	11.51
Niger	NER	2006	13.33	11.79	13.46	10.18	8.43	7.87
Niger	NER	2007	15.45	13.86	13.28	13.46	5.50	5.12
Niger	NER	2008	14.42	12.22	10.26	8.40	10.52	10.22
Niger	NER	2009	11.38	9.99	4.67	4.71	6.75	5.37
Niger	NER	2010	13.73	12.71	5.05	5.04	5.94	5.58
Niger	NER	2011	11.42	11.21	5.83	5.57	8.71	7.69
Niger	NER	2012	13.46	10.44	7.88	6.37	11.87	8.37
Niger	NER	2013	18.21	18.19	29.81	29.81	16.86	16.86
Niger	NER	2014	11.37	9.74	–	–	6.84	6.90
Niger	NER	2015	16.29	7.15	18.48	13.92	20.11	6.97
Niger	NER	2016	13.36	9.72	5.24	5.94	11.43	9.12
Niger	NER	2017	17.03	17.03	4.30	4.30	5.53	5.53
Niger	NER	2018	17.29	14.92	10.11	10.05	17.32	17.16
Niger	NER	2019	23.53	22.08	20.44	16.73	20.16	19.34
Niger	NER	2020	20.40	13.80	–	4.09	8.40	6.77
Nigeria	NGA	2000	31.12	31.12	21.37	21.37	31.87	31.87
Nigeria	NGA	2001	45.12	45.12	24.86	24.86	28.69	28.69
Nigeria	NGA	2002	32.70	32.70	9.67	9.67	34.74	34.74
Nigeria	NGA	2003	21.95	21.95	43.34	43.34	24.80	24.80
Nigeria	NGA	2005	10.24	10.24	5.22	5.21	5.68	5.78
Nigeria	NGA	2006	11.21	11.21	8.48	8.48	12.39	12.39
Nigeria	NGA	2008	23.15	23.15	5.32	5.32	25.72	25.72
Nigeria	NGA	2009	20.96	20.96	5.76	5.76	7.74	7.74
Nigeria	NGA	2010	16.64	15.97	10.10	8.70	11.71	10.20
Nigeria	NGA	2011	12.95	12.95	5.11	5.11	9.06	9.06
Nigeria	NGA	2013	15.84	15.03	5.68	5.35	10.88	9.82
Nigeria	NGA	2014	19.17	20.01	8.31	6.94	14.73	10.66
Nigeria	NGA	2015	12.88	5.77	5.14	0.91	6.43	17.99
Nigeria	NGA	2016	15.15	9.53	5.26	0.36	11.31	8.33
Nigeria	NGA	2020	17.69	17.56	3.49	3.51	6.63	6.65
Macedonia	MKD	2001	34.47	28.08	23.99	21.71	34.02	24.10
Macedonia	MKD	2002	29.61	29.56	143.21	143.09	27.08	27.08
Macedonia	MKD	2004	26.62	26.62	–	–	29.74	29.74

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Macedonia	MKD	2005	23.99	15.90	21.84	22.50	25.09	14.77
Macedonia	MKD	2006	23.69	15.96	12.75	13.16	22.25	13.84
Macedonia	MKD	2007	21.82	20.44	6.47	4.25	17.81	15.26
Macedonia	MKD	2008	21.53	13.73	–	–	24.38	15.75
Macedonia	MKD	2009	24.27	12.40	–	–	23.45	9.22
Macedonia	MKD	2010	22.47	17.04	–	11.82	17.11	9.81
Macedonia	MKD	2011	21.70	15.55	1.72	8.40	11.29	5.56
Macedonia	MKD	2012	22.11	16.15	–	0.88	15.94	11.08
Macedonia	MKD	2013	21.50	21.57	92.12	92.12	14.85	14.87
Macedonia	MKD	2014	18.31	15.58	–	4.37	14.19	9.05
Macedonia	MKD	2015	15.68	9.25	–	3.80	12.66	6.93
Macedonia	MKD	2016	16.29	10.26	47.47	44.87	12.58	9.04
Macedonia	MKD	2017	16.40	11.97	–	–	14.76	8.78
Macedonia	MKD	2018	14.00	10.08	–	–	11.40	8.18
Macedonia	MKD	2019	15.10	11.09	–	–	10.27	7.85
Macedonia	MKD	2020	21.61	15.08	–	–	11.76	7.14
Oman	OMN	2001	57.46	57.46	5.97	5.97	17.74	17.74
Oman	OMN	2002	50.93	50.93	138.31	138.31	34.33	34.33
Oman	OMN	2003	4.81	4.81	4.45	4.45	5.29	5.29
Oman	OMN	2004	4.85	4.85	3.83	3.83	5.05	5.05
Oman	OMN	2005	4.86	3.51	4.98	2.25	5.02	4.22
Oman	OMN	2006	4.89	3.05	5.00	1.98	5.01	3.57
Oman	OMN	2007	4.89	3.32	5.07	2.37	4.60	4.31
Oman	OMN	2008	4.89	2.55	4.88	1.80	3.65	2.86
Oman	OMN	2009	5.84	2.00	5.41	283.08	4.61	3.52
Oman	OMN	2012	4.89	13.18	5.06	3.18	4.77	13.25
Oman	OMN	2013	4.85	1.96	2.30	125.76	4.80	3.49
Oman	OMN	2014	4.85	4.04	4.59	4.78	4.92	2.76
Oman	OMN	2015	4.99	4.03	4.90	6.46	4.94	3.00
Oman	OMN	2016	4.96	2.13	55.08	55.71	1.96	0.68
Oman	OMN	2017	4.97	2.32	5.20	0.58	4.98	4.57
Oman	OMN	2018	5.08	4.20	5.28	–	4.66	–
Oman	OMN	2019	7.73	2.42	5.05	4.99	4.82	4.52
Oman	OMN	2020	4.39	2.32	3.92	0.32	1.63	0.56
Panama	PAN	2000	11.82	11.83	2.90	3.04	8.82	8.85
Panama	PAN	2001	11.46	11.47	3.03	3.03	3.42	3.42
Panama	PAN	2002	12.08	11.53	5.36	5.36	7.68	7.54
Panama	PAN	2003	12.35	12.34	4.94	4.94	8.42	8.40
Panama	PAN	2004	10.70	10.58	5.02	5.02	6.23	6.40
Panama	PAN	2005	10.39	9.29	3.91	2.01	7.28	6.87
Panama	PAN	2006	9.85	8.70	0.50	0.41	3.15	2.64
Panama	PAN	2007	9.66	8.28	3.21	3.53	3.47	2.74
Panama	PAN	2008	10.28	7.30	48.64	48.93	3.71	2.88
Panama	PAN	2009	9.92	9.52	144.71	144.71	0.80	0.80
Panama	PAN	2010	9.17	9.18	1.74	1.74	3.77	3.78
Panama	PAN	2011	9.51	9.52	4.04	4.04	8.44	8.45
Panama	PAN	2012	9.57	9.58	–	–	9.95	9.95
Panama	PAN	2013	9.79	9.50	0.47	0.35	2.39	2.31
Panama	PAN	2015	9.94	9.92	7.95	7.98	8.28	8.25
Panama	PAN	2020	9.10	8.89	0.24	0.24	2.46	2.48
Guinea	PNG	2000	26.22	26.22	10.29	10.29	–	–
Guinea	PNG	2001	27.44	27.44	21.01	21.01	12.85	12.85
Guinea	PNG	2002	22.72	22.72	11.34	11.34	–	–
Guinea	PNG	2003	16.14	16.14	4.00	4.00	0.67	0.67
Guinea	PNG	2004	22.85	22.83	35.94	35.94	9.48	9.48

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Guinea	PNG	2005	19.66	19.66	7.39	7.39	2.58	2.58
Guinea	PNG	2006	18.72	17.68	0.04	0.05	3.49	3.37
Guinea	PNG	2007	13.10	13.08	0.04	0.04	3.92	3.91
Guinea	PNG	2008	20.85	18.29	36.23	36.23	4.36	–
Guinea	PNG	2009	15.62	15.62	19.62	19.62	0.31	0.31
Guinea	PNG	2010	18.15	18.04	0.01	96.22	11.92	11.86
Guinea	PNG	2012	14.91	14.91	220.30	220.30	6.83	6.83
Guinea	PNG	2013	18.18	14.94	3.48	2.63	13.03	10.37
Guinea	PNG	2018	18.05	13.99	4.29	4.23	12.97	11.25
Guinea	PNG	2019	4.44	28.53	32.07	32.41	0.53	6.09
Guinea	PNG	2020	48.93	48.93	51.17	51.17	8.62	8.62
Paraguay	PRY	2000	17.09	15.27	9.60	9.56	14.58	13.84
Paraguay	PRY	2001	16.44	16.44	5.52	5.52	14.93	14.93
Paraguay	PRY	2002	15.23	15.23	3.83	3.89	17.03	17.03
Paraguay	PRY	2003	13.65	13.65	10.38	10.38	15.08	15.08
Paraguay	PRY	2004	13.26	11.25	7.78	7.71	15.23	12.02
Paraguay	PRY	2005	12.10	8.82	14.41	0.53	15.39	11.41
Paraguay	PRY	2006	13.37	10.43	11.23	10.41	12.42	10.07
Paraguay	PRY	2007	13.33	7.19	4.90	2.37	10.86	7.15
Paraguay	PRY	2008	13.06	9.71	–	–	12.42	10.33
Paraguay	PRY	2009	11.27	8.80	5.74	0.09	11.05	6.57
Paraguay	PRY	2010	11.04	6.28	4.66	2.37	11.69	10.06
Paraguay	PRY	2011	12.81	9.30	5.50	2.49	10.77	5.99
Paraguay	PRY	2012	11.76	6.94	13.60	4.25	7.88	4.22
Paraguay	PRY	2013	13.49	8.83	1.45	4.99	7.54	4.41
Paraguay	PRY	2014	14.16	8.93	7.13	3.50	8.68	4.57
Paraguay	PRY	2015	12.52	9.18	5.56	17.33	9.48	6.58
Paraguay	PRY	2016	12.75	9.05	30.03	0.10	10.30	4.94
Paraguay	PRY	2017	13.93	8.10	15.95	8.67	15.74	8.50
Paraguay	PRY	2018	12.68	9.93	37.63	37.22	9.64	6.79
Paraguay	PRY	2019	12.98	9.47	28.26	28.05	10.00	7.24
Paraguay	PRY	2020	13.16	6.06	1.63	1.44	8.74	6.14
Peru	PER	2000	14.60	14.60	12.32	12.32	14.38	14.38
Peru	PER	2001	14.60	14.61	–	–	15.44	16.50
Peru	PER	2002	12.39	12.39	40.03	40.03	18.61	18.61
Peru	PER	2003	12.34	12.34	25.84	25.84	18.61	18.61
Peru	PER	2004	12.23	12.00	9.06	9.14	18.56	15.57
Peru	PER	2005	12.16	12.05	20.53	19.94	15.62	14.68
Peru	PER	2006	12.11	11.92	–	–	18.55	15.86
Peru	PER	2007	14.18	10.51	5.45	3.40	17.00	9.62
Peru	PER	2008	9.17	8.83	110.13	108.12	14.15	11.68
Peru	PER	2009	9.04	8.80	48.51	28.90	13.81	7.52
Peru	PER	2010	9.10	8.03	–	–	12.06	10.70
Peru	PER	2011	6.31	4.61	41.89	43.61	9.17	8.43
Peru	PER	2013	6.08	3.29	–	–	6.69	4.51
Peru	PER	2014	5.45	4.12	–	–	6.28	5.69
Peru	PER	2015	2.18	2.14	0.42	328.83	5.41	5.39
Peru	PER	2017	1.95	1.42	–	–	5.39	4.27
Peru	PER	2018	4.49	2.92	12.11	6.71	5.06	3.89
Peru	PER	2019	4.53	2.53	6.82	4.49	5.05	3.42
Peru	PER	2020	1.83	1.12	3.85	1.75	4.63	3.64
Philippines	PHL	2000	7.66	7.67	0.88	0.86	2.86	2.86
Philippines	PHL	2001	7.66	7.69	–	–	3.34	3.34
Philippines	PHL	2002	6.38	6.33	0.17	0.17	1.86	1.86
Philippines	PHL	2003	5.07	5.02	18.07	18.08	1.41	1.41

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Philippines	PHL	2004	7.14	5.42	0.86	0.76	2.51	2.05
Philippines	PHL	2005	6.65	4.84	0.95	0.88	2.37	2.13
Philippines	PHL	2006	6.49	4.74	2.00	1.96	3.24	3.15
Philippines	PHL	2007	6.19	4.98	0.36	0.29	1.74	1.53
Philippines	PHL	2008	6.95	5.27	3.30	3.55	2.18	1.77
Philippines	PHL	2009	6.89	5.12	0.61	0.39	1.83	1.31
Philippines	PHL	2010	6.99	4.44	1.99	1.67	4.22	3.92
Philippines	PHL	2011	6.72	4.38	0.71	0.51	2.91	2.22
Philippines	PHL	2012	6.02	4.20	0.20	0.77	2.83	2.01
Philippines	PHL	2013	5.89	3.37	0.62	0.33	2.62	1.98
Philippines	PHL	2015	6.42	3.68	1.49	0.31	2.82	1.11
Philippines	PHL	2016	6.37	4.09	6.78	2.20	3.11	1.89
Philippines	PHL	2017	6.55	3.50	1.72	81.12	3.47	1.36
Philippines	PHL	2018	7.25	4.42	1.10	0.17	3.70	0.80
Philippines	PHL	2019	7.23	4.95	70.84	70.96	1.71	0.96
Philippines	PHL	2020	7.05	3.93	0.17	0.05	1.35	0.78
Poland	POL	2000	18.22	13.94	5.24	5.11	9.77	6.87
Poland	POL	2001	21.10	15.54	–	–	10.55	4.23
Poland	POL	2002	20.29	18.21	2.36	3.22	10.47	7.04
Poland	POL	2003	26.73	22.26	8.81	0.20	12.77	7.18
Poland	POL	2004	11.62	9.07	1.17	0.02	5.05	3.03
Poland	POL	2005	15.50	13.30	–	–	5.37	3.69
Poland	POL	2006	16.32	15.51	0.05	0.05	1.77	0.91
Poland	POL	2007	15.41	9.57	129.18	128.47	4.94	3.64
Poland	POL	2008	15.41	11.89	1.91	1.50	4.79	3.15
Poland	POL	2009	18.30	17.26	73.15	72.63	4.72	3.46
Poland	POL	2010	19.83	11.42	3.85	1.81	4.98	3.71
Poland	POL	2011	19.57	13.96	–	–	5.06	3.71
Poland	POL	2012	20.11	14.09	–	–	3.96	2.61
Poland	POL	2013	20.54	19.85	0.15	0.06	3.37	2.65
Poland	POL	2014	20.89	20.57	87.96	88.15	3.12	2.79
Poland	POL	2015	21.60	19.64	3.42	3.71	3.93	3.10
Poland	POL	2016	21.21	13.22	2.32	1.66	4.35	2.89
Poland	POL	2017	23.69	18.64	0.05	112.14	4.76	3.73
Poland	POL	2018	23.64	23.03	106.64	0.63	5.31	4.50
Poland	POL	2019	24.18	24.24	1.50	0.70	5.92	5.12
Poland	POL	2020	22.88	15.36	1.05	0.52	5.80	3.78
Portugal	PRT	2000	12.29	9.52	252.80	232.60	6.35	4.69
Portugal	PRT	2001	11.82	10.53	0.00	0.01	5.27	4.35
Portugal	PRT	2002	10.46	9.06	2.49	1.77	4.84	3.51
Portugal	PRT	2003	10.95	9.83	3.96	2.66	4.22	3.46
Portugal	PRT	2004	11.71	8.79	1.52	0.58	4.49	3.13
Portugal	PRT	2005	12.99	9.37	4.62	1.84	4.69	3.24
Portugal	PRT	2006	12.77	11.42	0.25	0.31	3.02	2.30
Portugal	PRT	2007	12.65	8.15	754.49	676.29	4.07	2.38
Portugal	PRT	2008	12.96	6.92	0.25	0.14	2.11	1.31
Portugal	PRT	2009	14.32	11.03	87.87	87.86	5.87	4.19
Portugal	PRT	2010	14.39	11.69	8.35	5.80	6.03	4.59
Portugal	PRT	2011	13.67	9.05	3.43	1.53	4.11	2.39
Portugal	PRT	2012	13.04	10.33	2.52	0.95	3.99	2.55
Portugal	PRT	2013	13.94	10.75	–	4.00	4.88	3.15
Portugal	PRT	2014	14.21	12.39	20.05	7.86	6.29	4.90
Portugal	PRT	2015	14.70	13.38	–	5.78	6.36	5.24
Portugal	PRT	2016	14.59	13.54	1.41	0.45	5.57	4.36
Portugal	PRT	2017	13.67	8.44	–	–	5.53	4.03

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Portugal	PRT	2018	13.51	8.33	4.59	2.24	5.82	3.62
Portugal	PRT	2019	13.47	7.81	3.68	2.18	5.51	4.10
Portugal	PRT	2020	12.28	7.43	–	–	4.68	3.49
Korea	KOR	2000	8.75	8.66	1.32	1.11	5.25	5.21
Korea	KOR	2001	8.78	8.77	35.62	6.49	4.16	3.92
Korea	KOR	2002	8.38	8.49	0.89	0.95	2.62	2.69
Korea	KOR	2003	8.64	8.48	0.73	0.74	2.61	2.51
Korea	KOR	2004	7.98	7.78	0.73	0.68	2.45	2.42
Korea	KOR	2005	7.99	7.44	0.54	0.51	3.36	3.32
Korea	KOR	2006	7.94	7.43	1.09	0.93	3.37	3.30
Korea	KOR	2007	7.82	6.76	56.59	57.18	3.64	3.13
Korea	KOR	2008	7.55	6.98	36.97	36.45	3.33	2.21
Korea	KOR	2009	7.84	7.42	–	4.27	2.68	2.18
Korea	KOR	2010	7.68	6.96	–	12.91	2.25	1.95
Korea	KOR	2011	7.53	7.12	4.10	–	2.35	1.90
Korea	KOR	2012	7.77	7.24	0.95	0.59	3.04	2.69
Korea	KOR	2013	7.60	6.33	0.45	0.06	2.58	2.06
Korea	KOR	2014	7.84	5.81	0.67	0.33	2.68	1.74
Korea	KOR	2015	8.31	6.37	0.47	0.34	2.46	1.63
Korea	KOR	2016	8.78	8.08	0.85	0.40	2.11	1.69
Korea	KOR	2017	8.70	6.14	3.81	3.48	5.88	1.42
Korea	KOR	2018	8.42	5.74	1.86	–	6.04	0.83
Korea	KOR	2020	9.56	6.83	3.63	4.06	6.29	2.59
Moldova	MDA	2000	7.45	7.49	19.62	19.62	10.92	10.89
Moldova	MDA	2001	8.74	8.74	0.11	0.11	12.77	12.77
Moldova	MDA	2003	8.68	8.71	8.01	8.01	10.37	10.37
Moldova	MDA	2004	7.61	7.60	97.46	97.46	10.65	10.65
Moldova	MDA	2005	7.94	7.95	0.20	0.20	35.43	35.43
Moldova	MDA	2006	8.85	4.90	0.91	1.07	13.58	10.30
Moldova	MDA	2007	7.16	7.15	181.84	181.84	8.55	8.55
Moldova	MDA	2008	8.69	6.39	4.37	4.02	9.93	8.40
Moldova	MDA	2009	7.77	7.76	4.80	0.02	11.10	11.05
Moldova	MDA	2010	9.89	5.88	4.65	4.44	9.93	6.20
Moldova	MDA	2011	8.10	8.08	351.64	351.62	9.23	9.23
Moldova	MDA	2012	9.43	5.49	136.62	140.60	9.79	7.53
Moldova	MDA	2013	16.97	12.94	–	1.12	12.20	7.89
Moldova	MDA	2014	18.62	4.37	4.53	4.57	10.12	5.22
Moldova	MDA	2015	17.98	16.15	1.55	3.02	9.65	6.49
Moldova	MDA	2016	20.38	11.19	40.85	9.00	9.97	5.24
Moldova	MDA	2020	10.01	6.54	0.63	0.18	7.77	6.40
Russia	RUS	2001	9.63	9.63	17.89	17.90	10.72	10.73
Russia	RUS	2002	9.44	9.44	0.53	6.18	11.82	11.09
Russia	RUS	2005	8.34	8.34	13.81	11.58	13.80	11.82
Russia	RUS	2007	8.06	6.11	6.97	6.95	13.02	12.76
Russia	RUS	2008	7.19	6.00	12.18	12.05	13.13	12.97
Russia	RUS	2009	7.88	7.02	10.57	10.37	12.63	12.02
Russia	RUS	2010	8.06	5.72	10.70	7.93	8.08	5.31
Russia	RUS	2011	7.14	4.79	10.92	6.06	8.03	5.86
Russia	RUS	2012	7.03	6.44	9.59	9.61	8.81	8.48
Russia	RUS	2013	6.96	5.12	52.19	10.09	8.28	7.97
Russia	RUS	2014	6.20	4.28	15.42	0.01	5.81	2.04
Russia	RUS	2015	5.70	4.94	7.56	264.21	7.29	0.89
Russia	RUS	2016	5.86	3.56	106.21	0.01	3.15	2.67
Russia	RUS	2017	5.96	4.33	0.82	1.00	3.04	2.29
Russia	RUS	2018	5.98	4.41	0.39	–	4.64	3.98

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Russia	RUS	2019	6.32	4.39	60.00	60.22	3.52	2.51
Russia	RUS	2020	6.66	3.73	8.09	5.13	3.96	2.99
Rwanda	RWA	2000	13.97	14.40	10.12	9.97	12.72	15.85
Rwanda	RWA	2001	13.01	11.95	14.59	14.59	19.90	19.92
Rwanda	RWA	2002	28.62	28.62	42.10	42.10	24.64	24.64
Rwanda	RWA	2003	16.48	15.20	1.63	0.25	14.31	14.13
Rwanda	RWA	2005	24.28	23.14	9.17	9.41	20.23	19.82
Rwanda	RWA	2006	23.77	19.66	85.83	86.04	19.36	16.01
Rwanda	RWA	2008	18.82	14.42	18.36	16.42	24.01	23.06
Rwanda	RWA	2009	24.18	14.87	10.83	0.50	19.45	17.73
Rwanda	RWA	2010	31.96	24.72	0.30	5.77	34.88	32.92
Rwanda	RWA	2011	24.50	22.62	30.56	30.60	32.77	32.73
Rwanda	RWA	2012	22.07	17.57	1.96	0.56	16.40	15.00
Rwanda	RWA	2013	47.21	39.52	–	–	38.76	38.36
Rwanda	RWA	2014	43.05	29.60	0.81	0.56	34.74	33.11
Rwanda	RWA	2015	37.60	29.76	58.62	50.32	30.18	34.08
Rwanda	RWA	2016	35.25	21.44	0.45	63.91	12.68	–
Rwanda	RWA	2018	29.31	23.54	0.35	0.36	2.66	1.87
Rwanda	RWA	2019	27.35	22.73	64.31	64.33	0.16	–
Rwanda	RWA	2020	35.91	14.65	0.30	10.10	3.51	10.11
KNA	KNA	2000	13.41	13.41	14.62	14.62	6.12	6.12
KNA	KNA	2001	11.55	11.55	5.12	5.12	5.81	5.81
KNA	KNA	2002	9.01	9.01	9.12	9.12	5.61	5.61
KNA	KNA	2003	12.87	12.87	1.37	1.37	5.38	5.38
KNA	KNA	2006	9.45	7.22	5.02	5.01	5.06	5.03
KNA	KNA	2007	10.97	6.92	5.40	5.45	5.11	5.02
KNA	KNA	2008	13.49	11.29	5.24	5.44	5.45	5.37
KNA	KNA	2009	14.77	14.77	1.27	1.27	9.50	9.50
KNA	KNA	2010	10.31	8.15	–	–	5.20	4.99
KNA	KNA	2011	11.28	8.78	0.72	0.52	5.19	5.12
KNA	KNA	2012	9.92	7.61	39.54	39.61	5.22	5.14
KNA	KNA	2013	9.42	7.39	5.00	5.00	5.02	5.00
KNA	KNA	2014	15.93	11.96	5.99	1.06	7.82	5.06
KNA	KNA	2015	16.38	9.00	21.22	–	8.28	6.52
KNA	KNA	2016	17.02	8.83	5.47	4.99	10.56	2.89
KNA	KNA	2017	19.54	15.84	0.62	1.84	7.85	4.70
KNA	KNA	2018	21.91	19.69	1.71	2.98	17.92	17.05
KNA	KNA	2019	16.71	13.42	4.76	4.97	13.86	11.66
KNA	KNA	2020	28.42	28.20	5.02	5.00	16.14	15.46
Saint Lucia	LCA	2000	33.14	33.14	23.54	23.54	32.93	32.93
Saint Lucia	LCA	2001	19.02	19.02	0.06	0.06	14.27	14.27
Saint Lucia	LCA	2002	20.29	20.48	8.13	9.24	19.81	23.32
Saint Lucia	LCA	2003	18.86	18.86	3.49	3.49	19.75	19.75
Saint Lucia	LCA	2005	20.51	20.70	3.51	3.82	18.32	22.05
Saint Lucia	LCA	2006	16.33	13.99	6.48	5.95	17.69	17.27
Saint Lucia	LCA	2007	18.25	13.28	3.00	2.45	11.69	9.31
Saint Lucia	LCA	2010	30.92	30.92	16.08	4.00	19.80	19.63
Saint Lucia	LCA	2011	19.42	19.42	4.87	4.87	15.52	15.52
Saint Lucia	LCA	2012	26.60	22.52	22.17	22.17	21.01	20.06
Saint Lucia	LCA	2013	28.09	15.65	20.28	11.66	23.48	18.03
Saint Lucia	LCA	2014	21.63	16.64	17.60	39.28	23.34	13.89
Saint Lucia	LCA	2015	21.85	12.63	20.56	13.38	17.60	5.95
Saint Lucia	LCA	2016	21.72	15.22	19.07	11.06	20.33	15.26
Saint Lucia	LCA	2019	31.27	28.43	2.10	0.36	28.04	28.29
Saint Lucia	LCA	2020	31.77	26.53	8.24	2.69	23.31	9.18

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
VCT	VCT	2000	21.72	21.72	20.16	20.16	20.20	20.20
VCT	VCT	2001	17.22	17.22	3.96	3.96	19.06	19.06
VCT	VCT	2002	16.55	16.55	0.01	0.03	16.67	16.67
VCT	VCT	2003	16.57	16.57	1.77	1.77	19.36	19.36
VCT	VCT	2006	14.87	12.72	13.91	8.83	11.01	9.79
VCT	VCT	2007	17.02	14.87	19.94	18.85	19.89	19.84
VCT	VCT	2010	17.23	17.23	15.67	15.67	17.63	17.63
VCT	VCT	2011	17.63	17.63	17.35	17.35	16.93	16.93
VCT	VCT	2012	18.55	13.65	21.43	0.03	19.37	12.08
VCT	VCT	2013	18.35	12.81	13.42	13.51	18.93	13.28
VCT	VCT	2015	18.32	12.23	8.31	8.32	14.65	10.52
VCT	VCT	2016	82.50	78.83	21.14	6.60	63.60	55.62
VCT	VCT	2017	78.16	75.00	3.27	–	43.28	36.62
VCT	VCT	2018	79.35	38.20	33.79	5.73	71.30	48.21
VCT	VCT	2019	80.77	9.74	–	8.61	72.18	10.30
VCT	VCT	2020	71.27	6.74	27.79	–	24.41	8.80
SAU	SAU	2000	12.75	12.66	9.98	9.98	11.58	11.58
SAU	SAU	2001	13.70	13.23	23.09	23.01	8.67	8.82
SAU	SAU	2002	11.23	11.08	7.47	7.47	9.24	9.24
SAU	SAU	2003	11.00	10.92	5.00	5.00	4.84	4.86
SAU	SAU	2004	10.90	10.83	2.69	2.69	4.21	4.21
SAU	SAU	2005	5.25	4.74	7.29	6.50	3.80	3.77
SAU	SAU	2006	5.14	4.78	3.08	4.28	3.53	3.38
SAU	SAU	2007	4.96	4.48	3.87	1.49	3.64	3.49
SAU	SAU	2008	4.75	3.17	0.89	0.85	1.41	1.41
SAU	SAU	2009	4.65	3.54	1.20	1.17	1.95	1.96
SAU	SAU	2011	5.12	5.11	31.48	31.49	2.80	2.80
SAU	SAU	2012	4.44	4.02	2.53	334.32	3.04	2.72
SAU	SAU	2013	4.47	3.91	2.67	2.47	3.10	3.05
SAU	SAU	2014	4.22	3.86	5.45	5.52	3.03	2.93
SAU	SAU	2015	5.09	5.08	5.17	5.17	3.95	3.93
SAU	SAU	2017	4.40	3.83	10.34	–	1.69	1.56
SAU	SAU	2020	6.40	4.20	11.94	12.14	1.23	0.65
Senegal	SEN	2001	13.84	13.89	19.90	19.90	11.83	11.88
Senegal	SEN	2002	16.57	16.57	14.26	14.25	15.42	15.42
Senegal	SEN	2003	17.61	17.60	5.71	5.73	12.93	12.91
Senegal	SEN	2004	15.42	15.39	5.03	5.03	6.96	6.94
Senegal	SEN	2005	14.93	14.90	7.74	7.75	6.65	6.62
Senegal	SEN	2006	14.81	12.67	5.93	5.92	7.39	7.13
Senegal	SEN	2007	15.42	10.09	0.18	0.18	2.44	2.44
Senegal	SEN	2008	16.14	13.20	5.02	5.02	6.26	6.15
Senegal	SEN	2009	17.50	17.44	6.66	8.46	12.59	12.52
Senegal	SEN	2010	17.22	15.73	11.23	11.25	11.57	10.61
Senegal	SEN	2011	18.03	17.35	10.05	10.04	11.55	11.19
Senegal	SEN	2012	17.81	17.65	10.21	6.65	14.20	12.30
Senegal	SEN	2013	17.93	17.66	5.96	5.80	11.07	10.54
Senegal	SEN	2014	17.50	17.36	6.13	6.15	13.30	13.21
Senegal	SEN	2015	18.15	17.35	5.02	5.01	5.88	5.65
Senegal	SEN	2016	18.28	8.28	5.08	4.87	8.92	7.15
Senegal	SEN	2017	19.12	19.11	5.19	5.10	13.75	13.16
Senegal	SEN	2018	19.17	18.84	5.35	3.88	13.20	13.01
Senegal	SEN	2019	18.93	17.73	10.25	5.30	12.34	9.68
Senegal	SEN	2020	19.23	18.41	9.27	3.86	15.48	15.83
Seychelles	SYC	2000	46.66	46.66	4.36	4.36	13.10	13.10
Seychelles	SYC	2001	73.71	73.71	–	–	76.74	76.74

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Seychelles	SYC	2005	3.06	3.06	7.08	7.08	10.19	10.19
Seychelles	SYC	2006	2.36	2.36	0.94	0.94	4.40	4.40
Seychelles	SYC	2007	3.62	3.62	12.04	12.04	16.89	16.89
Seychelles	SYC	2015	19.74	2.15	153.09	0.02	9.37	0.91
Seychelles	SYC	2016	18.76	18.76	75.96	75.96	2.48	2.48
Seychelles	SYC	2017	4.82	4.82	26.85	26.85	0.31	0.31
Seychelles	SYC	2018	12.18	0.25	15.83	15.46	2.75	0.15
Seychelles	SYC	2019	13.15	13.14	26.83	26.83	1.73	1.73
Seychelles	SYC	2020	9.03	9.02	74.77	74.77	0.30	0.30
Singapore	SGP	2000	1.56	1.58	0.00	0.00	–	–
Singapore	SGP	2001	1.66	1.66	0.00	0.00	–	–
Singapore	SGP	2002	1.71	1.72	0.00	0.00	–	–
Singapore	SGP	2003	1.75	0.71	0.00	0.00	–	–
Singapore	SGP	2004	0.00	0.00	67.61	67.61	0.01	0.01
Singapore	SGP	2005	1.32	0.00	0.00	61.67	0.06	0.01
Singapore	SGP	2006	1.22	1.22	0.00	0.00	0.02	0.02
Singapore	SGP	2007	1.51	1.51	0.00	0.00	0.03	0.03
Singapore	SGP	2008	0.14	0.14	133.30	133.30	0.01	0.01
Singapore	SGP	2009	0.00	0.00	87.61	87.61	0.00	0.00
Singapore	SGP	2010	0.21	0.00	146.86	146.86	0.01	0.00
Singapore	SGP	2011	0.00	0.00	125.89	125.89	0.00	0.00
Singapore	SGP	2012	0.00	0.00	161.36	161.36	0.00	0.00
Singapore	SGP	2013	0.33	0.00	166.19	166.19	0.02	0.00
Singapore	SGP	2014	0.32	0.00	161.53	161.53	0.01	0.00
Singapore	SGP	2015	0.00	0.00	165.71	165.71	0.00	0.00
Singapore	SGP	2016	0.35	0.09	166.60	166.60	0.01	0.00
Singapore	SGP	2017	0.00	0.27	151.66	151.66	0.00	0.01
Singapore	SGP	2018	0.26	0.08	14.98	0.00	0.02	0.01
Singapore	SGP	2019	0.29	0.00	25.79	25.79	0.03	0.00
Singapore	SGP	2020	0.00	0.00	34.33	34.33	0.00	0.00
Slovakia	SVK	2002	13.49	13.51	92.88	0.86	2.61	2.73
Slovakia	SVK	2004	11.57	10.55	2.13	0.69	6.32	4.10
Slovakia	SVK	2005	12.18	11.02	3.28	1.22	6.98	5.53
Slovakia	SVK	2006	12.72	11.91	2.15	0.46	4.19	2.25
Slovakia	SVK	2007	11.95	10.81	0.70	0.81	5.13	2.94
Slovakia	SVK	2008	11.09	9.94	0.06	0.44	4.89	3.65
Slovakia	SVK	2009	11.81	11.03	0.99	0.35	4.83	3.38
Slovakia	SVK	2010	12.13	11.33	0.35	0.25	4.38	2.95
Slovakia	SVK	2011	12.34	11.55	11.03	2.91	4.00	2.52
Slovakia	SVK	2012	11.54	10.34	8.05	3.56	3.24	2.15
Slovakia	SVK	2013	10.98	9.56	6.30	–	2.79	1.80
Slovakia	SVK	2014	10.52	9.65	43.82	0.08	3.35	2.05
Slovakia	SVK	2015	10.49	9.06	6.58	–	3.20	2.63
Slovakia	SVK	2016	10.67	9.02	–	–	3.26	2.13
Slovakia	SVK	2017	10.05	9.12	1.48	0.49	2.77	1.92
Slovakia	SVK	2018	9.56	7.51	0.44	–	3.17	2.66
Slovakia	SVK	2019	9.34	7.85	1.18	0.68	3.19	2.41
Slovakia	SVK	2020	6.59	5.19	–	–	3.39	2.84
Slovenia	SVN	2001	15.05	15.04	6.04	6.99	9.88	9.91
Slovenia	SVN	2002	16.08	8.40	11.17	1.85	5.63	2.75
Slovenia	SVN	2003	15.73	10.08	24.41	12.66	4.46	3.53
Slovenia	SVN	2004	8.40	7.46	21.81	21.62	3.99	3.75
Slovenia	SVN	2005	9.61	8.00	22.79	19.55	2.62	1.69
Slovenia	SVN	2006	9.58	8.74	3.50	2.88	4.66	3.81
Slovenia	SVN	2007	8.91	7.82	–	–	5.25	3.75

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Slovenia	SVN	2008	8.86	7.60	0.69	0.59	5.52	4.08
Slovenia	SVN	2009	9.27	7.83	5.53	4.42	5.65	3.77
Slovenia	SVN	2010	9.75	8.40	1.29	1.42	3.78	2.57
Slovenia	SVN	2011	9.88	8.49	205.55	187.55	3.79	2.59
Slovenia	SVN	2012	9.41	8.71	2.44	1.23	3.42	2.05
Slovenia	SVN	2013	10.16	7.16	0.15	0.41	3.31	2.39
Slovenia	SVN	2014	9.92	7.43	1.02	0.64	3.24	2.02
Slovenia	SVN	2015	9.88	7.63	4.58	0.03	3.52	2.26
Slovenia	SVN	2016	9.28	7.04	4.37	3.53	3.40	2.24
Slovenia	SVN	2017	10.76	9.26	7.20	4.75	3.06	1.92
Slovenia	SVN	2018	10.50	7.68	1.87	0.56	2.91	1.93
Slovenia	SVN	2019	11.49	6.17	0.13	0.36	3.90	1.27
Slovenia	SVN	2020	5.22	3.31	0.05	–	1.50	1.06
ZAF	ZAF	2000	16.85	16.15	3.70	3.65	16.88	15.93
ZAF	ZAF	2001	16.66	15.40	0.84	1.72	11.87	8.63
ZAF	ZAF	2002	15.34	10.65	0.68	0.86	12.13	11.12
ZAF	ZAF	2003	14.88	13.11	0.56	0.73	8.33	4.94
ZAF	ZAF	2004	14.91	14.43	1.77	1.74	8.14	7.96
ZAF	ZAF	2005	15.98	13.71	–	–	11.37	9.19
ZAF	ZAF	2006	14.98	8.78	26.92	27.19	6.99	5.63
ZAF	ZAF	2007	13.20	6.54	8.13	6.43	17.62	15.15
ZAF	ZAF	2008	14.38	11.16	329.53	316.30	9.74	7.62
ZAF	ZAF	2009	16.58	10.43	19.79	20.25	4.28	2.60
ZAF	ZAF	2010	15.96	9.69	32.76	15.79	14.02	7.23
ZAF	ZAF	2011	15.64	10.51	0.07	0.02	6.50	3.95
ZAF	ZAF	2012	16.05	10.45	3.15	2.36	10.24	7.09
ZAF	ZAF	2013	15.57	10.37	4.42	1.08	9.01	5.53
ZAF	ZAF	2014	15.52	10.77	0.15	0.72	4.61	4.05
ZAF	ZAF	2015	16.70	10.78	45.23	45.29	8.20	5.41
ZAF	ZAF	2016	17.90	13.40	48.15	48.15	1.66	1.28
ZAF	ZAF	2017	17.25	10.40	0.84	47.84	6.19	3.39
ZAF	ZAF	2018	17.15	12.46	0.01	0.02	3.36	1.94
ZAF	ZAF	2019	17.02	7.69	115.90	111.76	7.47	4.59
ZAF	ZAF	2020	15.27	11.23	0.22	1.86	5.58	2.64
Spain	ESP	2000	14.20	10.69	3.69	2.90	5.32	3.76
Spain	ESP	2001	14.74	12.56	71.86	71.51	4.94	4.37
Spain	ESP	2002	14.10	12.41	4.35	2.16	4.15	2.80
Spain	ESP	2003	13.29	11.32	0.37	0.65	4.48	3.22
Spain	ESP	2004	13.02	11.39	2.73	2.43	5.14	4.01
Spain	ESP	2005	12.35	10.11	348.91	342.29	2.46	1.72
Spain	ESP	2006	12.56	9.68	1.05	0.59	3.86	2.95
Spain	ESP	2007	11.84	10.07	0.97	0.68	2.30	1.79
Spain	ESP	2008	11.50	9.30	0.72	78.44	4.02	2.66
Spain	ESP	2009	11.69	9.66	1.33	0.72	3.33	2.24
Spain	ESP	2010	12.24	8.50	0.32	154.64	3.16	2.19
Spain	ESP	2011	12.56	10.09	0.90	0.26	2.94	1.94
Spain	ESP	2012	12.41	7.45	0.67	0.30	2.73	1.77
Spain	ESP	2013	11.84	9.16	0.82	0.55	2.61	1.92
Spain	ESP	2014	12.35	8.18	0.87	0.40	3.10	1.82
Spain	ESP	2015	11.98	8.75	0.92	0.53	3.12	2.03
Spain	ESP	2016	12.21	10.18	1.43	0.54	3.62	2.62
Spain	ESP	2017	11.49	9.27	1.27	0.40	3.69	2.80
Spain	ESP	2018	10.65	8.81	4.38	2.16	3.54	2.42
Spain	ESP	2019	11.40	9.34	0.99	0.45	2.96	1.99
Spain	ESP	2020	7.09	4.80	0.90	0.52	2.67	1.78

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Sri Lanka	LKA	2000	27.40	27.36	–	–	16.03	15.93
Sri Lanka	LKA	2001	32.64	32.53	0.60	0.57	16.53	16.46
Sri Lanka	LKA	2003	27.66	27.54	20.27	19.52	16.61	16.15
Sri Lanka	LKA	2004	28.54	28.33	18.03	17.93	19.64	18.85
Sri Lanka	LKA	2005	30.75	30.50	24.35	25.03	20.33	19.43
Sri Lanka	LKA	2006	33.11	31.51	–	–	19.83	18.88
Sri Lanka	LKA	2007	29.08	28.57	19.69	20.56	17.16	16.12
Sri Lanka	LKA	2008	31.34	30.88	12.44	12.11	23.66	23.04
Sri Lanka	LKA	2009	30.09	29.41	–	–	22.33	21.51
Sri Lanka	LKA	2010	29.95	29.38	–	–	17.87	16.46
Sri Lanka	LKA	2011	31.98	31.23	–	21.15	21.99	20.51
Sri Lanka	LKA	2012	30.66	29.97	–	16.52	21.78	19.77
Sri Lanka	LKA	2014	32.06	28.41	17.86	15.19	20.32	17.55
Sri Lanka	LKA	2015	30.93	30.66	–	19.83	24.38	23.39
Sri Lanka	LKA	2017	29.27	28.36	51.99	51.97	6.56	6.32
Sri Lanka	LKA	2018	34.69	32.73	13.28	60.28	15.69	13.93
Sri Lanka	LKA	2019	30.57	29.06	15.17	86.34	15.31	11.28
Sri Lanka	LKA	2020	37.21	35.16	14.53	41.55	15.64	9.16
Sudan	SDN	2002	14.99	24.56	15.18	14.61	10.23	–
Sudan	SDN	2006	16.58	9.44	9.92	–	7.02	8.96
Sudan	SDN	2008	13.42	9.91	10.01	4.56	–	–
Sudan	SDN	2009	29.39	7.12	10.15	3.22	8.62	3.14
Sudan	SDN	2010	16.01	11.49	9.99	0.09	9.97	–
Sudan	SDN	2011	18.71	12.39	11.61	8.31	11.02	4.05
Sudan	SDN	2012	20.35	15.97	10.28	9.97	7.47	–
Sudan	SDN	2013	21.97	11.13	8.20	5.95	7.96	6.68
Sudan	SDN	2017	25.17	9.96	3.27	3.04	–	–
Sudan	SDN	2020	26.84	0.00	8.95	181.50	15.76	1.22
Suriname	SUR	2000	34.88	34.88	–	–	47.18	47.18
Suriname	SUR	2007	31.49	28.87	5.02	4.98	5.65	5.19
Suriname	SUR	2009	15.99	16.31	4.93	4.93	16.83	16.81
Suriname	SUR	2010	10.83	6.67	3.56	3.86	5.15	5.02
Suriname	SUR	2011	14.39	14.39	4.41	4.41	10.19	10.19
Suriname	SUR	2012	16.82	18.06	1.36	1.67	26.50	26.61
Suriname	SUR	2013	16.50	4.93	7.43	8.11	20.40	–
Suriname	SUR	2018	8.20	7.34	5.32	4.68	7.10	5.50
Suriname	SUR	2019	12.13	7.57	1.72	94.30	16.66	16.36
Suriname	SUR	2020	8.08	8.11	4.25	331.48	13.35	13.20
Sweden	SWE	2000	6.98	5.65	1.33	1.30	1.92	1.39
Sweden	SWE	2001	9.15	8.64	1.36	1.38	2.05	1.66
Sweden	SWE	2002	6.85	4.62	2.50	163.36	2.47	1.04
Sweden	SWE	2003	6.69	5.13	1.65	1.70	2.49	1.64
Sweden	SWE	2004	7.83	6.60	0.72	0.46	2.13	1.75
Sweden	SWE	2005	7.23	5.45	0.84	0.56	2.09	1.39
Sweden	SWE	2006	7.48	5.96	60.62	60.52	1.98	1.66
Sweden	SWE	2007	7.58	5.85	1.20	149.21	1.95	1.55
Sweden	SWE	2008	7.24	5.00	1.05	0.21	2.28	1.07
Sweden	SWE	2009	7.26	5.43	29.16	29.11	1.67	1.22
Sweden	SWE	2010	7.82	6.42	1.07	0.87	1.87	1.56
Sweden	SWE	2011	7.56	5.94	0.50	0.38	2.30	1.55
Sweden	SWE	2012	7.55	6.14	1.10	0.20	2.20	1.42
Sweden	SWE	2013	6.99	5.44	4.39	4.28	3.72	3.53
Sweden	SWE	2014	7.60	5.81	161.62	161.06	3.48	3.05
Sweden	SWE	2015	7.86	5.93	2.03	1.60	3.26	2.66
Sweden	SWE	2016	8.39	6.86	1.11	0.64	2.26	1.50

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Sweden	SWE	2017	7.75	5.75	0.91	0.52	1.49	0.95
Sweden	SWE	2018	7.52	5.42	0.43	0.08	2.00	1.27
Sweden	SWE	2019	7.45	5.63	0.65	0.26	1.97	1.04
Sweden	SWE	2020	5.64	3.56	0.52	0.33	1.96	1.33
Turkey	TUR	2000	21.44	18.16	131.17	122.41	7.96	4.37
Turkey	TUR	2001	18.34	15.36	7.70	6.01	7.68	5.71
Turkey	TUR	2002	15.36	13.13	–	–	8.56	6.81
Turkey	TUR	2003	15.19	12.43	138.47	113.77	7.80	3.41
Turkey	TUR	2004	14.94	12.52	6.43	4.47	8.32	6.63
Turkey	TUR	2005	15.25	13.17	4.24	2.73	8.36	6.95
Turkey	TUR	2006	14.47	12.41	5.79	2.65	4.47	3.18
Turkey	TUR	2007	13.87	11.14	1.90	0.98	4.69	3.49
Turkey	TUR	2008	13.57	10.37	0.49	0.32	2.07	1.66
Turkey	TUR	2009	13.84	11.77	0.32	0.13	1.84	1.37
Turkey	TUR	2010	14.32	12.00	0.55	0.33	2.44	1.96
Turkey	TUR	2011	14.79	12.52	0.33	0.11	2.83	2.26
Turkey	TUR	2013	15.01	11.90	0.24	0.12	1.68	1.21
Turkey	TUR	2015	16.23	14.81	0.13	0.06	1.36	1.11
Turkey	TUR	2016	15.30	11.60	0.09	0.07	1.10	0.90
Turkey	TUR	2017	16.11	12.39	0.17	0.09	1.24	0.95
Turkey	TUR	2018	15.30	12.62	0.38	0.17	1.91	1.31
Turkey	TUR	2019	14.48	12.97	25.73	26.08	2.10	1.54
Turkey	TUR	2020	15.89	13.76	0.38	0.19	2.00	1.73
Thailand	THA	2000	33.64	33.65	5.35	5.35	11.50	11.51
Thailand	THA	2001	28.33	23.58	2.70	1.98	11.04	8.47
Thailand	THA	2003	28.65	23.81	–	26.46	9.88	6.93
Thailand	THA	2004	24.84	24.88	3.89	6.30	8.21	10.57
Thailand	THA	2005	24.79	18.92	3.37	1.73	10.62	7.72
Thailand	THA	2006	28.00	23.66	1.56	63.97	10.85	7.86
Thailand	THA	2007	24.12	24.19	2874.90	0.01	6.88	6.82
Thailand	THA	2008	24.46	24.43	2.95	2.94	8.73	8.58
Thailand	THA	2009	23.93	23.83	–	–	4.26	4.18
Thailand	THA	2010	25.52	25.47	0.22	0.22	5.15	5.06
Thailand	THA	2011	23.58	23.51	2.97	2.97	4.46	4.41
Thailand	THA	2013	39.77	39.75	429.08	429.09	5.63	5.65
Thailand	THA	2014	28.72	22.22	6.53	–	14.52	4.31
Thailand	THA	2015	30.37	24.27	32.99	32.47	9.58	7.77
Togo	TGO	2000	13.18	13.20	3.42	3.42	8.02	8.03
Togo	TGO	2001	13.74	13.73	–	–	10.12	10.11
Togo	TGO	2002	12.33	12.33	4.72	4.72	3.33	3.33
Togo	TGO	2003	12.96	12.96	20.28	20.28	13.26	13.26
Togo	TGO	2004	13.63	13.63	16.04	16.04	1.96	1.96
Togo	TGO	2005	13.85	13.44	18.10	17.67	13.70	13.33
Togo	TGO	2006	13.89	13.38	18.40	18.98	12.40	12.08
Togo	TGO	2007	14.12	13.66	12.29	11.90	12.22	13.05
Togo	TGO	2008	13.89	13.15	17.42	17.34	16.96	17.35
Togo	TGO	2009	14.45	13.28	20.08	–	19.32	7.53
Togo	TGO	2010	14.02	13.80	15.22	15.10	17.01	16.90
Togo	TGO	2011	14.33	13.38	0.79	–	18.13	15.81
Togo	TGO	2012	12.49	11.71	15.02	10.64	15.03	10.23
Togo	TGO	2013	12.70	11.23	10.15	10.07	14.36	12.27
Togo	TGO	2014	11.35	11.96	9.38	10.21	9.83	9.98
Togo	TGO	2015	13.40	11.35	10.12	9.91	21.95	21.69
Togo	TGO	2016	13.46	12.67	19.66	19.76	18.06	18.02
Togo	TGO	2017	18.64	18.64	28.71	28.71	23.55	23.54

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Togo	TGO	2018	19.49	13.00	–	–	25.37	10.04
Togo	TGO	2019	18.97	17.87	–	–	23.15	22.66
Togo	TGO	2020	18.08	9.92	–	–	22.26	11.68
TTO	TTO	2001	13.33	13.34	6.08	6.08	9.30	9.33
TTO	TTO	2002	14.22	14.23	12.14	12.14	14.36	14.36
TTO	TTO	2003	16.13	16.13	7.83	7.83	9.81	9.81
TTO	TTO	2004	14.98	13.74	3.02	2.97	8.14	7.81
TTO	TTO	2005	12.82	12.83	13.77	13.77	6.74	6.74
TTO	TTO	2006	12.17	11.09	2.06	1.15	13.34	12.84
TTO	TTO	2007	14.86	13.57	1.61	1.34	15.32	14.10
TTO	TTO	2008	18.80	17.27	0.21	0.19	17.63	16.64
TTO	TTO	2010	13.38	13.43	6.33	6.33	18.53	18.54
TTO	TTO	2011	14.82	14.78	9.94	9.94	18.52	18.51
TTO	TTO	2012	15.52	14.91	8.20	8.19	15.35	15.19
TTO	TTO	2013	15.45	14.97	5.81	5.80	18.46	18.28
Tunisia	TUN	2000	38.21	38.21	6.40	6.40	40.66	40.66
Tunisia	TUN	2002	38.38	38.32	10.95	10.87	40.29	40.29
Tunisia	TUN	2003	38.16	38.15	236.70	10.03	41.23	41.28
Tunisia	TUN	2004	54.41	37.76	7.27	11.49	41.37	41.21
Tunisia	TUN	2005	44.83	21.48	28.46	33.24	42.45	39.60
Tunisia	TUN	2006	48.13	46.28	19.86	11.09	40.21	36.55
Tunisia	TUN	2008	34.81	34.81	7.52	7.51	39.35	39.35
Tunisia	TUN	2009	29.82	29.78	–	–	33.99	33.99
Tunisia	TUN	2010	27.30	27.30	8.88	11.01	33.88	33.90
Tunisia	TUN	2011	24.03	23.99	18.87	17.97	27.90	27.89
Tunisia	TUN	2012	23.89	23.98	32.12	32.04	27.70	27.71
Tunisia	TUN	2013	23.78	15.11	14.26	6.29	26.77	13.93
Tunisia	TUN	2015	25.47	18.62	9.93	10.67	26.86	16.04
Tunisia	TUN	2016	19.53	19.53	28.28	28.28	13.43	13.43
Uganda	UGA	2000	14.23	8.20	7.00	6.95	7.36	6.79
Uganda	UGA	2001	13.08	7.66	3.19	3.18	7.45	4.52
Uganda	UGA	2002	13.14	8.21	5.53	5.45	5.57	4.28
Uganda	UGA	2003	13.07	6.73	0.31	0.23	1.24	0.55
Uganda	UGA	2004	12.30	11.82	0.20	0.18	5.02	4.61
Uganda	UGA	2005	27.36	14.04	1.41	1.48	17.88	3.03
Uganda	UGA	2006	28.70	10.76	0.88	0.82	5.21	4.60
Uganda	UGA	2007	29.76	19.27	0.00	20051.24	4.69	3.36
Uganda	UGA	2008	32.82	24.47	8.69	8.30	13.06	13.04
Uganda	UGA	2009	34.55	29.56	11.17	3.51	3.68	–
Uganda	UGA	2010	33.61	14.82	0.05	0.03	6.29	3.22
Uganda	UGA	2011	32.97	23.26	6.32	9.23	14.88	14.81
Uganda	UGA	2012	37.94	32.62	117.63	117.44	–	3.34
Uganda	UGA	2013	37.18	33.47	1.88	0.25	7.33	2.90
Uganda	UGA	2014	37.13	34.08	2.97	1.53	10.07	–
Uganda	UGA	2015	37.53	28.13	0.32	0.27	6.98	3.24
Uganda	UGA	2016	39.00	31.93	0.61	0.50	–	–
Uganda	UGA	2018	32.92	27.97	1.80	0.16	7.69	7.87
Uganda	UGA	2019	34.13	26.51	5.95	5.32	–	–
Uganda	UGA	2020	40.67	33.57	6.92	6.94	–	–
Ukraine	UKR	2002	8.28	8.25	2.32	10.13	7.58	7.59
Ukraine	UKR	2006	5.65	4.40	0.08	0.03	6.08	4.60
Ukraine	UKR	2007	5.98	5.96	1.30	0.99	4.93	4.91
Ukraine	UKR	2008	5.55	3.87	0.48	0.73	4.67	3.76
Ukraine	UKR	2009	6.47	4.47	6.11	9.22	3.40	3.35
Ukraine	UKR	2010	4.75	4.73	0.00	0.00	1.90	1.89

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Ukraine	UKR	2011	5.68	4.48	1.23	0.00	2.68	2.20
Ukraine	UKR	2012	6.37	5.50	71.26	0.02	2.00	1.33
Ukraine	UKR	2013	6.45	6.46	1.62	1.62	3.07	3.06
Ukraine	UKR	2014	7.00	5.21	1.04	0.46	3.52	2.87
Ukraine	UKR	2015	7.02	3.74	3.15	2.98	2.46	2.12
Ukraine	UKR	2016	9.23	9.24	3.83	3.83	3.31	3.31
Ukraine	UKR	2017	7.08	5.64	1.81	0.76	1.29	–
Ukraine	UKR	2018	7.67	6.15	0.78	0.54	2.44	–
Ukraine	UKR	2019	8.54	5.33	1.00	0.15	3.34	2.24
Ukraine	UKR	2020	8.94	3.52	1.86	0.30	3.57	2.01
UAE	ARE	2005	5.75	5.02	5.87	4.83	10.16	9.96
UAE	ARE	2007	5.18	4.35	5.60	11.90	4.97	4.27
UAE	ARE	2008	5.44	3.73	4.31	2.37	4.66	3.32
UAE	ARE	2009	5.23	3.78	4.94	3.33	4.70	3.64
UAE	ARE	2011	5.26	5.26	6.51	–	2.92	4.78
UAE	ARE	2012	6.69	5.98	4.52	1.41	5.56	4.26
UAE	ARE	2013	7.33	4.24	0.28	43.64	5.92	3.91
UAE	ARE	2014	5.76	4.37	220.40	215.63	5.20	4.00
UAE	ARE	2015	5.29	4.15	3.66	2.39	4.65	3.88
UAE	ARE	2016	5.31	4.24	4.40	3.03	4.75	4.31
UAE	ARE	2017	4.74	3.85	4.43	3.79	4.65	4.03
UAE	ARE	2018	5.29	3.08	3.79	2.64	4.54	3.91
UAE	ARE	2019	6.71	5.86	2.62	1.51	9.09	8.78
UAE	ARE	2020	5.60	3.68	2.76	2.52	4.66	3.98
Tanzania	TZA	2000	15.54	15.54	18.55	18.55	11.82	11.82
Tanzania	TZA	2003	15.79	13.34	2.58	2.00	8.86	7.86
Tanzania	TZA	2005	29.06	23.33	4.83	0.40	18.59	13.44
Tanzania	TZA	2006	26.69	17.11	0.95	0.63	8.94	4.94
Tanzania	TZA	2007	29.80	23.84	0.33	0.07	10.39	5.85
Tanzania	TZA	2008	29.43	19.48	0.55	0.28	17.62	15.12
Tanzania	TZA	2009	26.06	17.74	25.15	25.36	17.49	8.19
Tanzania	TZA	2010	28.81	24.46	14.95	7.45	12.21	10.17
Tanzania	TZA	2011	27.93	22.97	9.61	11.08	14.55	8.85
Tanzania	TZA	2012	28.52	24.33	40.89	40.84	32.03	31.96
Tanzania	TZA	2013	36.52	33.46	17.60	17.30	8.09	7.31
Tanzania	TZA	2014	32.63	29.07	–	1.83	17.39	3.64
Tanzania	TZA	2015	28.62	26.88	8.37	8.62	17.81	17.31
Tanzania	TZA	2016	28.97	15.75	238.45	237.54	13.22	12.47
Tanzania	TZA	2018	27.14	20.40	254.87	252.33	9.12	6.52
Tanzania	TZA	2019	29.23	16.63	0.03	0.03	3.24	1.81
Tanzania	TZA	2020	28.77	10.87	269.33	271.57	2.51	1.23
Uruguay	URY	2000	15.30	10.56	7.02	4.21	10.65	9.31
Uruguay	URY	2001	14.43	9.82	6.66	4.87	13.52	11.15
Uruguay	URY	2002	12.63	12.62	7.98	7.11	11.53	11.51
Uruguay	URY	2004	12.16	9.00	5.03	4.58	13.77	10.36
Uruguay	URY	2005	11.06	6.68	5.41	3.44	12.38	10.24
Uruguay	URY	2006	10.66	5.28	3.62	3.14	12.25	9.05
Uruguay	URY	2007	10.45	8.41	5.51	5.28	12.42	9.79
Uruguay	URY	2008	9.59	4.92	8.36	0.13	10.33	6.71
Uruguay	URY	2009	9.53	6.76	8.59	7.47	9.94	7.51
Uruguay	URY	2010	9.23	6.05	9.77	7.69	11.74	8.48
Uruguay	URY	2011	9.32	5.44	–	0.40	10.06	6.94
Uruguay	URY	2012	8.88	4.75	2.21	2.19	3.47	2.59
Uruguay	URY	2013	9.08	3.97	5.84	2.52	12.32	8.93
Uruguay	URY	2014	9.21	4.95	1.63	10.32	7.97	9.57

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Uruguay	URY	2015	9.38	4.08	10.10	3.49	9.98	4.12
Uruguay	URY	2016	8.29	5.48	5.35	4.68	10.84	6.51
Uruguay	URY	2017	8.65	4.83	2.70	1.57	9.39	6.61
Uruguay	URY	2018	8.39	4.77	10.03	2.03	9.98	2.29
Uruguay	URY	2019	8.72	4.90	–	1.84	8.68	6.81
Uruguay	URY	2020	8.99	4.95	1.68	1.72	8.92	8.84
USA	USA	2000	11.15	9.96	0.11	0.13	1.19	1.61
USA	USA	2001	11.37	11.72	0.60	0.39	1.62	3.31
USA	USA	2002	10.29	12.90	0.93	5.37	1.62	7.17
USA	USA	2003	10.17	9.57	97.22	1.12	2.77	2.29
USA	USA	2004	9.10	7.89	0.67	1.07	2.86	5.03
USA	USA	2005	6.78	6.54	174.20	1.96	1.95	7.88
USA	USA	2006	6.65	4.80	0.81	0.19	1.47	0.76
USA	USA	2007	5.83	5.74	0.63	0.06	1.42	0.56
USA	USA	2008	5.72	4.33	0.93	0.04	1.57	0.66
USA	USA	2009	5.66	3.78	1.24	0.14	1.77	0.71
USA	USA	2010	5.28	4.01	2.23	0.36	2.67	0.65
USA	USA	2011	5.09	3.65	1.56	1.56	1.84	1.79
USA	USA	2012	5.74	3.42	1.14	0.16	1.10	0.45
USA	USA	2013	6.28	4.53	0.87	0.81	0.90	0.82
USA	USA	2014	6.05	4.47	1.06	0.04	1.24	0.38
USA	USA	2015	6.18	5.20	0.84	0.04	1.26	0.49
USA	USA	2016	5.59	2.80	–	–	0.67	0.22
USA	USA	2017	5.97	4.25	113.67	113.67	1.31	1.25
USA	USA	2018	5.42	3.91	0.14	0.01	0.79	0.18
USA	USA	2019	6.36	4.58	0.58	0.16	1.11	0.54
USA	USA	2020	6.33	2.49	0.49	0.47	1.19	1.13
Venezuela	VEN	2000	15.09	15.09	8.03	7.96	12.48	12.41
Venezuela	VEN	2001	15.41	15.41	14.99	14.99	14.70	14.70
Venezuela	VEN	2002	17.11	17.10	8.64	8.64	12.67	12.69
Venezuela	VEN	2003	16.87	16.88	13.25	13.25	12.36	12.35
Venezuela	VEN	2004	14.35	11.99	6.66	2.75	12.68	10.65
Venezuela	VEN	2005	14.06	12.33	7.01	6.44	13.01	11.13
Venezuela	VEN	2006	13.40	10.89	8.76	7.86	14.05	10.42
Venezuela	VEN	2007	13.24	10.49	8.35	5.51	13.30	10.83
Venezuela	VEN	2008	12.10	7.95	9.80	5.08	13.02	9.95
Venezuela	VEN	2009	12.31	9.73	14.96	9.53	14.56	11.15
Venezuela	VEN	2010	13.56	11.73	10.61	8.05	13.77	8.94
Venezuela	VEN	2011	13.35	11.94	15.06	8.18	10.87	8.29
Venezuela	VEN	2012	12.66	10.71	12.31	7.04	10.16	6.89
Venezuela	VEN	2013	12.74	10.08	9.84	8.35	10.23	9.02
Venezuela	VEN	2014	10.21	7.98	14.69	10.66	12.67	10.54
Zambia	ZMB	2001	31.23	20.46	29.74	28.43	20.92	18.43
Zambia	ZMB	2002	21.85	21.86	–	–	33.47	33.47
Zambia	ZMB	2003	17.94	15.55	5.93	3.48	18.29	15.68
Zambia	ZMB	2004	18.00	17.20	16.43	16.43	19.60	19.60
Zambia	ZMB	2005	15.37	14.75	–	–	5.43	5.38
Zambia	ZMB	2006	35.14	20.22	25.71	25.70	21.29	21.26
Zambia	ZMB	2007	15.75	15.57	10.51	10.39	18.03	17.76
Zambia	ZMB	2008	15.78	5.31	8.04	0.35	20.15	17.33
Zambia	ZMB	2009	15.21	3.15	4.20	4.15	6.97	5.93
Zambia	ZMB	2010	16.11	11.28	13.39	11.18	6.73	5.30
Zambia	ZMB	2011	15.04	1.88	3.01	–	8.49	6.45
Zambia	ZMB	2012	16.49	13.15	23.13	–	22.04	20.52
Zambia	ZMB	2013	15.74	14.24	3.77	2.17	11.25	7.76

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Table A1: TRIs (Continued)

Country	ISO 3	Year	Small Country		Optimal Tariff		Free Trade	
			MFN	Non-MFN	MFN	Non-MFN	MFN	Non-MFN
Zambia	ZMB	2015	14.98	2.98	–	3.87	6.18	2.76
Zambia	ZMB	2016	17.78	13.12	–	–	5.63	5.08
Zambia	ZMB	2018	13.89	6.83	–	10.24	5.06	0.55
Zambia	ZMB	2020	16.05	9.70	0.22	0.17	8.19	6.19
Zimbabwe	ZWE	2000	22.86	22.86	–	–	13.55	13.55
Zimbabwe	ZWE	2001	18.04	18.04	9.97	9.97	10.43	10.43
Zimbabwe	ZWE	2002	23.35	23.35	10.01	10.01	13.00	13.00
Zimbabwe	ZWE	2003	27.71	27.71	2.49	2.49	12.34	12.33
Zimbabwe	ZWE	2007	29.98	8.92	9.74	10.04	11.25	10.34
Zimbabwe	ZWE	2010	31.37	31.37	9.63	9.63	11.65	11.65
Zimbabwe	ZWE	2011	28.85	5.21	9.69	0.58	11.18	–
Zimbabwe	ZWE	2012	28.43	28.43	9.63	9.63	11.79	11.79
Zimbabwe	ZWE	2015	29.06	28.50	9.83	9.85	10.67	10.62
Zimbabwe	ZWE	2016	11.27	4.31	9.98	1.44	10.05	1.99

Notes. “Small Country” refers to the TRI under the small country assumption. “Optimal Tariff” indicates the TRI level approximated from the optimal tariff points. “Free Trade” represents the TRI approximated from the free trade. All values are expressed as percentages.